# QUANTUM LEFSCHETZ THEOREM REVISITED

JUN WANG

ABSTRACT. Let $X$ be any smooth proper Deligne-Mumford stack with projective coarse moduli, and $Y$ be a smooth complete intersection in $X$ associated with a direct sum of semi-positive line bundles. We will introduce a useful and broad class known as admissible series for discussing quantum Lefschetz theorem. For any admissible series on the Givental's Lagrangian cone of $X$, we will show that a hypergeometric modification of the series lies on the Lagrangian cone of $Y$. This confirms a prediction from Coates-Corti-Iritani-Tseng about the genus zero quantum Lefschetz theorem beyond convexity. In our quantum Lefschetz theorem, we use extended variables to formulate the hypergeometric modification, which may be of self-independent interest.

CONTENTS



## 1. INTRODUCTION

*Gromov-Witten (GW) invariants* count the (virtual) number of stable maps from curves to a target variety/stack $X$ with prescribed conditions. *Quantum Lefschetz* is one of the central topics in Gromov-Witten theory and it compares the GW invariants of a complete intersection and its ambient space. On the other hand, in Givental's formalism [Giv04, CG07, Tse10], all information of genus-zero Gromov-Witten invariants is encoded in an *overruled Lagrangian cone* $\mathcal{L}_X$ sitting inside an infinite dimensional symplectic vector space. Consequently, a natural approach to the quantum Lefschetz problem in genus zero is to find an explicit slice on Givental's Lagrangian cone of the complete intersection using a given slice on Givental's Lagrangian cone of the ambient space. This approach is often referred to as a *mirror theorem* in the literature. The earliest quantum Lefschetz theorem can be traced back to the verification of the famous mirror conjecture for quintic threefolds [CDLOGP91, Giv96, LLY99]. Since then, more cases related to quantum Lefschetz have been proved. In genus zero, Givental-style quantum Lefschetz theorems can be divided into two categories:

---

*Date*: Friday 3rd July, 2026.
2020 *Mathematics Subject Classification.* Primary 14N35
Keywords: Gromov-Witten invariants; quantum Lefschetz; convexity; stack.

(1) The complete intersection $Y$ is associated with a direct sum of *convex* line bundles over the ambient space $X$, see e.g.,[Giv98, Lee01, CG07, CCIT19]. The reason for requiring the convexity condition is that we need to apply the so-called Quantum Lefschetz hyperplane principle proved in [KKP03]. This principle expresses the Gromov-Witten invariants of the complete intersection as an Euler class of a certain bundle over the moduli stack of genus-zero stable maps to $X$. See also [CGI+12] for some examples where this principle can fail if we drop the convexity condition, where the examples are associated with semi-positive line bundles in the sense of this paper. We note that convexity is a very rare condition when $X$ is a not a variety.

(2) When convexity fails, proving a quantum Lefschetz theorem is much more difficult. There has been significant progress on this case recently by solving the genus zero quasimap wall-crossing conjecture [Wan25, Zho22]. However, it's important to note that we require the ambient space to be a GIT quotient, which is a prerequisite for applying quasimap theory [CFKM14, CCFK15].

The main objective of this paper is to prove a genus-zero[1]quantum Lefschetz theorem that goes beyond the scope of the two categories mentioned previously, where the ambient space can be a non-GIT target and the convexity condition may not hold. Actually our scope of quantum Lefschetz theorem is quite broad, there are not many restrictions on the ambient space and we only assume that the complete intersection is associated with semi-positive line bundles, which is much weaker than convexity (see [CGI+12]). The proof relies on (and generalizes) the *recursive relation*[2] of a point on the Givental's Lagrangian cone discovered by the author in [Wan25], which was used to demonstrate the genus-zero quasimap wall-crossing conjecture for abelian GIT quotients. However, we need to carefully pick a space carried with a $\mathbb{C}^*-$action which could provide the recursive relation we want; in our case we will use a *root-stack modification of the space of deformation to the normal cone*. This is *different* from the spaces used in loc.cit and it has the advantage of generalizing the quantum Lefschetz theorem proved in loc.cit to non-GIT targets.

1.1. **Main theorem.** Let $X$ be a smooth proper Deligne-Mumford stack over $\mathbb{C}$ with projective coarse moduli. Let $\bar{I}_\mu X$ be the rigidified (cyclotomic) inertia stack of $X$, where a $\mathbb{C}-$point of $\bar{I}_\mu X$ can be written as a pair $(x, g)$ where $x$ is a $\mathbb{C}$-point of $X$ and $g$ is an element in the isotropy group $Aut(x)$. Denote by $\mathscr{C}$ the finite set of connected components of $\bar{I}_\mu X$ with $\bar{I}_c X$ being the component corresponding to the element $c \in \mathscr{C}$. The involution on $\bar{I}_\mu X$ by sending $(x, g)$ to $(x, g^{-1})$ also induces an involution on $\mathscr{C}$, we write $c^{-1}$ to be image of $c$ under the involution.

The *overruled Lagrangian cone* $\mathcal{L}_X$ introduced by A.Givental is comprised by the so-called (big) $J$-function:

$$J^X(q, \mathbf{t}(z), -z) := -z\mathbb{1}_X + \mathbf{t}(z) + \sum_{\beta \in \mathrm{Eff}(X)} \sum_{m \geq 0} \frac{q^\beta}{m!} \phi^\alpha \langle \mathbf{t}(\bar{\psi}_1), \cdots, \mathbf{t}(\bar{\psi}_m), \frac{\phi_\alpha}{-z - \bar{\psi}_\star} \rangle^X_{0,[m]\cup\star,\beta}, \tag{1.1}$$

where the input[3] $\mathbf{t}(z) \in H^*_{CR}(X, \mathbb{C})[z]$ is a polynomial in $z$ with coefficients in the Chen-Ruan cohomology $H^*_{CR}(X, \mathbb{C}) := H^*(\bar{I}_\mu X, \mathbb{C})$, the notation $q^\beta$ stands for the Novikov variable corresponding to the degree $\beta$ in the cone $\mathrm{Eff}(X)$ of effective curve classes of $X$. We note that such a choice of

[1]Recently, there have been significant advance in the high genus case, where the failure of convexity provides one main obstacle to the computation of high genus GW invariants, see e.g., [Zin08, GJR17, GJR18, CGL21, CJR22, LR22, CJR21] and their references therein.

[2]Our approach towards the main Theorem 4.8 doesn't relay on the quantum Lefschetz principle mentioned in [CGI+12], so the counter-example considered in [CGI+12], which show the quantum Lefschetz principle can fail beyond convexity, doesn't contradict our main Theorem 4.8 here.

[3]We will also use the notation $\mu(z)$ in this paper.

the input $\mathbf{t}(z)$ usually lead to some (rather mild) convergence issue about $J-$function. One way to handle this is using formal geometry; let $N$ be a nonnegative integer, we will choose our input $\mathbf{t}$ in the $\mathbb{Z}_2-$*graded space*

$$(q,t_1,\cdots,t_N)H^*(\bar{I}_\mu X,\mathbb{C})[z][\![t_1,\cdots,t_N]\!][\![\mathrm{Eff}(X)]\!]\,,$$

where each variable $t_i$ is associated with a grading in $\mathbb{Z}_2 := \mathbb{Z}/2\mathbb{Z}$. More precisely, $\mathbf{t}(z)$ can be written in the form of

$$\sum_{\substack{\beta\in\mathrm{Eff}(X)\\ \vec{k}=(k_1,\cdots,k_N)\in\mathbb{Z}^N_{\geq 0}}} q^\beta t_1^{k_1}\cdots t_N^{k_N} f_{\beta,\vec{k}}(z)\,, \tag{1.2}$$

where $f_{\beta,\vec{k}}(z)\in H^*(\bar{I}_\mu X,\mathbb{C})[z]$, $f_{0,\vec{0}}(z)=0$ and $t_1^{k_1}\cdots t_N^{k_N} f_{\beta,\vec{k}}(z)$ is of even grading. See §2 for more details.

Let $Y\subset X$ be a smooth complete intersection associated with a direct sum of semi-positive line bundles $\oplus_{j=1}^r L_j$, i.e., the pairing $\beta(L_j) := (c_1(L_j),\beta)\geq 0$ for any degree $\beta\in\mathrm{Eff}(X)$. Then the age function $age_g(L_j|_x)$ for each $(x,g)\in\bar{I}_c X$ is constant on each connected component $\bar{I}_c X$ of $\bar{I}_\mu X$ and we will denote the constant value to be $age_c(L_j)$. To state the main theorem in this paper, we will introduce a useful category for discussing the quantum Lefschetz known as *admissible series*. This concept summarizes a common feature of all previous proved $J-$functions about quantum Lefschetz. Here we will describe the content about admissible series needed in the statement of the main theorem, see §2 for a more general definition of admissible series.

Endow each variable $t_i$ and line bundle $L_j$ with a weight $w_{ij}\in\mathbb{Q}_{\geq 0}$. We will call a tuple $(\beta,\vec{k}=(k_1,\cdots,k_N),c)\in\mathrm{Eff}(X)\times\mathbb{Z}^N_{\geq 0}\times\mathscr{C}$ an *admissible triple* if

$$age_c(L_j)\equiv\beta(L_j)+\sum_{i=1}^N w_{ij}k_i \mod \mathbb{Z}$$

for all line bundles $L_j$. Denote by $\mathrm{Adm}_X$ the set of all admissible triples. Write each component $f_{\beta,\vec{k}}(\mathbf{t},z)$ in (1.2) as a sum

$$f_{\beta,\vec{k}}(z)=\sum_{c\in\mathscr{C}} f_{\beta,\vec{k},c}(z)$$

where $f_{\beta,\vec{k},c}(z)$ belongs to the space $H^*(\bar{I}_{c^{-1}}X,\mathbb{C})[z]$. Then we call the input $\mathbf{t}(z)$ an *admissible (power) series* if $\mathbf{t}(z)$ can be written as in (1.2) and $f_{\beta,\vec{k},c}(z)=0$ whenever $(\beta,\vec{k},c)$ is *not* an admissible triple.

Let $J^X(q,\mathbf{t},-z)$ be a point on the Lagrangian cone $\mathcal{L}_X$ of $X$ with $\mathbf{t}(z)$ being an admissible (power) series. Following the tradition in the literature, to prove a quantum Lefschetz theorem, it's more convenient to flip the sign of $z$ in the function $J^X(q,\mathbf{t},-z)$; in our case, $J^X(q,\mathbf{t},z)$ will always have an asymptotic expansion in variables $q^\beta t_1^{k_1}\cdots t_N^{k_N}$ as:

$$z+\sum_{(\beta,\vec{k},c)\in\mathrm{Adm}_X} q^\beta t_1^{k_1}\cdots t_N^{k_N} J^X_{\beta,\vec{k},c}(\mathbf{t},z)\,,$$

where $J^X_{\beta,\vec{k},c}(\mathbf{t},z)\in H^*(\bar{I}_{c^{-1}}X,\mathbb{C})[z,z^{-1}]]$. Define $J^{X,tw}(q,\mathbf{t},z)$ to be the hypergeometric modification of $J^X(q,\mathbf{t},z)$:

$$\begin{aligned} J^{X,tw}(q,\mathbf{t},z) &:= z+\sum_{(\beta,\vec{k},c)\in\mathrm{Adm}_X} q^\beta t_1^{k_1}\cdots t_N^{k_N} J^X_{\beta,\vec{k},c}(\mathbf{t},z)\\ &\cdot\prod_{j=1}^r\prod_{0\leq m<\beta(L_j)+\sum_i w_{ij}k_i}\Big(c_1(L_j)+(\beta(L_j)+\sum_i w_{ij}k_i-m)z\Big)\,. \end{aligned} \tag{1.3}$$

Our main theorem in this paper will be the following:

**Theorem 1.1** (=Theorem 4.8)**.** *Let $i : \bar{I}_\mu Y \to \bar{I}_\mu X$ be the inclusion of rigidified inertia stacks, then the series $i^* J^{X,tw}(q, \mathbf{t}, -z)$ lies on the Lagrangian cone $\mathcal{L}_Y$ of $Y$.*

Here we are suppressing a change about the $J$−function defining the Lagrangian cone $\mathcal{L}_Y$ of $Y$ by using Novikov variables from $\mathrm{Eff}(X)$ (or $\mathrm{Eff}(W)$ where $W$ is a space containing $X$) rather than $\mathrm{Eff}(Y)$, see §3.4 for more details.

Even when the convexity holds, our theorem still have some new results compared to previous quantum Lefschetz theorems. Our selection of the variables $t_i$ with their respective weights $w_{ij}$ plays the same role as the Novikov variables $q^\beta$ with the corresponding numbers $\beta(L_j)$, which resembles much similarity with the usage of extended degrees for $S$-extended $I$−functions of toric stacks in [CCIT15, CCIT19]. Therefore we will call the pair $(\beta, \vec{k})$ an *extended degree* and call $t_i$ *extended variables*. However the appearance of extended variables beyond toric cases is new. We note that introducing extended variables in loc.cits is a vital aspect to establish a Laudau-Ginzburg mirror for the big (equivariant) quantum ring of toric stacks (see [CCI+20]). We expect our theorem will have the similar mirror result (at least for ambient cohomology) and we will return to this question in the future. It's worth noting that the method of introducing extended variables in loc.cits cannot generalized to general cases as it relies on the $S$−extended fan structure for toric stacks in loc.cits (see also [Jia08]). This reflects a novel aspect of our method.

1.2. **Relation to twisted Gromov-Witten invariants.** We will discuss a relationship of our quantum Lefschetz theorem to twisted Gromov-Witten invariants, in particular a conjecture of Coates-Corti-Iritani-Tseng.

Consider the $\mathbb{C}^*$−action on the vector bundle $E := \oplus_{j=1}^r L_j$ via scaling the fibers. The $\mathbb{C}^*$−equivariant Euler class of $E$ can be written in term of the Chern roots $c_1(L_j)$ as

$$e^{\mathbb{C}^*}(E) := \prod_j (\kappa + c_1(L_j)) ,$$

where $\kappa$ is the equivariant parameter corresponding to the standard representation of $\mathbb{C}^*$. Then one can define a *twisted Lagrangian cone* $\mathcal{L}_X^{tw}$ using $(e^{\mathbb{C}^*}, E)$-*twisted Gromov-Witten invariants* (See [CG07] and [Tse10] for more details).

The series $J^X(q, \mathbf{t}, z)$ in our main theorems are related to points on the twisted Lagrangian cone $\mathcal{L}_X^{tw}$ in the following way: we can associate $J^X$ a series defined by

$$\begin{aligned} I^{X,tw} &:= z + \sum_{(\beta,\vec{k},c)\in \mathrm{Adm}_X} q^\beta t_1^{k_1} \cdots t_N^{k_N} J^X_{\beta,\vec{k},c}(\mathbf{t}, z) \\ &\cdot \prod_{j=1}^r \prod_{0\le m<\beta(L_j)+\sum_i w_{ij}k_i} \kappa + \left(c_1(L_j) + (\beta(L_j) + \sum_i w_{ij}k_i - m)z\right) . \end{aligned} \tag{1.4}$$

Then we have

$$\lim_{\kappa\to 0} I^{X,tw} = J^{X,tw} .$$

Moreover, using the same idea in [CCIT19, Theorem 22] where we view our $t_i$ here as their $x_i$ in loc.cit, one can prove that $I^{X,tw}$ lies on the twisted Lagrangian cone $\mathcal{L}_X^{tw}$. In general, Coates-Corti-Iritani-Tseng make the following conjecture (see [OP18, Conejcture 5.2]):

**Conjecture 1.2.** *Let $X$ be a smooth proper Deligne-Mumford stack with projective coarse moduli and $Y$ is a smooth complete intersection*[4] *of $X$. Let $I^{tw}$ be a point on the twisted Lagrangian cone $\mathcal{L}_X^{tw}$ of $X$, if the limit $\lim_{\kappa\to 0} i^* I^{tw}$ exits, then $\lim_{\kappa\to 0} i^* I^{tw}$ is a point on the Lagrangian cone $\mathcal{L}_Y$ of $Y$.*

[4]Note that the conjecture doesn't require the line bundles defining the complete intersection to be semi-positive.

This conjecture is known to be false in general, where a counterexample is found in [SW22]. However we can still take this as a guiding principle towards genus zero quantum Lefschetz theorem in general case and we are able to verify this conjecture under mild hypotheses as in Theorem 4.8.

1.3. **Sketch of the main idea of the proof.** The proof will first reduce to the hypersurface case. Set $\mu^{X,tw} := [J^{X,tw}(q,\mathbf{t},z) - z]_+$ to be the truncation in nonnegative $z-$powers. Note that the $J-$function $J^Y(q,\mathbf{t},z)$ is of the form $z+\mathbf{t}+\mathcal{O}(z^{-1})$, we observe that, to prove the main theorem, we need to show that

$$J^Y(q, i^*\mu^{X,tw}, z) = i^*J^{X,tw}(q,\mathbf{t},z)\,,$$

for which we only need to consider their *negative* $z-$*powers*. We aim to demonstrate that both sides of the negative $z-$powers of the above equation satisfy the *same recursive relations* in each degree corresponding to $q^\beta t_1^{k_1}\cdots t_N^{k_N}$ except a special consideration in low degrees (see Lemma 4.9). First we will consider two auxiliary spaces carried with $\mathbb{C}^*-$actions, which are root-stack modifications of a certain orbi-$\mathbb{P}^1$ bundle over Y (see §3.1) and the space of deformation to the normal cone (see §4.1). We then apply virtual localization to express two auxiliary cycles (see (3.17) and (4.5)) corresponding to the two spaces in graph sums and extract $\lambda^{-1}$ coefficient and $\lambda^{-2}$ coefficient respectively (where $\lambda$ is an equivariant parameter). Finally, the polynomiality of the two auxiliary cycles implies that the coefficients must vanish, leading to the same type of recursive relations (see also Proposition 3.8 and Theorem 4.4).

1.4. **Outline.** The rest of this paper is organized as follows. In §2, we introduce the concept of admissible series in more details than the in the introduction and collect some background on Gromov-Witten theory. In §3, we will give a recursive relation of an admissible series on the Lagrangian cone. Based on this recursive relation, in §4, we will prove our main theorem. In the Appendix, we carry out a detailed computation about edge contributions needed in the localization formula in our proof.

1.5. **Notation and convention.** We work over the field $\mathbb{C}$. Let $D$ be a divisor of a smooth stack $X$, we will denote the $\mathcal{O}(D)$ or $\mathcal{O}_X(D)$ to be the line bundle associated with $D$.

For any positive integer $i$, we will use the notation $\boldsymbol{\mu}_i$ to mean the finite cyclic subgroup of $\mathbb{C}^*$ of order $i$, and use the notation $[i]$ to mean the set $\{1,2,\cdots,i\}$. For any rational number $a$, we will use the notation $\mathrm{e}^{\,a}$ for to mean the exponential $\exp(2\pi a\sqrt{-1})$.

Let $\lambda$ be the equivariant class in $H^2_{\mathbb{C}^*}(\{\mathrm{Spec}(\mathbb{C})\})$ corresponding to the standard representation of $\mathbb{C}^*$. For any rational number $q$, we will write $\mathbb{C}_{q\lambda}$ to be the trivial line bundle with a $\mathbb{C}^*-$action of weight $q$.

## 2. Background on Gromov-Witten theory

2.1. **Admissible series.** Let $X$ be a smooth proper DM stack over $\mathbb{C}$ with projective coarse moduli and $\bar{I}_\mu X := \sqcup_{a\geq 1}\bar{I}_{\mu_a}X$ be the associated rigidified (cyclotomic) inertia stack. For our purpose, a finer decomposition of the rigidified inertia stack $\bar{I}_\mu X$ is needed. Given a set of line bundles $L_1,\cdots,L_r$ over $X$, we will choose a decomposition of the rigidified inertia stack into open-closed components

$$\bar{I}_\mu X := \bigsqcup_{c\in\mathscr{C}} \bar{I}_c X$$

where $\mathscr{C}$ is a index set for the decomposition which satisfies the following assumption:

**Assumption 2.1.** *We require that there are only finite elements in $\mathscr{C}$ such that $\bar{I}_c X$ is nonempty. For each $c\in\mathscr{C}$ and every $\mathbb{C}$-point $(x,g)\in\bar{I}_c X$, where $x$ is a $\mathbb{C}$-point of $X$ and $g$ is an element in the* *isotropy group $Aut(x)$, we will assume that the age function*[5] $age_g L_j|_x$ *and the order function* $ord(g)$ *are all constant functions on* $\bar{I}_c X$. *Then we can define* $age_c L_j := age_g L_j|_x$ *and* $a(c) := ord(g)$, *which are well-defined as they are independent of the choice of the point* $(x,g) \in \bar{I}_c X$.

*Moreover, we require that the index set* $\mathscr{C}$ *has an involution which is compatible with the involution* $\iota : \bar{I}_\mu X \to \bar{I}_\mu X$ *sending* $(x,g)$ *to* $(x,g^{-1})$ *for any* $\mathbb{C}$*-point* $(x,g)$ *of* $\bar{I}_\mu X$*; namely, for each index* $c \in \mathscr{C}$, *there exists a unique index in* $\mathscr{C}$ *and we will denote it to be* $c^{-1}$ *such that* $\iota(\bar{I}_c X) = \bar{I}_{c^{-1}} X$.

We will call an index set satisfying the above assumption a *good index set*, and call $c \in \mathscr{C}$ a *multiplicity*. We will sometimes use $\mathscr{C}$ to index the components of the inertia stack as well.

**Remark 2.2.** We note that a good index set $\mathscr{C}$ always exists (but may not unique), e.g., we can decompose $\bar{I}_\mu X$ into connected components, which corresponds to the biggest good index set if we require that all components $\bar{I}_c X$ are non-empty. When $X$ is a quotient Deligne-Mumford stack $[W/G]$ where $W$ is a quasi-projective variety and $G$ is a linear algebraic group, then we can choose the conjugacy class $\mathrm{Conj}(G)$ of $G$ to be the index set $\mathscr{C}$. When the component $\bar{I}_c X$ is empty for some $c \in \mathscr{C}$, we take the convention that the cohomology $H^*(\bar{I}_c X)$ is zero, the corresponding terms $\mu^X_{\beta,\vec{k},c}, J^X_{\beta,\vec{k},c}$, etc introduced in this paper are understood as zero, and the pull-back morphism $f^*$ on the cohomology groups induced from a morphism $f$ where $\bar{I}_c X$ appears as a domain or a codomain of $f$ is understood as the zero morphism.

Let $Y$ be a substack of $X$, we will also use $\mathscr{C}$ to index $\bar{I}_\mu Y$ such that $\bar{I}_c Y := \bar{I}_\mu Y \cap \bar{I}_c X$. Then $\mathscr{C}$ is also a good index set for $Y$ with respect to the restriction of line bundles $L_1, \dots, L_r$ to $Y$.

Let $m$ be a nonnegative integer, for an (ordered) tuple $\vec{m} = (c_1, \cdots, c_m) \in \mathscr{C}^m$, we will denote

$$\mathcal{K}_{g,\vec{m}}(X,\beta) := \mathcal{K}_{g,m}(X,\beta) \cap \bigcap_{i=1}^m ev_i^{-1}(\bar{I}_{c_i} X)\,.$$

Here $\mathcal{K}_{g,m}(X,\beta)$ is the moduli stack of genus $g$ (twisted) stable maps to $X$ with $m$ (not necessarily trivialized) gerby-marked points of degree $\beta$ as defined in [AV02] and $ev_i : \mathcal{K}_{g,m}(X,\beta) \to \bar{I}_\mu X$ is the evaluation map at the $i$th marking. Then $\mathcal{K}_{g,m}(X,\beta)$ can be written as a disjoint union

$$\mathcal{K}_{g,m}(X,\beta) = \bigsqcup_{\vec{m} \in \mathscr{C}^m} \mathcal{K}_{g,\vec{m}}(X,\beta)\,.$$

Sometimes, we will use the notation $[m] \cup \star$ (resp. $\vec{m} \cup \star$) to mean $[m+1]$ (resp. $\overrightarrow{m+1}$) to distinguish the $m+1$-th marking, which we denote to be $\star$.

Now we introduce *admissible triples*:

**Definition 2.3.** *Given a set of line bundles* $L_1, \cdots, L_r$ *over* $X$, *a nonnegative integer* $N$ *and* $r$ *weights* $\overrightarrow{w_j} = (w_{1j}, \cdots, w_{Nj}) \in \mathbb{Q}^N_{\geq 0}$ *corresponding to each* $L_j$. *Let* $\mathscr{C}$ *be a good index set. Let* $c \in \mathscr{C}$, $\vec{k} = (k_1, \cdots, k_N) \in \mathbb{Z}^N_{\geq 0}$ *and* $\beta \in \mathrm{Eff}(X)$, *where* $\mathrm{Eff}(X)$ *is the cone of effective curve classes of* $X$. *We will call* $(\beta, \vec{k}, c)$ *an admissible triple if and only if* $\bar{I}_c X$ *is nonempty, and*

$$age_c(L_j) \equiv \beta(L_j) + (\overrightarrow{w_j}, \vec{k}) \mod \mathbb{Z}$$

*for all line bundles* $L_j$, *where* $(\overrightarrow{w_j}, \vec{k}) = \sum_i w_{ij} k_i$. *We will denote* $\mathrm{Adm}_X$ *to be the set of all admissible triples.*

**Definition 2.4.** *Let* $t_1, \cdots, t_N$ *be* $N$ *variables with a* $\mathbb{Z}_2(= \mathbb{Z}/2\mathbb{Z})$*-grading on each variable* $t_i$, *which we denote the grading to be* $\bar{t}_i$. *Note that we also require the super-commutativity:*

$$t_i t_j = (-1)^{\bar{t}_i \cdot \bar{t}_j} t_j t_i\,.$$

[5] Recall that if $g$ acts on the fiber of $L_j$ over $x$ by multiplication by the number $e^q$, where $q \in \mathbb{Q}$, then we define the age $age_g(L_j|_x)$ to be $\langle q \rangle$, which is the fractional part of $q$.

*Let*

$$f(z)=\sum_{(\beta,\vec{k},c)\in \mathrm{Eff}(X)\times\mathbb{Z}^N_{\geq 0}\times\mathscr{C}} q^\beta t_1^{k_1}\cdots t_N^{k_N} f_{\beta,\vec{k},c}(z)\,,$$

*be a formal series in* $H^*(\bar{I}_\mu X,\mathbb{C})[z,z^{-1}]][[t_1,\cdots,t_N]][[\mathrm{Eff}(X)]]$ *in which*

$$f_{\beta,\vec{k},c}\in H^*(\bar{I}_{c^{-1}}X,\mathbb{C})[z,z^{-1}]]\,.$$

*We will put a* $\mathbb{Z}_2$*-grading on* $H^*(\bar{I}_\mu X,\mathbb{C})$*, which is induced from its ordinary cohomological grading, and put even grading on Novikov variables* $q^\beta$ *and* $z$*.*

*We will call* $f(z)$ *an admissible series if* $f(z)$ *satisfies the following conditions.*

(1) $f_{\beta,\vec{k},c}(z)=0$ *whenever* $(\beta,\vec{k},c)$ *is not an admissible triple for* $X$*.*
(2) $f_{0,\vec{0},c}(z)=0$ *for all* $c\in\mathscr{C}$*.*
(3) $f_{\beta,\vec{k},c}(z)$ *belongs to the space* $H^{\sum_i k_i\bar{t}_i}(\bar{I}_{c^{-1}}X,\mathbb{C})[z,z^{-1}]]$*, i.e., when* $\sum_i k_i\bar{t}_i=0\in\mathbb{Z}_2$*,* $H^{\sum_i k_i\bar{t}_i}(\bar{I}_{c^{-1}}X,\mathbb{C})$ *means the even degree part of the cohomology group of* $\bar{I}_{c^{-1}}X$*; when* $\sum_i k_i\bar{t}_i=1\in\mathbb{Z}_2$*,* $H^{\sum_i k_i\bar{t}_i}(\bar{I}_{c^{-1}}X,\mathbb{C})$ *means the odd degree part of the cohomology group of* $\bar{I}_{c^{-1}}X$*.*

*Furthermore, let* $g\in\mathbb{C}[z]$*, we say a formal series* $g+f(z)$ *is an admissible series near* $g$ *if* $f(z)$ *is an admissible series as above.*

*When* $Y$ *is a substack of* $X$*, we use the same index set* $\mathscr{C}$ *to index* $\bar{I}_\mu Y$ *as explained before. Let*

$$f(z)=\sum_{(\beta,\vec{k},c)\in \mathrm{Eff}(X)\times\mathbb{Z}^N_{\geq 0}\times\mathscr{C}} q^\beta t_1^{k_1}\cdots t_N^{k_N} f_{\beta,\vec{k},c}(z)\,,$$

*be a formal series in* $H^*(\bar{I}_\mu Y,\mathbb{C})[z,z^{-1}]][[t_1,\cdots,t_N]][[\mathrm{Eff}(X)]]$ *in which*

$$f_{\beta,\vec{k},c}\in H^*(\bar{I}_{c^{-1}}Y,\mathbb{C})[z,z^{-1}]]\,.$$

*Analogously, we call* $f(z)$ *(or* $g+f(z)$*) an admissible series (near* $g$*) if* $f(z)$ *satisfies the three conditions above where we only replace* $H^*(\bar{I}_{c^{-1}}X,\mathbb{C})$ *by* $H^*(\bar{I}_{c^{-1}}Y,\mathbb{C})$ *in the third condition.*

For any pair $(\beta,\vec{k})$ in $\mathrm{Eff}(X)\times\mathbb{Z}^N_{\geq 0}$, we will write $q^\beta\mathbf{t}^{\vec{k}}:=q^\beta t_1^{k_1}\cdots t_N^{k_N}$, and call $(\beta,\vec{k})$ an extended degree and call $t_i$ extended variables.

**Remark 2.5.** Examples of admissible series include $J-$functions[Giv04] and $I-$functions in quasimap theory[CCFK15, CFK14, Web18, Web21, Wan25]. Therefore admissible series consists of a large class of objects studied in Gromov-Witten theory.

2.2. **Background on orbifold Gromov-Witten theory.** Now assume that $X$ carries a algebraic torus $T$ action (can be trivial), we can define the so-called *Chen-Ruan cohomology* of $X$. Given any two elements $\alpha_1,\alpha_2$ in the $T-$equivariant

$$H^*_{\mathrm{CR},T}(X,\mathbb{C}):=H^*_T(\bar{I}_\mu X,\mathbb{C})\,,$$

We can define the Poincaré pairing in the *non-rigidified* inertia stack $I_\mu X$ of $X$:

$$\langle\alpha_1,\alpha_2\rangle_{\mathrm{orb}}:=\int_{\sum_{c\in\mathscr{C}}a(c)^{-1}[\bar{I}_c X]}\alpha_1\cdot\iota^*\alpha_2\,.$$

Here $\iota$ is the involution of $\bar{I}_\mu X$ obtained from the inversion automorphisms of the band. Therefore, the diagonal class $[\Delta_{\bar{I}_c X}]$ obtained via push-forward of the fundamental class by $(\mathrm{id},\iota):\bar{I}_c X\to\bar{I}_c X\times\bar{I}_{c^{-1}}X$ can be written as

$$\sum_{c\in\mathscr{C}}a(c)[\Delta_{\bar{I}_c X}]=\sum_\alpha\phi_\alpha\otimes\phi^\alpha \text{ in } H^*_T(\bar{I}_\mu X\times\bar{I}_\mu X,\mathbb{C}),$$

where $\{\phi_\alpha\}$ is a graded basis of $H^*_{\mathrm{CR},T}(X,\mathbb{C})$ with $\{\phi^\alpha\}$ the dual basis with respect to the Poincaré pairing defined above, i.e., $\langle\phi^\alpha,\phi_\beta\rangle_{\mathrm{orb}}=\delta_{\alpha,\beta}$, where $\delta_{\alpha,\beta}=1$ if $\alpha=\beta$, and $\delta_{\alpha,\beta}=0$ otherwise.

Denote by $\bar\psi_i$ the first Chern class of the universal cotangent line bundle whose fiber at $[f:(\mathcal{C},\vec{\mathbf{q}}:=(q_1,...,q_m))\to X]$ is the cotangent space of the coarse moduli $\underline{C}$ of $C$ at $i$-th marking $\underline{q}_i$. For non-negative integers $a_i$ and classes $\alpha_i\in H^*_T(\bar I_\mu X,\mathbb{C})$, using the virtual cycle $[\mathcal{K}_{g,\vec m}(X,\beta)]^{\mathrm{vir}}$ defined in [LT98, BF97, AGV08], we write the Gromov-Witten invariant:

$$\langle\alpha_1\bar\psi^{a_1},...,\alpha_m\bar\psi^{a_m}\rangle^X_{g,\vec m,\beta}:=\int_{[\mathcal{K}_{g,\vec m}(X,\beta)]^{\mathrm{vir}}}\prod_i ev_i^*(\alpha_i)\bar\psi_i^{a_i}\,.$$

When the tuple $(g,\vec m,\beta)$ gives rise to an empty stack $\mathcal{K}_{g,\vec m}(X,\beta)$, we define the above integral to be zero.

We will also need the stable-map Chen-Ruan classes

$$(\widetilde{ev}_j)_*=\iota_*(\boldsymbol{r}_j(ev_j)_*),\tag{2.1}$$

where $\boldsymbol{r}_j$ is the order function of the band of the gerbe structure at the marking $q_j$. Define a class in $H_*^T(\bar I_\mu X)\cong H_T^*(\bar I_\mu X)$ by

$$\begin{aligned}\langle\alpha_1\bar\psi^{a_1},...,\alpha_m\bar\psi^{a_m},-\rangle^X_{0,\beta}&:=(\widetilde{ev}_{m+1})_*\left(\prod_{i=1}^m ev_i^*\alpha_i\bar\psi_i^{a_i}\cap[\mathcal{K}_{0,m+1}(X,\beta)]^{\mathrm{vir}}\right)\\&=\sum_\alpha\phi^\alpha\langle\alpha_1\bar\psi^{a_1},...,\alpha_m\bar\psi^{a_m},\phi_\alpha\rangle^X_{0,m+1,\beta}\,.\end{aligned}$$

Sometimes we will also write $(\widetilde{ev}_j)_*$ as $(\widetilde{ev_j^X})_*$ to emphasis the role of the target $X$, this notation is used in the proof of Lemma 4.9.

## 3. A recursive relation about the Lagrangian cone

3.1. **A root-stack modification of the twisted graph space.** Let $Y$ be any smooth proper Deligne-Mumford stack with projective coarse moduli. Let $L$ be a semi-positive line bundle over $Y$, i.e., $\beta(L)\geq 0$ for any degree $\beta\in\mathrm{Eff}(Y)$. Denote by $\mathbb{P}Y_{r,s}$ the root stack of the projective bundle (also named twisted graph space[6]) $\mathbb{P}_Y(L^\vee\oplus\mathbb{C})$ over $Y$ by taking $s-$th root of the zero section $D_0:=\mathbb{P}(0\oplus\mathbb{C})$ and $r-$th root of the infinity section $D_\infty:=\mathbb{P}(L^\vee\oplus 0)$. Let $\mathcal{D}_0$ and $\mathcal{D}_\infty$ be the corresponding root divisors of $D_0$ and $D_\infty$ respectively, then the zero section $\mathcal{D}_0\subset\mathbb{P}Y_{r,s}$ is isomorphic to the root stack $\sqrt[s]{L^\vee/Y}$ with normal bundle being isomorphic to the root bundle $(L^\vee)^{\frac{1}{s}}$, and the infinity section $\mathcal{D}_\infty\subset\mathbb{P}Y_{r,s}$ is isomorphic to the root stack $\sqrt[r]{L/Y}$ with normal bundle being isomorphic to the root bundle $L^{\frac{1}{r}}$.

There are two morphisms

$$q_0,q_\infty:\mathbb{P}Y_{r,s}\to\mathbb{B}\mathbb{C}^*\,,$$

associated to the line bundles $\mathcal{O}(\mathcal{D}_0)$ and $\mathcal{O}(\mathcal{D}_\infty)$ respectively such that $\mathcal{O}(\mathcal{D}_0)\cong q_0^*(\mathbb{L})$ and $\mathcal{O}(\mathcal{D}_\infty)\cong q_\infty^*(\mathbb{L})$, where $\mathbb{L}$ is the universal line bundle over $\mathbb{B}\mathbb{C}^*$. This will induce morphisms on their inertia stack counterparts:

$$q_0,q_\infty:I_\mu\mathbb{P}Y_{r,s}\to I\mathbb{B}\mathbb{C}^*\,,$$

Recall that $I\mathbb{B}\mathbb{C}^*\cong[\mathbb{C}^*/\mathbb{C}^*]\cong\mathbb{C}^*\times\mathbb{B}\mathbb{C}^*$, where $\mathbb{C}^*$ acts on $\mathbb{C}^*$ by conjugation. Using this isomorphism, for any $c\in\mathbb{C}^*$, we denote $I_c\mathbb{B}\mathbb{C}^*$ to be the closed substack $\{c\}\times\mathbb{B}\mathbb{C}^*$ of $I\mathbb{B}\mathbb{C}^*$. A $\mathbb{C}$-point of $I_c\mathbb{B}\mathbb{C}^*$ corresponds to a pair $(1_{\mathbb{C}},c)$, where $1_{\mathbb{C}}$ is the trivial principal $\mathbb{C}$-bundle over $\mathrm{Spec}(\mathbb{C})$ and $c\in\mathbb{C}^*$ is in the automorphism group $\mathrm{Aut}_{\mathbb{C}}(1_{\mathbb{C}})\cong\mathbb{C}^*$. Let $(y,g)$ be a $\mathbb{C}-$point of $I_\mu\mathbb{P}Y_{r,s}$, where

[6]The terminology of "twisted graph space" is taken from [CJR17a, CJR17b], see loc.cits for some other applications of the twisted graph space in Gromov-Witten theory.

$y \in Ob(\mathbb{P}Y_{r,s}(\mathbb{C}))$ and $g \in \mathrm{Aut}(y)$, if $q_0((y,g)) = (1_{\mathbb{C}}, c)$, then $g$ acts on the fiber $\mathcal{O}(\mathcal{D}_0)|_y$ via the multiplication by $c$. We have a similar relation for $q_\infty$.

Define

$$S = \{c \in \mathbb{C}^* | \text{There exists} (y,g) \in I\mathbb{P}Y_{r,s} \text{such that } q_0(y,g) = (1_{\mathbb{C}}, c) \; or \; q_\infty(y,g) = (1_{\mathbb{C}}, c)\}\,.$$

Then $S$ is a finite set[7] closed under taking inversion, and the image of $q_0, q_\infty$ is contained $\bigsqcup_{s\in S} I_s \mathbb{B}\mathbb{C}^*$. Now we have an induced morphism on

$$q_0, q_\infty \,:\, I_\mu \mathbb{P}Y_{r,s} \to \bigsqcup_{s\in S} I_s \mathbb{B}\mathbb{C}^*\,.$$

Let $\mathrm{pr}_{r,s} \,:\, \mathbb{P}Y_{r,s} \to Y$ be the projection to the base, which is a composition of deroot-stackification and projection from $\mathbb{P}_Y(L^\vee \oplus \mathbb{C})$ to $Y$.

Now choose a good index set $\mathscr{C}$ for $\bar{I}_\mu Y$ satisfying the assumption 2.1. For each $\mathfrak{c} \in \mathscr{C}, c_0, c_\infty \in S$, we will define the inertia component $I_{(\mathfrak{c},c_0,c_\infty)}\mathbb{P}Y_{r,s}$ to be

$$\mathrm{pr}_{r,s}^{-1}(I_{\mathfrak{c}} Y) \cap q_0^{-1}(I_{c_0}\mathbb{B}\mathbb{C}^*) \cap q_\infty^{-1}(I_{c_\infty}\mathbb{B}\mathbb{C}^*)\,.$$

Set $(\mathfrak{c}, c_0, c_\infty)^{-1} = (\mathfrak{c}^{-1}, c_0^{-1}, c_\infty^{-1})$. Each $I_{(\mathfrak{c},c_0,c_\infty)}\mathbb{P}Y_{r,s}$ is an open-closed substack of $I_\mu \mathbb{P}Y_{r,s}$. Denoted by $\bar{I}_{(\mathfrak{c},c_0,c_\infty)}\mathbb{P}Y_{r,s}$ the corresponding rigidified inertia component of $\bar{I}_\mu \mathbb{P}Y_{r,s}$. If one of $c_0, c_\infty$ is not in $S$, we set $\bar{I}_{(\mathfrak{c},c_0,c_\infty)}\mathbb{P}Y_{r,s} = \emptyset$. Then $\bar{I}_\mu \mathbb{P}Y_{r,s}$ decomposes as a disjoint union:

$$\bar{I}_\mu \mathbb{P}Y_{r,s} = \bigsqcup_{(\mathfrak{c},c_0,c_\infty)\in \mathscr{C}\times\mathbb{C}^*\times\mathbb{C}^*} \bar{I}_{(\mathfrak{c},c_0,c_\infty)}\mathbb{P}Y_{r,s}\,.$$

First we state a useful fact about $\bar{I}_\mu \mathbb{P}Y_{r,s}$:

**Lemma 3.1.** *Let $m = (\mathfrak{c}, a, b) \in \mathscr{C} \times \mathbb{C}^* \times \mathbb{C}^*$. Assume that $\bar{I}_m \mathbb{P}Y_{r,s}$ is nonempty. Then at least one of $a, b$ is equal to one. If $a \neq 1$ (resp. $b \neq 1$), then $\bar{I}_m \mathbb{P}Y_{r,s} = \bar{I}_m \mathcal{D}_0$ (resp. $\bar{I}_m \mathbb{P}Y_{r,s} = \bar{I}_m \mathcal{D}_\infty$). Furthermore, $a, b$ are roots of unity and we have $\mathrm{e}^{age_{\mathfrak{c}} L} = a^{-s}$ when $a \neq 1$ and $\mathrm{e}^{age_{\mathfrak{c}} L} = b^r$ when $b \neq 1$.*

*Proof.* Let $(x, g)$ be a $\mathbb{C}-$point of $\bar{I}_m \mathbb{P}Y_{r,s}$ where $x \in \mathbb{P}Y_{r,s}(\mathbb{C})$ and $g$ is an element of the isotropy group of $x$. If $a, b$ are not equal to one simultaneously, $g$ acts on the fibers $\mathcal{O}(\mathcal{D}_0)|_x$ and $\mathcal{O}(\mathcal{D}_\infty)|_x$ nontrivially; but this implies that $x \in \mathcal{D}_0 \cap \mathcal{D}_\infty$, which is impossible as $\mathcal{D}_0$ and $\mathcal{D}_\infty$ are disjoint. This shows that at least one of $a, b$ is equal to one. Then if $a \neq 1$, we have $x \in \mathcal{D}_0$ and hence $\bar{I}_m \mathbb{P}Y_{r,s} = \bar{I}_m \mathcal{D}_0$; similarly if $b \neq 1$, we have $x \in \mathcal{D}_\infty$ and hence $\bar{I}_m \mathbb{P}Y_{r,s} = \bar{I}_m \mathcal{D}_\infty$ .

As $\mathbb{P}Y_{r,s}$ is a Deligne-Mumford stack, then $m$ must be of finite order, hence $a, b$ must be roots of unity. Finally since $L_0 := \mathrm{pr}_{r,s}^* L \otimes \mathcal{O}(s\mathcal{D}_0)$ is a trivial line bundle over $\mathcal{D}_0$, thus for any $\mathbb{C}-$point $(x, g) \in \bar{I}_m \mathcal{D}_0$, $g$ acts on the fiber $L_0|_x$ trivially and therefore we have $\mathrm{e}^{age_{\mathfrak{c}} L} = a^{-s}$ if $a \neq 1$. Similarly since $L_\infty := \mathrm{pr}_{r,s}^* L^\vee \otimes \mathcal{O}(r\mathcal{D}_\infty)$ is a trivial line bundle over $\mathcal{D}_\infty$, thus for any $\mathbb{C}-$point $(x, g) \in \bar{I}_m \mathcal{D}_\infty$, $g$ acts on the fiber $L_\infty|_x$ trivially and therefore $\mathrm{e}^{age_{\mathfrak{c}} L} = b^r$ if $b \neq 1$. □

**Definition 3.2.** *For any degree $\beta \in \mathrm{Eff}(Y)$ and rational number $\delta \in \mathbb{Q}_{\geq 0}$, we say a stable map $f \,:\, C \to \mathbb{P}Y_{r,s}$ is of degree $(\beta, \frac{\delta}{r})$ if $(\mathrm{pr}_{r,s} \circ f)_*[C] = \beta$ and $deg(f^*\mathcal{O}(\mathcal{D}_\infty)) = \frac{\delta}{r}$. We will denote $\mathcal{K}_{0,m}(\mathbb{P}Y_{r,s}, (\beta, \frac{\delta}{r}))$ to be the corresponding moduli stack of stable maps to $\mathbb{P}Y_{r,s}$ of degree $(\beta, \frac{\delta}{r})$. Note that the line bundle $\mathcal{O}(\mathcal{D}_\infty)$ is semi-positive.*

[7]Indeed, let e be the integer which is divided by all the orders of isotropy groups of $C-$points of $\mathbb{P}Y_{r,s}$, we have $c^{\mathrm{e}} = 1$ for all $c \in S$.

3.2. **Localization analysis.** Consider the $\mathbb{C}^*-$action on the projection bundle $\mathbb{P}_Y(L^\vee \oplus \mathbb{C})$ by scaling the fiber such that the normal bundle of $D_0$ (resp. $D_\infty$) in $\mathbb{P}_Y(L^\vee \oplus \mathbb{C})$ is of $\mathbb{C}^*-$weight 1 (resp. $-1$). This $\mathbb{C}^*-$action induces a $\mathbb{C}^*-$action on $\mathbb{P}Y_{r,s}$ such that the normal bundle of $\mathcal{D}_0$ (resp. $\mathcal{D}_\infty$) in $\mathbb{P}Y_{r,s}$ is of $\mathbb{C}^*-$weight $\frac{1}{s}$ (resp. $-\frac{1}{r}$). Then we have an induced $\mathbb{C}^*-$action on the moduli $\mathcal{K}_{0,\vec{m}}(\mathbb{P}Y_{r,s},(\beta,\frac{\delta}{r}))$ of twisted stable maps to $\mathbb{P}Y_{r,s}$. We will apply the virtual localization formula of Graber–Pandharipande [GP99] to $\mathcal{K}_{0,\vec{m}}(\mathbb{P}Y_{r,s},(\beta,\frac{\delta}{r}))$, for which introduce the notation of decorated graph so that we index the components of $\mathbb{C}^*-$fixed loci of $\mathcal{K}_{0,\vec{m}}(\mathbb{P}Y_{r,s},(\beta,\frac{\delta}{r}))$ by decorated graphs (trees). First, a decorated graph $\Gamma$ consists of vertexes, edges and legs with the following decorations:

- Each vertex $v$ is associated with an index $j(v) \in \{0,\infty\}$, and a degree $\beta(v) \in \mathrm{Eff}(Y)$.
- Each edge $e$ consists of a pair of half-edges $\{h_0,h_\infty\}$, and $e$ is equipped with a degree $\delta(e) \in \mathbb{Q}_{>0}$. Here we call $h_0$ and $h_\infty$ half edges, and $h_0$ (resp. $h_\infty$) is attached to a vertex labeled by 0 (resp. $\infty$).
- Each half-edge $h$ and each leg $l$ has a multiplicity $m(h)$ or $m(l)$ in $\mathscr{C}\times\mathbb{C}^*\times\mathbb{C}^*$.
- The legs are labeled with the numbers $\{1,\dots,m\}$ and each leg is incident to a unique vertex.

By the "valence" of a vertex $v$, denoted $\mathrm{val}(v)$, we mean the total number of incident half-edges and legs.

For each $\mathbb{C}^*-$fixed stable map $f : (C;\vec{\mathbf{q}} := (q_1,\dots,q_m)) \to \mathbb{P}Y_{r,s}$ in $\mathcal{K}_{0,\vec{m}}(\mathbb{P}Y_{r,s},(\beta,\frac{\delta}{r}))$, we can associate a decorated graph $\Gamma$ in the following way.

- Each edge $e$ corresponds to a genus-zero irreducible component $C_e$ which maps constantly to the base $Y$ and intersects $\mathcal{D}_0$ and $\mathcal{D}_\infty$ properly. The degree of the restriction $f|_{C_e}$ is $(0,\frac{\delta(e)}{r})$; then $\delta(e)$ satisfies that $r\cdot deg(f|_{C_e}^*\mathcal{O}(\mathcal{D}_\infty)) = s\cdot deg(f|_{C_e}^*\mathcal{O}(\mathcal{D}_0)) = \delta(e)$.
- Each vertex $v$ for which $j(v)=0$ (with unstable exceptional cases noted below) corresponds to a maximal sub-curve $C_v$ of $C$ which maps totally into $\mathcal{D}_0$, then the restriction of $f$ to $C_v$ defines a twisted stable map in
$$\mathcal{K}_{0,val(v)}(\sqrt[s]{L^\vee/Y},\beta(v))\,.$$
Each vertex $v$ for which $j(v)=\infty$ (again with unstable exceptions) corresponds to a maximal sub-curve which maps totally into $\mathcal{D}_\infty$, then the restriction of $f$ to $C_v$ defines a twisted stable map in
$$\mathcal{K}_{0,val(v)}(\sqrt[r]{L/Y},\beta(v))\,.$$
The label $\beta(v)$ denotes the degree coming from the restriction $\mathrm{pr}_{r,s}\circ f|_{C_v} : C_v \to Y$.
- A vertex $v$ is *unstable* if stable twisted maps of the type described above do not exist. In this case, we have $\beta(v)=0$ and $v$ corresponds to a single point of the component $C_e$ for each incident edge $e$, which may be a node at which $C_e$ meets another edge curve $C_{e'}$, a marked point of $C_e$, or an unmarked point. We will also use $C_v$ to denote the corresponding point.
- Each leg $l$ labeled by $i$ corresponds to the $i$th marking $q_i$, and it's incident to the vertex $v$ if $q_i$ lies on $C_v$. The index $m(l)$ on a leg $l$ indicates the rigidified inertia stack Component $\bar{I}_{m(l)}\mathbb{P}Y_{r,s}$ of $\mathbb{P}Y_{r,s}$ on which the gerby marked point corresponding to the leg $l$ is evaluated.
- A half-edge $h$ of an edge $e$ corresponds a ramification point $q \in C_e$, i.e., there are two distinguished points(we call them ramification points) $q_0$ and $q_\infty$ on $C_e$ satisfying that $q_0$ maps to $\mathcal{D}_0$ and $q_\infty$ maps to $\mathcal{D}_\infty$, respectively. We associate half-edges $h_0$ and $h_\infty$ to $q_0$ and $q_\infty$ respectively. Then $m(h)$ indicates the rigidified inertia component $\bar{I}_{m(h)}\mathbb{P}Y_{r,s}$ of $\mathbb{P}Y_{r,s}$ on which the ramification point $q$ associated with $h$ is evaluated. We will also write the edge $e$ as $\{h_0,h_\infty\}$.

Moreover, the decorated graph from a $\mathbb{C}^*-$fixed stable map should satisfy the following:

(1) (Monodromy constraint for edge) For each edge $e = \{h_0, h_\infty\}$, we have that $m(h_0) = (\mathcal{c}^{-1}, e^{\frac{\delta(e)}{s}}, 1)$ and $m(h_\infty) = (\mathcal{c}, 1, e^{\frac{\delta(e)}{r}})$ for some $\mathcal{c} \in C$ by Lemma A.1. Moreover the choice of $c$ should satisfy that
$$\delta(e) = age_{\mathcal{c}}(L) \bmod \mathbb{Z}\ ,$$
which follows from the fact that the ages of line bundles $f|_{C_e}^* \mathcal{O}(s\mathcal{D}_0)$ and $f|_{C_e}^*(\mathcal{O}(r\mathcal{D}_\infty) \otimes \mathrm{pr}_{r,s}^* L^\vee)$ at $q_\infty$ should be equal, as the two line bundles on $C_e$ are isomorphic.

(2) (Monodromy constraint for vertex) For each vertex $v$, let $I_v \subset [m]$ be the set of incident legs to $v$, and $H_v$ be the set of incident half-edges to $v$. For each $i \in I_v$, write $m(l_i) = (c_i, a_i, b_i)$, and for each $h \in H_v$, write $m(h) = (c_h, a_h, b_h)$. If $v$ is labeled by 0, we have that all $b_i$ and $b_h$ are equal to 1, and
$$e^{\frac{\beta(v)(L)}{s}} \times \prod_{i \in I_v} a_i \times \prod_{h \in H_v} a_h^{-1} = 1\ . \tag{3.1}$$
If $v$ is labeled by $\infty$, then we have all $a_i$ and $a_h$ are all equal to 1 and
$$e^{-\frac{\beta(v)(L)}{r}} \times \prod_{i \in I_v} b_i \times \prod_{h \in E_v} b_h^{-1} = 1\ . \tag{3.2}$$

(3) (Degree constraint) We have
$$\sum_v \beta(v) = \beta\ , \tag{3.3}$$
and
$$\sum_e \delta(e) + \sum_{v:\, j(v)=\infty} \beta(v)(L) = \beta(L)\ . \tag{3.4}$$

For each leg $l$, write $m(l) = (c_l, a_l, b_l) \in \mathscr{C} \times \mathbb{C}^* \times \mathbb{C}^*$. By Lemma 3.1, we can assume that $a_l, b_l$ are roots of unity, then we can write all $a_l$ (resp. $b_l$) in the exponential forms $e^{\frac{\delta_a}{s}}$ (resp. $e^{\frac{\delta_b}{r}}$) where $\delta_a$ (resp. $\delta_b$) is a rational number; we also require that $0 \le \delta_a < s$ (resp. $0 \le \delta_b < r$), then $\delta_a$ (resp. $\delta_b$) are uniquely determined.

*Now, we will always assume that $r, s$ are sufficiently large primes,* more precisely, $r, s$ are sufficiently large comparing to $\beta(L)$, $\delta$ and the rational numbers $\delta_a, \delta_b$ introduced above.

*We will only consider decorated graphs coming from $\mathbb{C}^*$−fixed stable maps.* Let $f : (\mathcal{c}, \vec{\mathbf{q}}) \to \mathbb{P}Y_{r,s}$ be a $\mathbb{C}^*$−fixed stable map with $\Gamma$ being its decorated graph. We get a normalization exact sequence coming from resolving nodes imposed by the decorated graph $\Gamma$ (cf. [GP99, §4])
$$0 \to \mathcal{O}_C \to \bigoplus_{vertices} \mathcal{O}_{C_v} \oplus \bigoplus_{edges} \mathcal{O}_{C_e} \to \bigoplus_{flags} \mathcal{O}_{q_{(e,v)}} \to 0\ . \tag{3.5}$$
Here, a flag is a pair $(e, v)$ such that the vertex $v$ is incident to the edge $e$, and we denote $q_{(e,v)}$ to be the corresponding ramification point on $C_e$. When $r, s$ are sufficiently large primes, we can show that the set of flags coincides with the set of all the ramification points on the edges using the Lemma below.

**Lemma 3.3.** *When $r$ (resp. $s$) is a sufficiently large prime, there is no unstable vertex labeled by $\infty$ (resp. 0) linking two edges in a decorated graph $\Gamma$.*

*Proof.* Let's only deal with the case when the unstable vertex is labeled by 0; the case when the the unstable vertex is labeled by $\infty$ will be proved similarly. Assume the contrary, there is an unstable vertex $v$ linking two edges $e_1$ and $e_2$. Let $h_{01}$ be the half-edge of $e_1$ over 0 and $h_{02}$ be the half-edge of $e_2$ over 0. Assume that the multiplicity $m(h_{01})$ associated to $h_{01}$ is equal to $(\mathcal{c}_1, e^{\frac{\delta(e_1)}{s}}, 1)$ and the multiplicity $m(h_{02})$ associated to $h_{02}$ is equal to $(c_2, e^{\frac{\delta(e_2)}{s}}, 1)$; note that here $\delta(e_1)$ and $\delta(e_2)$ are the decorated degrees of $e_1$ and $e_2$ respectively. As the node corresponding to $v$ is a balanced node, we have $m(h_{01}) = m(h_{02})^{-1}$ and thus we have

$$\delta(e_1) + \delta(e_2) \equiv 0 \bmod s\,.$$

However we also have $0 < \delta(e_1) + \delta(e_2) \leq \delta \ll s$ by (3.4), we see that $\delta(e_1) + \delta(e_2) \equiv 0 \bmod s$ is impossible. Then the case when the unstable vertex is labeled by 0 is proved. □

**Remark 3.4.** The above Lemma also holds if we drop the semi-positivity assumption on $L$. See [JPPZ20, Lemma 12] for a proof.

For each decorated graph $\Gamma$, let $M_\Gamma$ be the component of $\mathbb{C}^*$−fixed loci of $\mathcal{K}_{0,\vec{m}}(\mathbb{P}Y_{r,s}, (\beta, \frac{\delta}{r}))$ such that all the $\mathbb{C}$−points of $\mathcal{K}_{0,\vec{m}}(\mathbb{P}Y_{r,s}, (\beta, \frac{\delta}{r}))$ are associated with the decorated graph $\Gamma$. Now let's describe an explicit étale cover of $M_\Gamma$. First we will associate each vertex $v$ (resp. edge $e$) a moduli space $\mathcal{M}_v$ (resp. $\mathcal{M}_e$) over which there is a family of $\mathbb{C}^*$−stable maps to $\mathbb{P}Y_{r,s}$ with the corresponding decorated degree.

(1) For each stable vertex $v$, we note that the decorations at each stable vertex $v$ yield a tuple

$$\overrightarrow{val}(v) \in (\mathscr{C} \times \mathbb{C}^* \times \mathbb{C}^*)^{\mathrm{val}(v)}$$

recording the multiplicities at every special point of $C_v$.[8] Then we define

$$\mathcal{M}_v := \mathcal{K}_{0,\overrightarrow{val}(v)}(\mathcal{D}_{j(v)}, \beta(v))\,.$$

(2) For each unstable vertex $v$, $v$ is incident to a unique half-edge $h$. If $v$ is labeled by 0, we denote $\mathcal{M}_v := \bar{I}_{m(h)^{-1}}\mathcal{D}_0$, and if $v$ is labeled by $\infty$, we denote $\mathcal{M}_v := \bar{I}_{m(h)^{-1}}\mathcal{D}_\infty$.

(3) For each edge $e = \{h_0, h_\infty\}$ (with the associated decoration) of $\Gamma$, we denote

$$\mathcal{K}_e := \mathcal{K}_{0,2}(\mathbb{P}Y_{r,s}, (0, \frac{\delta(e)}{r})) \cap ev_1^{-1}(\bar{I}_{m(h_0)}\mathbb{P}Y_{r,s}) \cap ev_2^{-1}(\bar{I}_{m(h_\infty)}\mathbb{P}Y_{r,s})\,,$$

and denote $\mathcal{K}_e^{\mathbb{C}^*}$ to be the $\mathbb{C}^*$−fixed substack of $\mathcal{K}_e$. As $r, s$ are sufficiently large primes, by Lemma A.1, we can show that all domain curves of the stable maps in $\mathcal{K}_e^{\mathbb{C}^*}$ are irreducible.

Using Lemma 3.3 and the normalization sequence (3.5), the fiber product

$$\widetilde{M}_\Gamma := \prod_{v:j(v)=0} \mathcal{M}_v \times_{(\bar{I}_\mu\mathcal{D}_0)^{|E|}} \prod_{e\in E} \mathcal{K}_e^{\mathbb{C}^*} \times_{(\bar{I}_\mu\mathcal{D}_\infty)^{|E|}} \prod_{v:j(v)=\infty} \mathcal{M}_v\,,$$

will be a finite étale cover of degree $|\mathrm{Aut}(\Gamma)|$ over $M_\Gamma$, where the fiber product is taken by gluing the two branches at each node.[9] However $\mathcal{K}_e^{\mathbb{C}^*}$ is not explicit to us enough (together with the associated family stable maps); instead we will use a more concrete space called $\mathcal{M}_e$. The space $\mathcal{M}_e$ is isomorphic to the root gerbe $\sqrt[a_e s\delta(e)]{L^\vee/I_c Y}$ (see more details in §3.2.2), and we can prove in Lemma A.7 that $\mathcal{M}_e$ is a finite étale cover of $\mathcal{K}_e^{\mathbb{C}^*}$ of degree $\frac{1}{a_e s}$. Using the space $\mathcal{M}_e$, denote by $F_\Gamma$ the space

$$\prod_{v:j(v)=0} \mathcal{M}_v \times_{(\bar{I}_\mu\mathcal{D}_0)^{|E|}} \prod_{e\in E} \mathcal{M}_e \times_{(\bar{I}_\mu\mathcal{D}_\infty)^{|E|}} \prod_{v:j(v)=\infty} \mathcal{M}_v\,,$$

[8] For each node of $C_v$, let $h$ be the incident half-edge, then we define the multiplicity at the branch of node at $C_v$ to be $m(h)^{-1}$ as nodes of a twisted stable map are balanced.

[9] Here we also allow a node can link a unstable vertex and an edge, hence there are two corresponding nodes on each edge.

where the fiber product is taken by gluing the two branches at each node (see also §3.2.4 for an alternative construction of $F_\Gamma$). Then $F_\Gamma$ is a finite étale cover of $M_\Gamma$. By a generalization of the virtual localization formula [GP99] (see Appendix §B), we can write

$$[\mathcal{K}_{0,\vec{m}}(\mathbb{P}Y_{r,s},(\beta,\frac{\delta}{r}))]^{\mathrm{vir}}\,,$$

in terms of contributions from each decorated graph $\Gamma$:

$$[\mathcal{K}_{0,\vec{m}}(\mathbb{P}Y_{r,s},(\beta,\frac{\delta}{r}))]^{\mathrm{vir}}=\sum_{\Gamma}\frac{1}{\mathbb{A}_\Gamma}\iota_{\Gamma *}\left(\frac{[F_\Gamma]^{\mathrm{vir}}}{e^{\mathbb{C}^*}(N_\Gamma^{\mathrm{vir}})}\right). \tag{3.6}$$

Here, for each graph $\Gamma$, $[F_\Gamma]^{\mathrm{vir}}$ is obtained via the $\mathbb{C}^*$-fixed part of the restriction to the $\mathbb{C}^*-$fixed loci of the obstruction theory of $\mathcal{K}_{0,\vec{m}\cup\star}(\mathbb{P}Y_{r,s},(\beta,\frac{\delta}{r}))$, and $N_\Gamma^{\mathrm{vir}}$ is the equivariant Euler class of the $\mathbb{C}^*$-moving part of this restriction. Besides, $\mathbb{A}_\Gamma$ is the automorphism factor for the graph $\Gamma$, which represents the degree of $F_\Gamma$ into the corresponding open and closed $\mathbb{C}^*$-fixed substack $i_\Gamma(F_\Gamma)$ in $\mathcal{K}_{0,\vec{m}\cup\star}(\mathbb{P}Y_{r,s},(\beta,\frac{\delta}{r}))$. In our case, $\mathbb{A}_\Gamma$ is the product of the size of the automorphism group $\mathrm{Aut}(\Gamma)$ of $\Gamma$ and the degrees from each edge moduli $\mathcal{M}_e$ into the corresponding $\mathbb{C}^*-$fixed substack $\mathcal{K}_e^{\mathbb{C}^*}$.

We will do an explicit computation for the contributions of each graph $\Gamma$ in the following. As for the contribution of a graph $\Gamma$ to (3.6), one can first apply the normalization exact sequence

$$0\to\mathcal{O}_C\to\bigoplus_{vertices}\mathcal{O}_{C_v}\oplus\bigoplus_{edges}\mathcal{O}_{C_e}\to\bigoplus_{flags}\mathcal{O}_{q_{(e,v)}}\to 0$$

to the perfect *relative* obstruction theory of deforming the maps for $\mathcal{K}_{0,\vec{m}\cup\star}(\mathbb{P}Y_{r,s},(\beta,\frac{\delta}{r}))$ over the moduli of twisted curves $\mathcal{M}^{tw}_{0,\vec{m}}$ (see [AGV08, §4.5]), which decomposes the contribution from $\Gamma$ to (3.6) into contributions from vertex, edge, and node factors, This includes all but the automorphisms and deformations within moduli of curves $\mathcal{M}^{tw}_{0,\vec{m}}$. The latter are distributed in the vertex, edge, and node factors as deformations of the vertex components, deformations of the edge components, and deformations of smoothing the nodes, respectively.

3.2.1. *Vertex contribution.* Assume that $v$ is a stable vertex. If the vertex is labeled by $\infty$, we define the vertex moduli $\mathcal{M}_v$ to be $\mathcal{K}_{0,\overrightarrow{val(v)}}(\mathcal{D}_\infty,\beta(v))$ and the fixed part of perfect obstruction theory gives rise to $[\mathcal{K}_{0,\overrightarrow{val(v)}}(\mathcal{D}_\infty,\beta(v))]^{\mathrm{vir}}$. Let $\pi:\mathcal{C}\to\mathcal{K}_{0,\overrightarrow{val(v)}}(\mathcal{D}_\infty,\beta(v))$ be the universal curve and $f:\mathcal{C}\to\mathcal{D}_\infty$ be the universal map, then the moving part of the perfect obstruction theory yields the *inverse of the Euler class* of the virtual normal bundle which is equal to

$$e^{\mathbb{C}^*}((-R^\bullet\pi_*f^*L^{\frac{1}{r}})\otimes\mathbb{C}_{-\frac{\lambda}{r}})\,.$$

When $r$ is sufficiently large, following a generalization of [JPPZ20] to the orbifold case, the above Euler class has a representation

$$\sum_{d\geq 0}c_d(-R^\bullet\pi_*f^*L^{\frac{1}{r}})(\frac{-\lambda}{r})^{|E(v)|-1-d}\,.$$

Here the virtual bundle $-R^\bullet\pi_*f^*L^{\frac{1}{r}}$ has virtual rank $|E(v)|-1$, where $|E(v)|$ is the number of edges incident to the vertex $v$.

If the vertex $v$ is labeled by 0, we define the vertex moduli $\mathcal{M}_v$ to be $\mathcal{K}_{0,\overrightarrow{val(v)}}(\mathcal{D}_0,\beta(v))$ and the fixed part of perfect obstruction theory gives rise to $[\mathcal{K}_{0,\overrightarrow{val(v)}}(\mathcal{D}_0,\beta(v))]^{\mathrm{vir}}$. Let $\pi:\mathcal{C}\to\mathcal{K}_{0,\overrightarrow{val(v)}}(\mathcal{D}_0,\beta(v))$ be the universal curve and $f:\mathcal{C}\to\mathcal{D}_0$ be the universal map, then the moving part of the perfect obstruction theory yields the *inverse of the Euler class* of the virtual normal bundle which is equal to

$$e^{\mathbb{C}^*}((-R^\bullet\pi_*f^*(L^\vee)^{\frac{1}{s}})\otimes\mathbb{C}_{\frac{\lambda}{s}})\,.$$

When $s$ is sufficiently large, following a generalization of [JPPZ20] to the orbifold case, the above Euler class has a representation

$$\sum_{d\geq 0}c_d(-R^\bullet\pi_*f^*(L^\vee)^{\frac{1}{s}})(\frac{\lambda}{s})^{|E(v)|-1-d}\,.$$

Here the virtual bundle $-R^\bullet\pi_*f^*(L^\vee)^{\frac{1}{s}}$ has virtual rank $|E(v)|-1$, where $|E(v)|$ is the number of edges incident to the vertex $v$.

Now assume that there is only one edge $e$ incident to the vertex $v$ labeled by 0 (we will only consider this case in our later localization computation). By the monodromy condition for the edge (see 1), the multiplicity $m(h_0)$ corresponding to the incident half-edge $h_0$ corresponding to $e$ is equal to $(c_e,\mathrm{e}^{\frac{\delta(e)}{s}},1)$ for some $c_e\in\mathscr{C}$. By [Wan25, Lemma 5.2, Remark 5.3], the above Euler class is equal to 1 when $\beta(v)(L)>0$, we note here the semi-positivity of $L$ is essentially used in the proof of [Wan25, Lemma 5.2]. On the other hand, when $\beta(v)(L)=0$, the above Euler class is still equal to 1. Indeed, by semi-positivity of $L$, for any stable map $f:C\to\mathcal{D}_0$ in $\mathcal{K}_{0,\overrightarrow{val(v)}}(\mathcal{D}_0,\beta(v))$, we see that the line bundle $f^*(L^\vee)^{\frac{1}{s}}$ restricts to be a line bundle of degree zero on every irreducible component of curve $C$; we claim that $f^*(L^\vee)^{\frac{1}{s}}$ can't be a trivial line bundle as the isotropy group of the node corresponding to the incident edge $e$ acts nontrivially on $f^*(L^\vee)^{\frac{1}{s}}$, which can be seen from the fact that the multiplicity $m$ of the node at the branch of $\mathscr{C}$ is equal to $m(h_0)^{-1}$ and it satisfies that $age_m(L^\vee)^{\frac{1}{s}}=1-\frac{\delta(e)}{s}\neq 0$ as $0<\delta(e)\leq\delta\ll s$. Then we see that $f^*(L^\vee)^{\frac{1}{s}}$ has no global section and therefore $R^0\pi_*f^*(L^\vee)^{\frac{1}{s}}=0$ and

$$-R^\bullet\pi_*f^*(L^\vee)^{\frac{1}{s}}=0$$

in the Grothendieck $K$-group as $-R^\bullet\pi_*f^*(L^\vee)^{\frac{1}{s}}$ has virtual rank $|E(v)|-1=1-1=0$.

When $v$ is an unstable vertex over 0 (resp. $\infty$), let $h$ be the half-edge incident to $v$, the vertex moduli $\mathcal{M}_v$ is defined to be $\bar{I}_{m(h)^{-1}}\mathcal{D}_0$ (resp. $\bar{I}_{m(h)^{-1}}\mathcal{D}_\infty$) with $[\mathcal{M}_v]^{\mathrm{vir}}=[\mathcal{M}_v]$ and zero virtual normal bundle. In this case, we define the identify map of $\mathcal{M}_e$ to be the evaluation map at the vertex moduli appearing in $ev_{nodes}$ in §3.2.4.

3.2.2. *Edge contribution.* Let $e=\{h_0,h_\infty\}$ be an edge in $\Gamma$ with decorated degree $\delta(e)\in\mathbb{Q}_{>0}$ and decorated multiplicities $m(h_0)$ and $m(h_\infty)$. By the monodromy condition for the edge (see 1), we can write $m(h_\infty)=(c,1,\mathrm{e}^{\frac{\delta(e)}{r}})$ for some $c\in\mathscr{C}$, denote $a_e:=a(c)$ be the number as in Assumption 2.1. We define the edge moduli $\mathcal{M}_e$ to be the root gerbe $\sqrt[a_es\delta(e)]{L^\vee/I_cY}$ over $I_cY$, then the virtual cycle $[\mathcal{M}_e]^{vir}$ coming from the fixed part of the obstruction theory is equal to the fundamental class $[\mathcal{M}_e]$ of $\mathcal{M}_e$ and *inverse of the Euler class* of virtual normal bundle is equal to 1 when $r$ is a sufficiently large prime. We note that $\mathcal{M}_e$ allows a finite étale map into the corresponding fixed-loci in $\mathcal{K}_{0,2}(\mathbb{P}Y_{r,s},(0,\frac{\delta(e)}{r}))$ of degree $\frac{1}{a_es}$. See appendix A.2 for more details.

3.2.3. *Node contributions.* The deformations in $\mathcal{K}_{0,\vec{m}}(\mathbb{P}Y_{r,s},(\beta,\frac{\delta}{r}))$ smoothing a node contribute to the Euler class of the virtual normal bundle as the first Chern class of the tensor product of the two cotangent line bundles at the branches of the node. For nodes at which a component $C_e$ meets a component $C_v$ over the stable vertex labeled by $X_0$, this contributes

$$\frac{\lambda - c_1(L)}{a_e s\delta(e)} - \frac{\bar{\psi}_v}{a_e s}$$

to the *inverse* of the Euler class of the virtual normal bundle.

For nodes at which a component $C_e$ meets a component $C_v$ over the stable vertex labeled by $\mathcal{D}_\infty$, this contributes

$$\frac{-\lambda + c_1(L)}{a_e r\delta(e)} - \frac{\bar{\psi}_v}{a_e r}$$

to the *inverse* of the Euler class of the virtual normal bundle.

We will not need node contributions from other types of nodes as the above types suffice the need in this paper.

3.2.4. *Total contribution.* For any decorated graph $\Gamma$, we define $F_\Gamma$ to be the fiber product

$$\prod_{v:j(v)=0} \mathcal{M}_v \times_{(\bar{I}_\mu \mathcal{D}_0)^{|E|}} \prod_{e\in E} \mathcal{M}_e \times_{(\bar{I}_\mu \mathcal{D}_\infty)^{|E|}} \prod_{v:j(v)=\infty} \mathcal{M}_v$$

of the following diagram:

$$\begin{tikzcd} F_\Gamma \arrow[r] \arrow[d] & \prod_{v:j(v)=0} \mathcal{M}_v \times \prod_{e\in E} \mathcal{M}_e \times \prod_{v:j(v)=\infty} \mathcal{M}_v \arrow[d, "ev_{nodes}"] \\ \prod_E \bar{I}_\mu \mathcal{D}_0 \times \bar{I}_\mu \mathcal{D}_\infty \arrow[r, "(\Delta^0\times\Delta^\infty)^{|E|}"] & \prod_E (\bar{I}_\mu \mathcal{D}_0)^2 \times (\bar{I}_\mu \mathcal{D}_\infty)^2 , \end{tikzcd}$$

where $\Delta^0 = (id,\iota)$(resp. $\Delta^\infty = (id,\iota)$) is the diagonal map of $\bar{I}_\mu\mathcal{D}_0$ (resp. $\bar{I}_\mu\mathcal{D}_\infty$). Here the right-hand vertical map $ev_{nodes}$ is the product of the evaluation maps at the two branches of each node. We note that there are two nodes corresponding to $h_0$ and $h_\infty$ for each edge $e = \{h_0, h_\infty\}$. Then we can define $[F_\Gamma]^{\mathrm{vir}}$ as the Gysin pull-back:

$$\left((\Delta^0\times\Delta^\infty)^{|E|}\right)^! \Big( \prod_{v:j(v)=0} [\mathcal{M}_v]^{\mathrm{vir}} \times \prod_{e\in E} [\mathcal{M}_e]^{\mathrm{vir}} \times \prod_{v:j(v)=\infty} [\mathcal{M}_v]^{\mathrm{vir}} \Big) .$$

Then the contribution of the decorated graph $\Gamma$ to the virtual localization is:

$$Cont_\Gamma = \frac{\prod_{e\in E} a_e s}{|\mathrm{Aut}(\Gamma)|} (\iota_\Gamma)_* \left( \frac{[F_\Gamma]^{\mathrm{vir}}}{e^{\mathbb{C}^*}(N_\Gamma^{\mathrm{vir}})} \right) . \tag{3.7}$$

Here $\iota_F : F_\Gamma \to \mathcal{K}_{0,\vec{m}}(\mathbb{P}Y_{r,s}, (\beta, \frac{\delta}{r}))$ is a finite étale map of degree $\frac{|\mathrm{Aut}(\Gamma)|}{\prod_{e\in E} a_e s}$ into the corresponding $\mathbb{C}^*$-fixed substack. The Euler class of the virtual normal bundle $e^{\mathbb{C}^*}(N_\Gamma^{\mathrm{vir}})$ is the product of the Euler class of virtual normal bundles coming from vertex contributions, edge contributions and node contributions.

We also analyze an important class of localization graph $\Gamma$ which will be one of the cases we need to consider in Proposition 3.8 and give some hints on simplifying the localization computation.

**Remark 3.5.** (An important special case) Let's assume that the graph $\Gamma$ has only one leg, two vertexes and one edge $e = \{h_0, h_\infty\}$, where the leg is marked by $\star$ and it is incident to the vertex labeled by $\infty$ (denoted by $v_\star$). We denote by $v_0$ the vertex labeled by 0. We also assume that the two vertexes are stable. Let $\beta_0$ (resp. $\beta_\star$) be the decorated degree of $v_0$ (resp. $v_\star$). Assume that the corresponding multiplicity $m(h_\infty)$ (resp. $m(h_0)$) of the edge $e$ is equal to $(c, 1, \mathrm{e}^{\frac{\delta(e)}{r}})$ (resp. $(c^{-1}, \mathrm{e}^{\frac{\delta(e)}{s}}, 1)$) for some $c \in \mathscr{C}$, where $\delta(e)$ is the decorated degree of $e$.

Let $f : (\not c, \star) \to \mathbb{P}Y_{r,s}$ be $\mathbb{C}^*-$fixed stable map in $F_\Gamma$. Then we can decompose $(\not c, \star)$ into three parts:

$$(c_0, \bullet), \quad (C_e, (\bullet, \check{\bullet})), \quad (c_\infty, (\check{\bullet}, \star)) .$$

Here, $\bullet$ and $\check{\bullet}$ are the two gluing nodes over 0 and $\infty$ respectively. Let $m(\bullet) = (\not c, \mathrm{e}^{-\frac{\delta(e)}{s}}, 1)$ and $m(\check{\bullet}) = (\not c^{-1}, 1, \mathrm{e}^{-\frac{\delta(e)}{r}})$. Then $m(\bullet) = m(h_0)^{-1}$ is the multiplicity associated to the node $\bullet$ at the branch of $c_0$, and $m(\check{\bullet}) = m(h_\infty)^{-1}$ is the multiplicity associated to the node $\check{\bullet}$ at the branch of $c_\infty$. Denote $\mathcal{M}_{v_0} := \mathcal{K}_{0,\bullet}(\mathcal{D}_0, \beta_0)$ and $\mathcal{M}_{v_\star} := \mathcal{K}_{0,\{\check{\bullet},\star\}}(\mathcal{D}_\infty, \beta_\star)$.

For any target $\spadesuit$, where $\spadesuit$ can be any component of $\bar{I}_\mu Y$, $\bar{I}_\mu \mathcal{D}_0$, $\bar{I}_\mu \mathcal{D}_\infty$ or $\mathcal{M}_e$, by an abuse of notation, we will use $c_1(L)$ to mean the first Chern class of the pullback line bundle of $L$ along the natural map $\spadesuit \to Y$ which factors through the structural map $\bar{I}_\mu Y \to Y$ sending $(x, g)$ to $x$.

Now using the localization formula, the localization contribution

$$Cont_\Gamma\Big( \int_{[\mathcal{K}_{0,\star}(\mathbb{P}Y_{r,s}, (\beta, \frac{\delta}{r}))]^{\mathrm{vir}}} \mathbb{1} \Big)$$

is equal to

$$\begin{aligned}(3.8)\qquad & \frac{a_e s}{|\mathrm{Aut}(\Gamma)|} \int_{[\mathcal{M}_{v_0}]^{\mathrm{vir}} \times [\mathcal{M}_e]^{vir} \times [\mathcal{M}_{v_\star}]^{\mathrm{vir}}} \frac{(ev_{nodes})^*([\Delta_{\bar{I}_{m(\bullet)}\mathcal{D}_0}] \times [\Delta_{\bar{I}_{m(\check{\bullet})}\mathcal{D}_\infty}])}{\big(\frac{\lambda - p_2^* c_1(L)}{a_e s \delta(e)} - \frac{p_1^* \bar{\psi}_\bullet}{a_e s}\big) \cdot \big(-\frac{\lambda - p_2^* c_1(L)}{a_e r \delta(e)} - \frac{p_3^* \bar{\psi}_{\check{\bullet}}}{a_e r}\big)} \\ & \cdot \sum_{d=0}^{\infty} p_3^* c_d(-R^\bullet \pi_* f^* L^{\frac{1}{r}}) (\frac{-\lambda}{r})^{-d} .\end{aligned}$$

Here $|\mathrm{Aut}(\Gamma)| = 1$; $p_1$, $p_2$ and $p_3$ are the projections of $\mathcal{M}_{v_0} \times \mathcal{M}_e \times \mathcal{M}_{v_\star}$ to $\mathcal{M}_{v_0}$, $\mathcal{M}_e$ and $\mathcal{M}_{v_\star}$ respectively. Note that we will also write $p_2^* c_1(L)$ as $c_1(L)$ for simplicity (the same simplification applies to $p_1^* \bar{\psi}_\bullet$, $p_3^* \bar{\psi}_{\check{\bullet}}$ and $p_3^* c_d(\cdots)$) if no confusion occurs.

Let $\gamma_k$ be a (graded) basis of $H^*(\bar{I}_{m(\bullet)}\mathcal{D}_0, \mathbb{C})$. Then the diagonal class $[\Delta_{\bar{I}_{m(\bullet)}\mathcal{D}_0}]$ obtained via push-forward of the fundamental class by $\Delta^0 := (\mathrm{id}, \iota)$ can be written as

$$[\Delta_{\bar{I}_{m(\bullet)}\mathcal{D}_0}] = \sum_k \gamma_k \otimes \gamma^k \text{ in } H^*(\bar{I}_{m(\bullet)}\mathcal{D}_0 \times \bar{I}_{m(\bullet)^{-1}}\mathcal{D}_0, \mathbb{C}),$$

where $\gamma^k \in H^*(\bar{I}_{m^{-1}(\bullet)}\mathcal{D}_0, \mathbb{C})$ is the dual basis satisfying that

$$\int_{[\bar{I}_{m(\bullet)}\mathcal{D}_0]} \iota^*(\gamma^k) \cdot \gamma_j = \delta_{kj} ,$$

where $\delta_{kj} = 1$ if $k = j$, and $\delta_{kj} = 0$ otherwise. Note that we have $\langle \gamma^k, \gamma_j \rangle_{orb} = \frac{1}{a_e} \delta_{kj}$, where $\langle , \rangle_{orb}$ is the orbifold pairing as defined in §2.2. As we have two natural isomorphisms (cf. Lemma 4.3)

$$\bar{\theta}_0 : \bar{I}_{m(h_0)}\mathcal{D}_0 \to \bar{I}_{\not c^{-1}} Y$$

and

$$\bar{\theta}_\infty : \bar{I}_{m(h_\infty)}\mathcal{D}_\infty \to \bar{I}_{\not c} Y$$

induced from the natural maps $\theta_0 : \mathcal{D}_0 \to Y$ and $\theta_\infty : \mathcal{D}_\infty \to Y$. We have induced isomorphisms $H^*(\bar{I}_{m(\bullet)}\mathcal{D}_0) \cong H^*(\bar{I}_{\not c} Y) \cong H^*(\bar{I}_{m(\check{\bullet})^{-1}}\mathcal{D}_\infty)$, from which, by an abuse of notation, we can also use $\gamma_k$ to mean a basis of $H^*(\bar{I}_{m(\check{\bullet})^{-1}}\mathcal{D}_\infty)$ or $H^*(\bar{I}_{\not c} Y)$ with dual basis $\gamma^k$ satisfying that

$$\int_{[\bar{I}_{m(\check{\bullet})^{-1}}\mathcal{D}_\infty]} \iota^*(\gamma^k) \cdot \gamma_j = \int_{[\bar{I}_{\not c} Y]} \iota^*(\gamma^k) \cdot \gamma_j = \delta_{kj} .$$

Let $ev_{h_0}$ and $ev_{h_\infty}$ be the evaluation maps of $\mathcal{M}_e$ corresponding to the half edge $h_0$ and $h_\infty$ respectively (see §A.2 for more details). We have that $ev_{node(\bullet)} = ev_\bullet \times ev_{h_0}$ (resp. $ev_{node(\check{\bullet})} = ev_{h_\infty} \times ev_{\check{\bullet}}$)

is the product of evaluation maps at the two branches of the node $\bullet$ (resp. $\check{\bullet}$). Using the geometrical expansion[10]

$$\frac{1}{\frac{\lambda-c_1(L)}{a_e s\delta(e)}-\frac{\bar{\psi}_{\bullet}}{a_e s}} = a_e s\sum_{m=0}^{\infty}\bar{\psi}_{\bullet}^{m}\Big(\frac{\lambda-c_1(L)}{\delta(e)}\Big)^{-m-1}.$$

and

$$\frac{1}{-\frac{\lambda-c_1(L)}{a_e r\delta(e)}-\frac{\bar{\psi}_{\check{\bullet}}}{a_e r}} = a_e r\sum_{n=0}^{\infty}\bar{\psi}_{\check{\bullet}}^{n}\Big(-\frac{\lambda-c_1(L)}{\delta(e)}\Big)^{-n-1},$$

equation (3.8) is equal to

$$\begin{aligned}
& a_e s\sum_k\sum_j\sum_{m=0}^{\infty}\sum_{n=0}^{\infty}\int_{[\mathcal{K}_{0,\bullet}(\mathcal{D}_0,\beta_0)]^{\mathrm{vir}}} a_e s\cdot\bar{\psi}_{\bullet}^{m}\cdot ev_{\bullet}^{*}\gamma_k \\
&\cdot\int_{[\mathcal{M}_e]} ev_{h_0}^{*}(\gamma^k)\cdot ev_{h_\infty}^{*}(\gamma_j)\cdot\Big(\frac{\lambda-c_1(L)}{\delta(e)}\Big)^{-m-1}\cdot\Big(-\frac{\lambda-c_1(L)}{\delta(e)}\Big)^{-n-1} \\
&\cdot\int_{[\mathcal{K}_{0,(\check{\bullet},\star)}(\mathcal{D}_\infty,\beta_\star)]^{\mathrm{vir}}} a_e r\cdot ev_{\check{\bullet}}^{*}(\gamma^j)\cdot\bar{\psi}_{\check{\bullet}}^{n}\cdot\sum_{d=0}^{\infty}c_d(-R^{\bullet}\pi_*f^*L^{\frac{1}{r}})\Big(\frac{-\lambda}{r}\Big)^{-d} \\
(3.9)\qquad &= a_e s\sum_k\sum_j\sum_{m=0}^{\infty}\sum_{n=0}^{\infty}\int_{[\mathcal{K}_{0,\bullet}(\mathcal{D}_0,\beta_0)]^{\mathrm{vir}}} a_e s\cdot\bar{\psi}_{\bullet}^{m}\cdot ev_{\bullet}^{*}\gamma_k \\
&\cdot\frac{1}{a_e^2 s\delta(e)}\int_{[\bar{I}_c Y]}\iota^*(\gamma^k)\cdot\gamma_j\cdot\Big(\frac{\lambda-c_1(L)}{\delta(e)}\Big)^{-m-1}\cdot\Big(-\frac{\lambda-c_1(L)}{\delta(e)}\Big)^{-n-1} \\
&\cdot\int_{[\mathcal{K}_{0,(\check{\bullet},\star)}(\mathcal{D}_\infty,\beta_\star)]^{\mathrm{vir}}} a_e r\cdot ev_{\check{\bullet}}^{*}(\gamma^j)\cdot\bar{\psi}_{\check{\bullet}}^{n}\cdot\sum_{d=0}^{\infty}c_d(-R^{\bullet}\pi_*f^*L^{\frac{1}{r}})\Big(\frac{-\lambda}{r}\Big)^{-d}
\end{aligned}$$

Here, in the first equality, we push the edge contribution along the natural morphism $\omega_e:\mathcal{M}_e\to\bar{I}_c Y$. Then the induced morphism

$$(\omega_e)_*: H^*(\mathcal{M}_e,\mathbb{C})\to H^*(\bar{I}_c Y,\mathbb{C})$$

is an isomorphism which sends the fundamental class $[\mathcal{M}_e]$ of $\mathcal{M}_e$ to the class $\frac{1}{a_e^2 s\delta(e)}[\bar{I}_c Y]$; then, by push-pull formula, we can replace $ev_{h_0}^{*}(\gamma^k)$ (resp. $ev_{h_\infty}^{*}(\gamma_j)$ by $\iota^*(\gamma^k)$ (resp. $\gamma_j$) as $\bar{\theta}_0\circ ev_{h_0}=\iota\circ\omega_e$ (resp. $\bar{\theta}_\infty\circ ev_{h_\infty}=\omega_e$) shown in §A.2.

Observe that (cf. (2.1))

$$(3.10)\qquad \begin{aligned}
&\sum_k\gamma^k\int_{[\mathcal{K}_{0,\bullet}(\mathcal{D}_0,\beta_0)]^{\mathrm{vir}}}\bar{\psi}_{\bullet}^{m}\cdot ev_{\bullet}^{*}\gamma_k \\
&=(\iota\circ ev_{\bullet})_*(\bar{\psi}_{\bullet}^{m}\cap[\mathcal{K}_{0,\bullet}(\mathcal{D}_0,\beta_0)]^{\mathrm{vir}}),
\end{aligned}$$

[10]This is actually a finite sum as the psi-class $\bar{\psi}_{\bullet}$ (or $\bar{\psi}_{\check{\bullet}}$) is nilpotent.

and

$$\begin{aligned}
&\sum_{m=0}^{\infty}(\iota\circ ev_{\bullet})_*(\bar{\psi}_{\bullet}^m\cap[\mathcal{K}_{0,\bullet}(\mathcal{D}_0,\beta_0)]^{\mathrm{vir}})\cdot(\frac{\lambda-c_1(L)}{\delta(e)})^{-m-1}\\
(3.11)\qquad &=\sum_{m=0}^{\infty}(\iota\circ ev_{\bullet})_*(\bar{\psi}_{\bullet}^m\cup(\frac{\lambda-(\iota\circ ev_{\bullet})^*c_1(L)}{\delta(e)})^{-m-1}\cap[\mathcal{K}_{0,\bullet}(\mathcal{D}_0,\beta_0)]^{\mathrm{vir}})\\
&=(\iota\circ ev_{\bullet})_*\Big(\frac{1}{\frac{\lambda-ev_{\bullet}^*c_1(L)}{\delta(e)}-\bar{\psi}_{\bullet}}\cap[\mathcal{K}_{0,\bullet}(\mathcal{D}_0,\beta_0)]^{\mathrm{vir}}\Big)\,,
\end{aligned}$$

where we use the fact $c_1(L)=\iota^*c_1(L)$ in $H^*(\bar{I}_\mu\mathcal{D}_0)$. Denote

$$f(z):=a_e(\iota\circ ev_{\bullet})_*\big(\frac{1}{z-\bar{\psi}_{\bullet}}\cap[\mathcal{K}_{0,\bullet}(\mathcal{D}_0,\beta_0)]^{\mathrm{vir}}\big)\,.$$

By (B.1), (3.10) and (3.11), (3.8) is equal to

$$\begin{aligned}
&a_e s\sum_j\sum_{n=0}^{\infty}\frac{s}{a_e^2 s\delta(e)}\int_{[\bar{I}_c Y]}\iota^*f(\frac{\lambda-ev_{\bullet}^*c_1(L)}{\delta(e)})\cdot\gamma_j\cdot(-\frac{\lambda-c_1(L)}{\delta(e)})^{-n-1}\\
(3.12)\qquad &\cdot\int_{[\mathcal{K}_{0,(\dot{\bullet},\star)}(\mathcal{D}_\infty,\beta_\star)]^{\mathrm{vir}}}a_e r\cdot ev_{\dot{\bullet}}^*(\gamma^j)\cdot\bar{\psi}_{\dot{\bullet}}^n\cdot\sum_{d=0}^{\infty}c_d(-R^{\bullet}\pi_*f^*L^{\frac{1}{r}})(\frac{-\lambda}{r})^{-d}
\end{aligned}$$

Note that, for any $\alpha\in H^*(\bar{I}_{c^{-1}}Y)$, we have

$$\alpha=\sum_j\gamma^j\int_{[\bar{I}_c Y]}\iota^*\alpha\cdot\gamma_j\,.$$

Then (3.12) is equal to

(3.13)

$$\begin{aligned}
&a_e s\sum_{n=0}^{\infty}\int_{[\mathcal{K}_{0,(\dot{\bullet},\star)}(\mathcal{D}_\infty,\beta_\star)]^{\mathrm{vir}}}a_e r\cdot ev_{\dot{\bullet}}^*(\frac{1}{a_e^2\delta(e)}f(z)(-z)^{-n-1})|_{z=\frac{\lambda-c_1(L)}{\delta(e)}}\cdot\bar{\psi}_{\dot{\bullet}}^n\cdot\sum_{d=0}^{\infty}c_d(-R^{\bullet}\pi_*f^*L^{\frac{1}{r}})(\frac{-\lambda}{r})^{-d}\\
&=a_e s\int_{[\mathcal{K}_{0,(\dot{\bullet},\star)}(\mathcal{D}_\infty,\beta_\star)]^{\mathrm{vir}}}\frac{\frac{1}{a_e^2\delta(e)}(ev_{\dot{\bullet}}^*f(z))|_{z=\frac{\lambda-c_1(L)}{\delta(e)}}}{-\frac{\lambda-ev_{\dot{\bullet}}^*c_1(L)}{a_e r\delta(e)}-\frac{\bar{\psi}_{\dot{\bullet}}}{a_e r}}\cdot\sum_{d=0}^{\infty}c_d(-R^{\bullet}\pi_*f^*L^{\frac{1}{r}})(\frac{-\lambda}{r})^{-d}
\end{aligned}$$

Denote

$$J_{\beta_0}(z):=s(\widetilde{ev}_1)_*\Big(\frac{1}{z-\bar{\psi}_1}\cap\epsilon'_*[\mathcal{K}_{0,\bullet}(\mathcal{D}_0,\beta_0)]^{\mathrm{vir}}\Big)\,,$$

where

$$\epsilon':\mathcal{K}_{0,\bullet}(\mathcal{D}_0,\beta_0)\to\mathcal{K}_{0,\vec{1}}(Y,\beta_0)$$

is the natural structural morphism induced by forgetting root of $\mathcal{D}_0\cong\sqrt[s]{L^\vee/Y}$ (c.f.[TT21]) and $\vec{1}:=c\in\mathscr{C}$. By canceling out the coefficients $a_e$ everywhere, (3.13) is further simplified to

$$(3.14)\qquad \int_{[\mathcal{K}_{0,(\dot{\bullet},\star)}(\sqrt[r]{L/Y},\beta_\star)]^{\mathrm{vir}}}\frac{ev_{\dot{\bullet}}^*(\frac{1}{\delta(e)}J_{\beta_0}(z)|_{z=\frac{\lambda-c_1(L)}{\delta(e)}})}{-\frac{\lambda-ev_{\dot{\bullet}}^*c_1(L)}{r\delta(e)}-\frac{\bar{\psi}_{\dot{\bullet}}}{r}}\cdot\sum_{d=0}^{\infty}c_d(-R^{\bullet}\pi_*f^*L^{\frac{1}{r}})(\frac{-\lambda}{r})^{-d}\,,$$

Note here we treat $J_{\beta_0}(z)$ as an element in $H^*(\bar{I}_{(c^{-1},1,\mu_r^{-\delta(e)})}\sqrt[r]{L/Y},\mathbb{Q})[z,z^{-1}]$ using the natural isomorphism

$$\bar{I}_{(c^{-1},1,\mu_r^{-\delta(e)})}\sqrt[r]{L/Y}\cong\bar{I}_{(c^{-1},1,\mu_r^{-\delta(e)})}\mathbb{P}Y_{r,s}\cong\bar{I}_{c^{-1}}Y$$

when $r$ is a sufficient large prime (see Lemma 4.3).

3.3. **Recursive relation.** Take $L_1 = L$ and $r = 1$ in 2.4 (with target $Y$) and write $\vec{w} = \vec{w_1}$ to be the weight. Let $\mu(z) = \sum_{\beta,\vec{k},c} q^{\beta}\mathbf{t}^{\vec{k}}\mu_{\beta,\vec{k},c}(z)$ be an admissible (power) series as in 2.4. We further require that $\mu_{\beta,\vec{k},c} \in H^*(\bar{I}_{c^{-1}}Y,\mathbb{C})[z]$ is a polynomial in $z$. For any admissible triple $(\beta,\vec{k},c)$ in $\mathrm{Eff}(Y)\times\mathbb{Z}^N_{\geq 0}\times\mathscr{C}$, define $J^Y_{\beta,\vec{k},c}(\mu,z)$ to be

$$
\begin{aligned}
&\mu_{\beta,\vec{k},c}(z) + \mathbf{Coeff}_{\mathbf{t}^{\vec{k}}}\Bigg[\sum_{m\geq 0}\sum_{\substack{(\beta_j,\vec{k_j},c_j)_{j=1}^m\in \mathrm{Adm}_Y^m,\beta_\star\in\mathrm{Eff}(Y)\\ \beta_1+\cdots+\beta_m+\beta_\star=\beta\\ \vec{k_1}+\cdots+\vec{k_m}=\vec{k}}} \\
&\frac{1}{m!}\phi^{\alpha}\langle \mathbf{t}^{\vec{k_1}}\mu_{\beta_1,\vec{k_1},c_1}(-\bar{\psi}_1),\cdots,\mathbf{t}^{\vec{k_m}}\mu_{\beta_m,\vec{k_m},c_m}(-\bar{\psi}_m),\frac{\phi_\alpha}{z-\bar{\psi}_\star}\rangle^Y_{0,\vec{m}\cup\star,\beta_\star}\Bigg],
\end{aligned}
\tag{3.15}
$$

where $\vec{m}\cup\star = (c_1^{-1}\cdots,c_m^{-1},c) \in C^{m+1}$, $\vec{k_j} = (k_{j1},\cdots,k_{jN})$ and $\vec{k_1}+\cdots+\vec{k_m} = \vec{k}$ means the component-wise addition in $\mathbb{Z}^N_{\geq 0}$.

**Remark 3.6.** We can show that when $(\beta,\vec{k},c)$ is not an admissible triple and we define $J^Y_{\beta,\vec{k},c}(\mu,z)$ in the same way as above, we have $J^Y_{\beta,\vec{k},c}(\mu,z) = 0$. Moreover $J_{0,\vec{0},c} = 0$. This implies that $J$−function

$$J^Y(q,\mu,z) = z + \sum_{\beta,\vec{k},c} q^{\beta}\mathbf{t}^{\vec{k}}J^Y_{\beta,\vec{k},c}(\mu,z)$$

associated with the input $\mu(z)$ above is an admissible series near $z$.

**Definition 3.7.** *Let $m,n$ be two nonnegative integers and $(\beta,\vec{k},c)$ be an admissible triple. We denote $\Lambda^Y_{\beta,\vec{k},c,m,n}$ to be the set of tuples*

$$\left(c,\beta_\star,((\beta_1,\vec{k_1},c_1),\cdots,(\beta_{m+n},\overrightarrow{k_{m+n}},c_{m+n}))\right)\in\mathscr{C}\times\mathrm{Eff}(Y)\times\mathrm{Adm}_Y^{m+n},$$

*where we require that $\beta_\star + \sum_{i=1}^{m+n}\beta_i = \beta$, $\sum_{i=1}^{m+n}\vec{k_i} = \vec{k}$, $\beta_i(L)+(\vec{w},\vec{k_i}) > 0$ for $1\leq i\leq m$ and $\beta_i(L)+(\vec{w},\vec{k_i}) = 0$ for $m+1\leq i\leq m+n$. We call an element of $\Lambda^Y_{\beta,\vec{k},c,m,n}$ stable if $\beta_\star\neq 0$ or $m+n\geq 2$ when $\beta_\star = 0$.*

We note that $\Lambda^Y_{\beta,\vec{k},c,m,n}$ is a finite set. Indeed, let e be a positive integer which is divided by all the orders of stablizer groups of $\mathbb{C}$−points of $Y$, then we can show that the coarsification map $\pi : Y\to \underline{Y}$ induces an injection

$$\pi_* : \mathrm{Eff}(Y)\to \frac{1}{\mathrm{e}}\mathrm{Eff}(\underline{Y}).$$

Let $M$ be a line bundle over $Y$ which is a pullback of an ample line bundle over the coarse moduli $\underline{Y}$ of $Y$, then we have that, for any positive number $b$, the set $\{\beta\in\mathrm{Eff}(Y)|\beta(M) < b\}$ is finite (cf. [Laz17, Example 1.4.31]).

For any admissible triple $(\beta,\vec{k},c)$ with $\beta(L)+(\vec{w},\vec{k}) > 0$. We have the following recursive characterization about $J^Y_{\beta,\vec{k},c}(\mu,z)$.

**Proposition 3.8.** *For any admissible triple $(\beta,\vec{k},\mathcal{c})\in \mathrm{Adm}_Y$ with $\beta(L)+(\vec{w},\vec{k})>0$. Assume that $r$ is a sufficiently large prime, for any nonnegative integer $b$, we have the following recursive relation:*

$$
\begin{aligned}
&[J^Y_{\beta,\vec{k},\mathcal{c}}(\mu,z)]_{z^{-b-1}}\\
&=\Big[\sum_{m=0}^{\infty}\sum_{n=0}^{\infty}\sum_{\substack{\Gamma\in\Lambda^Y_{\beta,\vec{k},\mathcal{c},m,n}\\ \Gamma \text{ is stable}}}\frac{1}{m!n!}(\widetilde{ev_\star})_*\Big(\prod_{i=1}^{m}\frac{ev_i^*(\frac{1}{\delta_i}\mathbf{t}^{\vec{k_i}}J^Y_{\beta_i,\vec{k_i},\mathcal{c}_i}(\mu,z)|_{z=\frac{\lambda-c_1(L)}{\delta_i}})}{\frac{\lambda-ev_i^*c_1(L)}{r\delta_i}+\frac{\bar{\psi}_i}{r}} \qquad (3.16)\\
&\cup\prod_{i=m+1}^{m+n}ev_i^*(\mathbf{t}^{\vec{k_i}}\mu_{\beta_i,\vec{k_i},\mathcal{c}_i}(-\bar{\psi}_i))\cup\bar{\psi}_\star^b\\
&\cap\sum_{d=0}^{\infty}\epsilon_*\big(c_d(-R^\bullet\pi_*f^*L^{\frac{1}{r}})(\frac{\lambda}{r})^{-1+m-d}(-1)^d\cap[\mathcal{K}_{0,\overrightarrow{m+n}\cup\star}(\sqrt[r]{L/Y},\beta_\star)]^{\mathrm{vir}}\Big)\Big]_{\mathbf{t}^{\vec{k}}\lambda^{-1}}.
\end{aligned}
$$

*Here $\delta_i=\beta_i(L)+(\vec{w},\vec{k_i})$, and $\epsilon:\mathcal{K}_{0,\overrightarrow{m+n}\cup\star}(\sqrt[r]{L/Y},\beta_\star)\to\mathcal{K}_{0,\overrightarrow{m+n}\cup\star}(Y,\beta_\star)$ is the natural structural morphism induced by forgetting the root structure of $\sqrt[r]{L/Y}$ (c.f.[TT21]), where we choose the tuple $\overrightarrow{m+n}\cup\star$ for $\mathcal{K}_{0,\overrightarrow{m+n}\cup\star}(\sqrt[r]{L/Y},\beta_\star)$ to be*

$$\big((\mathcal{c}_1^{-1},1,\mathrm{e}^{\frac{-\delta_1}{r}}),\cdots,(\mathcal{c}_{m+n}^{-1},1,\mathrm{e}^{\frac{-\delta_{m+n}}{r}}),(\mathcal{c},1,\mathrm{e}^{\frac{\delta}{r}})\big)\in(\mathscr{C}\times\mathbb{C}^*\times\mathbb{C}^*)^{m+n+1},$$

*and choose the tuple $\overrightarrow{m+n}\cup\star$ for $\mathcal{K}_{0,\overrightarrow{m+n}\cup\star}(Y,\beta_\star)$ to be*

$$(\mathcal{c}_1^{-1},\cdots,\mathcal{c}_{m+n}^{-1},\mathcal{c})\in\mathscr{C}^{m+n+1}.$$

The proof of the above proposition is based on applying virtual localization to the following integral.

$$
\begin{aligned}
&\sum_{m=0}^{\infty}\sum_{\substack{(\beta_j,\vec{k_i},\mathcal{c}_i)_{i=1}^m\in\mathrm{Adm}_Y^m,\beta_\star\in\mathrm{Eff}(Y)\\ \beta_\star+\sum_{i=1}^m\beta_i=\beta\\ \vec{k_1}+\cdots+\vec{k_m}=\vec{k}}}\frac{1}{m!}(\widetilde{EV_\star})_*\Big(\prod_{i=1}^{m}ev_i^*(\mathrm{pr}_{r,s}^*(\mathbf{t}^{\vec{k_i}}\mu_{\beta_i,\vec{k_i},\mathcal{c}_i}(-\bar{\psi}_i))) \qquad (3.17)\\
&\cup\bar{\psi}_\star^b\cap[\mathcal{K}_{0,\vec{m}\cup\star}(\mathbb{P}Y_{r,s},(\beta_\star,\frac{\delta}{r}))]^{\mathrm{vir}}\Big).
\end{aligned}
$$

Here an explanation of the notations is in order:

(1) For degrees $\beta_\star,\beta_1,\cdots,\beta_m$ in $\mathrm{Eff}(Y)$ and tuples $\vec{k_1},\cdots,\vec{k_m}$ with $\sum_{i=1}^m\beta_i+\beta_\star=\beta$ and $\vec{k_1}+\cdots+\vec{k_m}=\vec{k}$, let $(\mathcal{c}_1,\cdots,\mathcal{c}_m)\in\mathscr{C}^m$ such that $(\beta_i,\vec{k_i},\mathcal{c}_i)$ are admissible triples. Write $\delta_i=\beta_i(L)+(\vec{w},\vec{k_i})$ and $\delta=\beta(L)+(\vec{w},\vec{k})$, we define $\vec{m}\cup\star$ to be the $(m+1)$–tuple

$$\big((\mathcal{c}_1^{-1},\mathrm{e}^{\frac{\delta_i}{s}},1),\cdots,(\mathcal{c}_m^{-1},\mathrm{e}^{\frac{\delta_m}{s}},1),(\mathcal{c},1,\mathrm{e}^{\frac{\delta}{r}})\big).$$

(2) The morphism $EV_\star$ is a composition of the following maps:

$$\mathcal{K}_{0,\vec{m}\cup\star}(\mathbb{P}Y_{r,s},(\beta_\star,\frac{\delta}{r}))\xrightarrow{ev_\star}\bar{I}_\mu\mathbb{P}Y_{r,s}\xrightarrow{\mathrm{pr}_{r,s}}\bar{I}_\mu Y,$$

and $(\widetilde{EV_\star})_*$ is defined by

$$\iota_*(r_\star(EV_\star)_*).$$

Note here $r_\star$ is the order of the band from the gerbe structure of $\bar{I}_\mu Y$ but not $\bar{I}_\mu\mathbb{P}Y_{r,s}$.

First we state a vanishing lemma regarding applying localization formula to (3.17).

**Lemma 3.9.** *Assume $r, s$ is sufficiently large. If the localization graph $\Gamma$ has more than one vertex labeled by $\infty$, then the corresponding $\mathbb{C}^*$−fixed loci moduli $F_\Gamma$ is empty; therefore it will contribute zero to* (3.17).

*Proof.* For any twisted stable map $f : C \to \mathbb{P}Y_{r,s}$ in $\mathcal{K}_{0,\vec{m}\cup\star}(\mathbb{P}Y_{r,s}, (\beta, \frac{\delta}{r}))$, denote $N := f^*(\mathcal{O}(-\mathcal{D}_\infty))$, first we show that $H^1(C, N) = 0$. Indeed, using orbifold Riemann-Roch, we have

$$\chi(N) = 1 + deg(N) - age(N|_{q_\star}) = 0\,,$$

as $deg(N) = -\frac{\delta}{r}$, and $age(N|_{q_\star}) = 1 - \frac{\delta}{r}$, then showing $H^1(C, N) = 0$ is equivalent to show $H^0(C, N) = 0$. As the degree of $N$ is negative on $C$, it remains to show that the degree of the restriction of the line bundle $N$ to every irreducible component $E$ of $C$ is non-positive. Observe that the degree $deg(N|_E)$ is equal to intersection number $-([E], [\mathcal{D}_\infty])$. If the image of an irreducible component of $C$ via $f$ isn't contained in $\mathcal{D}_\infty$, the restricted degree of $N$ to $E$ is obviously non-positive. Otherwise, observe that $N$ is isomorphic to $(L^{\frac{1}{r}})^\vee$ over $\mathcal{D}_\infty$. As $L$ is semi-positive, the claim follows. This finishes the proof of the part $H^1(C, N) = 0$.

Now assume by contradiction that the moduli of fixed-loci $F_\Gamma$ is nonempty. By the connectedness of the graph $\Gamma$, there is at least one vertex of the graph $\Gamma$ labeled by 0 with at least two edges attached. Suppose $f : C \to \mathbb{P}Y_{r,s}$ belongs to the moduli $F_\Gamma$ of $\mathbb{C}^*$−fixed loci. Assume that $C_0 \cup C_1 \cup C_2$ is part of curve $C$, where $C_0$ is mapped by $f$ to $\mathcal{D}_0$ and $C_1, C_2$ are edges meeting with $C_0$ at $b_1$ and $b_2$. Then in the normalization sequence (cf.(3.5) ) for $R^\bullet\pi_* N$:

$$\begin{aligned} 0 \to &H^0(C, N) \to \bigoplus_{vertices} H^0(C_v, N) \oplus \bigoplus_{edges} H^0(C_e, N) \to \bigoplus_{flags} H^0(q_{(e,v)}, N) \\ \to &H^1(C, N) \to \bigoplus_{vertices} H^0(C_v, N) \oplus \bigoplus_{edges} H^0(C_e, N) \to 0 \end{aligned} \tag{3.18}$$

which contains the part

$$\begin{aligned} &H^0(c_0, N) \oplus H^0(C_1, N) \oplus H^0(C_2, N) \\ \to &H^0(b_1, N) \oplus H^0(b_2, N) \\ \to &H^1(C, N)\,. \end{aligned}$$

Note that there is a natural $\mathbb{C}^*$−action on the line bundle $\mathcal{O}(-\mathcal{D}_\infty)$ over $\mathbb{P}Y_{r,s}$ so that $\mathbb{C}^*$ acts on the fibers over $D_\infty$ with weight 1 and $\mathbb{C}^*$ acts on the fibers over $\mathcal{D}_0$ with weight 0. This also induces $\mathbb{C}^*$−actions on the cohomology groups above. In particular, we have $H^0(b_1, N)$, $H^0(b_2, N)$, and $H^0(C_0, N)$ are isomorphic to the one-dimensional trivial $\mathbb{C}^*$−representation. Note that $H^0(C_1, N) = H^0(C_2, N) = 0$. Hence there is one of the weight-0 pieces in $H^0(b_1, N) \oplus H^0(b_2, N) \cong \mathbb{C}^2$ that is canceled with a weight-0 piece of $H^0(C_0, N)$, and the other is mapped injectively into $H^1(C, N)$, but this contradicts that $H^1(C, N) = 0$. So $F_\Gamma$ is empty. □

Now we prove Proposition 3.8.

*Proof of Proposition 3.8.* The proof is almost identical to the proof in [Wan25, §6.2], we here sketch the main steps for the convenience of readers.

By Lemma 3.9, we only need to consider the decorated graph $\Gamma$ that has only one vertex labeled by $\infty$. Note that the marking $q_\star$ corresponding to $\star$ is incident to the vertex $v_\star$ due to the choice of multiplicity at the marking $q_\star$, then the vertex $v_\star$ can't be a node linking two edges. Such decorated graph is *star-shaped*, i.e., the vertex set $V$ allows a decomposition

$$V = \{v_\star\} \bigsqcup V_0\,.$$

where the vertex $v_\star$ is labeled by $\infty$, and all the vertexes in $V_0$ are labeled by 0. Each vertex labeled by 0 is linked to $v_\star$ by a unique edge. There are three types of star-shaped decorated graphs:

(1) (Type I) The vertex $v_\star$ is unstable and there is only one edge incident to $v_\star$ in $\Gamma$, the unique vertex $v$ labeled by 0 is unstable;
(2) (Type II) The vertex $v_\star$ is unstable and there is only one edge incident to $v_\star$ in $\Gamma$, the unique vertex $v$ labeled by 0 is stable;
(3) (Type III) The vertex $v_\star$ is stable.

Denote by $\beta_\star$ the degree of the unique vertex $v_\star$ labeled by $\infty$. Now let's compute the localization contribution from the above three types of graphs:

(1) If the graph $\Gamma$ is of type I, the vertex over 0 corresponds to a marked point with input $\mu_{\beta,\vec{k},c}$, then the graph $\Gamma$ contributes
$$\mathbf{t}^{\vec{k}}\frac{\mu_{\beta,\vec{k},c}(\frac{\lambda-c_1(L)}{\delta})}{\delta}\cdot(\frac{\lambda-c_1(L)}{\delta})^b$$
to (4.5). Here we use the fact that the restriction of the psi-class $\bar{\psi}_\star$ to $\mathcal{M}_e$ is equal to $\frac{\lambda-c_1(L)}{\delta}$(see Remark A.4).

(2) If the graph $\Gamma$ is of type II, $\Gamma$ has the same description with type I except that the vertex over 0 is stable. Then this type of graphs contributes
$$\sum_{m=0}^{\infty}\sum_{\substack{(\beta_j,\vec{k_j},c_j)_{j=1}^m\in\mathrm{Adm}_Y^m,\beta_\star\in\mathrm{Eff}(Y)\\ \beta_1+\cdots+\beta_m+\beta_\star=\beta\\ \vec{k_1}+\cdots+\vec{k_m}=\vec{k}}}\frac{1}{m!}(\widetilde{ev_\star})_*\Big(\prod_{i=1}^m ev_i^*(\mathbf{t}^{\vec{k_i}}\mu_{\beta_i,\vec{k_i},c_i}(-\bar{\psi}_i))$$
$$\cup\frac{\frac{1}{\delta}(\frac{\lambda-ev_\star^*c_1(L)}{\delta})^b}{\frac{\lambda-ev_\star^*c_1(L)}{s\delta}-\frac{\bar{\psi}_\star}{s}}\cap\epsilon'_*([\mathcal{K}_{0,\vec{m}_s\cup\star}(\sqrt[s]{L^\vee/Y},\beta_\star)]^{\mathrm{vir}})\Big)$$
to (3.17), where $\vec{m}_s\cup\star=\big((c_1^{-1},e^{\frac{\delta_1}{s}},1),\cdots,(c_m^{-1},e^{\frac{\delta_m}{s}},1),(c,e^{-\frac{\delta}{s}},1)\big)\in(\mathscr{C}\times\mathbb{C}^*\times\mathbb{C}^*)^{m+1}$, $\vec{m}\cup\star=\big(c_1^{-1},\cdots,c_m^{-1},c\big)\in C^{m+1}$ and
$$\epsilon':\mathcal{K}_{0,\vec{m}_s\cup\star}(\sqrt[s]{L^\vee/Y},\beta_\star)\to\mathcal{K}_{0,\vec{m}\cup\star}(Y,\beta_\star)$$
is the natural structure morphism by forgetting root of $\sqrt[s]{L^\vee/Y}$ (c.f.[TT21]). Note here we implicitly cancel out $a_e$ factors from the node contribution and automorphic factor everywhere and use the fact that the Euler class of virtual normal bundle for the vertex over 0 is equal to 1. It's proved that
$$\epsilon'_*([\mathcal{K}_{0,\vec{m}_s\cup\star}(\sqrt[s]{L^\vee/Y},\beta_\star)]^{\mathrm{vir}})=\frac{1}{s}[\mathcal{K}_{0,\vec{m}\cup\star}(Y,\beta_\star)]^{\mathrm{vir}},$$
in [TT21], then the above formula is equal to
$$\sum_{m\geq 0}\sum_{\substack{(\beta_j,\vec{k_j},c_j)_{j=1}^m\in\mathrm{Adm}_Y^m,\beta_\star\in\mathrm{Eff}(Y)\\ \beta_1+\cdots+\beta_m+\beta_\star=\beta\\ \vec{k_1}+\cdots+\vec{k_m}=\vec{k}}}\frac{1}{m!}\phi^\alpha\langle\mu_{\beta_1,\vec{k_1},c_1}(-\bar{\psi}_1),\cdots,\mu_{\beta_m,\vec{k_m},c_m}(-\bar{\psi}_m),\frac{\frac{1}{\delta}(\frac{\lambda-c_1(L)}{\delta})^b\phi_\alpha}{\frac{\lambda-c_1(L)}{\delta}-\bar{\psi}_\star}\rangle_{0,\vec{m}\cup\star,\beta_\star}.$$

(3) If the graph $\Gamma$ is of type III, then $v_\star$ is incident to the distinguished leg corresponding to the marking $q_\star$ and $m$ edges ($m$ can be 0) and $n$ legs. We can assume that each leg is associated with an input $\mathbf{t}^{\vec{k'}}\mu_{\beta',\vec{k'},c'}(z)$ with $\beta'(L)+(\vec{w},\vec{k'})=0$, otherwise the support of

the twisted sector $\bar{I}_{c'^{-1}}Y$ will avoid $\mathcal{D}_\infty$. Choose a labeling[11] of the $m$ edges attached to the vertex $v_\star$ by $[m] := \{1, \cdots, m\}$. Let $\delta_i$ be the decorated degree associated with the $i$th edge $e_i$. Let $v_i$ be the vertex over 0 incident to $e_i$, then $v_i$ can't be a unstable vertex of valence 1 as the corresponding ramification point $q_0$ is a stacky point, or it can't be a node linking two edges by Lemma 3.9. Therefore $v_i$ corresponds to either a marking point or a stable vertex. Assume that there are $l$ legs ($l$ can be zero) incident to $v_i$. let's label the legs incident to $v_i$ by $\{i1, \cdots, il\} \subset [\ |\{\text{Legs of }\Gamma\}|\ ]$. Note that when $v_i$ is unstable, $l = 1$.

Assume that the vertex $v_i$ is decorated by the degree $\beta_{i0}$, and the insertion at the marking $q_{ij}$ on the curve[12] $C_{v_i}$ corresponds to $\mathbf{t}^{\overrightarrow{k_{ij}}}\mu_{\beta_{ij},\overrightarrow{k_{ij}},c_{ij}}(-\bar{\psi}_{ij})$ in (3.17). Let's say the leg for $q_{ij}$ contributes a *virtual extended degree* $(\beta_{ij}, \overrightarrow{k_{ij}})$ to the vertex $v_i$. Denote by $(\beta_i, \overrightarrow{k_i})$ the summation of $(\beta_{i0}, 0)$ and all the virtual extended degrees from the legs incident to $v_i$. We call $(\beta_i, \overrightarrow{k_i})$ a *total extended degree* at the vertex $v_i$. From (3.17), one has

$$\beta_\star + \sum_{i=1}^{m} \beta_i = \beta, \quad \sum_{i=1}^{m} \overrightarrow{k_i} = \vec{k}\,.$$

Assume the multiplicity of the half edge incident to $C_{v_i}$ is equal to $(c_i^{-1}, e^{\frac{\delta_i}{s}}, 1)$, here $\delta_i$ is the decorated degree associated with the edge $e_i$ and $\delta_i = age_c(L) \bmod \mathbb{Z}$ (see 1). Observe that to ensure such a graph $\Gamma$ exists, one must have

$$\beta_i(L) + (\vec{w}, \overrightarrow{k_i}) = \delta_i\,. \tag{3.19}$$

Indeed, by orbifold Riemann-Roch Theorem, one has

$$deg(f^*\mathcal{O}(\mathcal{D}_0)|_{C_{v_i}}) = -\frac{\beta_{i0}(L)}{s} = (1 - \frac{\delta_i}{s}) + \sum_{j=1}^{l} \frac{\beta_{ij}(L) + (\vec{w}, \overrightarrow{k_{ij}})}{s} \quad mod \quad \mathbb{Z}\,.$$

Here the first term on the right-hand is the age of $f^*\mathcal{O}(\mathcal{D}_0)$ at the node of $C_{c_i}$, and the second term on the right is the sum of the ages of $f^*\mathcal{O}(\mathcal{D}_0)$ at the marked points on $C_{c_i}$. As $s$ is sufficiently large, one must have

$$\frac{\delta_i}{s} = \frac{\beta_{i0}(L)}{s} + \sum_{j=1}^{l} \frac{\beta_{ij}(L) + (\vec{w}, \overrightarrow{k_{ij}})}{s}\,,$$

which implies that $\beta_i(L) + (\vec{w}, \overrightarrow{k_i}) = \delta_i$. Note that this also imply that $(\beta_i, \overrightarrow{k_i}, c_i)$ is an admissible triple.

Now we can group the edge-labeled decorated graphs by the set $\Lambda^{Y}_{\beta,\vec{k},c,m,n}$ defined in 3.7. For any element of $\Lambda^{Y}_{\beta,\vec{k},c,m,n}$, we can naturally associate a set of edge-labeled star-shaped decorated graphs such that the vertex incident to the edge labeled by $i$ has total extended degree $(\beta_i, \overrightarrow{k_i})$ and the multiplicity $m(h_i)$ at the half-edge $h_i$ incident to $v_i$ over 0 is equal to $(c_i^{-1}, e^{\frac{\delta_i}{s}}, 1)$. We may call an element $\phi := (c, \beta_\star, ((\beta_1, \overrightarrow{k_1}, c_1), \cdots, (\beta_{m+n}, \overrightarrow{k_{m+n}}, c_{m+n})))$ of $\Lambda^{Y}_{\beta,\vec{k},c,m,n}$ a *meta graph*, and denote all the (star-shaped) edge-labeled decorated graphs associated to $\phi$ by $\Gamma_\phi$.

Now we use the localization formula in §3.2.4 to compute the contribution from $\Gamma_\phi$ to (3.17). Note that we have the natural isomorphism $\bar{I}_{c_i^{-1}}Y \cong \bar{I}_{m_{h_i}}\mathcal{D}_0$ when $s$ is a sufficiently large prime (c.f. Lemma 4.3). Summing over the vertex contribution of the vertex $v_i$ and

[11] Such a labeling is not unique, but we will divide $m!$ to offset the labeling in the end.

[12] When $v_i$ is unstable, $v_i$ corresponds to $q_{i1}$.

the node contribution of the node corresponding to the half-edge $h_i$ incident to $v_i$ from all the graphs in $\Gamma_\phi$, and pushing forward to $\bar{I}_{c_i^{-1}}Y \cong \bar{I}_{m(h_i)}\mathcal{D}_0$ along $\iota\circ(ev_{h_i})_*$(recall that $\iota$ is the inversion map on Chen-Ruan cohomology $H^*(\bar{I}_\mu Y)$), it yields

$$\mathbf{t}^{\vec{k}_i}\mu_{\beta_i,\vec{k}_i,c_i}(\frac{\lambda-c_1(L)}{\delta_i})+\sum_{m=0}^{\infty}\sum_{\substack{(\beta_j,\vec{k}_j,c_j)_{j=1}^m\in\mathrm{Adm}_Y^m,\beta_\star\in\mathrm{Eff}(Y)\\ \beta_\star+\sum_{j=1}^m\beta_j=\beta_i\\ \vec{k}_1+\cdots+\vec{k}_m=\vec{k}}}\frac{1}{m!}$$
$$(\widetilde{ev_\star})_*\Big(\prod_{j=1}^m ev_j^*(\mathbf{t}^{\vec{k}_j}\mu_{\beta_j,\vec{k}_j,c_j}(-\bar{\psi}_j))\cup\frac{1}{\frac{\lambda-ev_\star^*c_1(L)}{\delta_j s}-\frac{\bar{\psi}_\star}{s}}\cap\epsilon'_*\big([\mathcal{K}_{0,\vec{m}_s\cup\star}(\sqrt[s]{L^\vee/Y},\beta_\star)]^{\mathrm{vir}}\big)\Big),$$

which is equal to $\mathbf{t}^{\vec{k}_i}J^Y_{\beta_i,\vec{k}_i,c_i}(\mu,z)|_{z=\frac{\lambda-c_1(L)}{\delta_i}}$. Note here we use the fact the Euler class of the virtual normal bundle for $v_i$ is equal to 1. Observe that all the edge-labeled decorated graphs in $\Gamma_\phi$ have the same vertex (edge, and node) localization contributions from the unique vertex $v_\star$ labeled by $\infty$ (the edge $e_i$, and the node over $\infty$ incident to $e_i$ respectively). Moreover the localization formula for any graph in $\Gamma_\phi$ depends multi-linearly on the localization contributions of vertexes over 0, then the localization from all meta graphs in $\Lambda^Y_{\beta,\vec{k},c,m,n}$ to (3.17) yields the summation:

(3.20)
$$\begin{aligned}&\sum_{m=0}^{\infty}\sum_{n=0}^{\infty}\sum_{\substack{\Gamma\in\Lambda^Y_{\beta,\vec{k},c,m,n}\\ \Gamma\text{ is stable}}}\frac{1}{m!n!}(\widetilde{ev_\star})_*\Big(\prod_{i=1}^m\frac{ev_i^*(\frac{1}{\delta_i}\mathbf{t}^{\vec{k}_i}J^Y_{\beta_i,\vec{k}_i,c_i}(\mu,z)|_{z=\frac{\lambda-c_1(L)}{\delta_i}})}{-\frac{\lambda-ev_i^*c_1(L)}{r\delta_i}-\frac{\bar{\psi}_i}{r}}\\ &\cup\prod_{i=m+1}^{m+n}ev_i^*(\mathbf{t}^{\vec{k}_i}\mu_{\beta_i,\vec{k}_i,c_i}(-\bar{\psi}_i))\cup\bar{\psi}_\star^b\\ &\cap\sum_{d=0}^{\infty}\epsilon_*\big(c_d(-R^\bullet\pi_*f^*L^{\frac{1}{r}})(\frac{-\lambda}{r})^{-1+m-d}\cap[\mathcal{K}_{0,\overrightarrow{m+n}\cup\star}(\sqrt[r]{L/Y},\beta_\star)]^{\mathrm{vir}}\big)\Big).\end{aligned}$$

By the discussion above, we can write (3.17) in the following way:

$$
\begin{aligned}
&\mathbf{t}^{\vec{k}}\frac{\mu_{\beta,\vec{k},c}(\frac{\lambda-c_1(L)}{\delta})}{\delta}\cdot(\frac{\lambda-c_1(L)}{\delta})^b+\sum_{m=0}^{\infty}\sum_{\substack{(\beta_j,\vec{k_j},c_j)_{j=1}^m\in\mathrm{Adm}_Y^m,\beta_\star\in\mathrm{Eff}(Y)\\ \beta_1+\cdots+\beta_m+\beta_\star=\beta\\ \vec{k_1}+\cdots+\vec{k_m}=\vec{k}}}\frac{1}{m!}(\widetilde{ev_\star})_*\\
&\Big(\prod_{i=1}^m ev_i^*(\mathbf{t}^{\vec{k_i}}\mu_{\beta_i,\vec{k_i},c_i}(-\bar{\psi}_i))\cup\frac{\frac{1}{\delta}(\frac{\lambda-ev_\star^*c_1(L)}{\delta})^b}{\frac{\lambda-ev_\star^*c_1(L)}{\delta}-\bar{\psi}_\star}\cap[\mathcal{K}_{0,\vec{m}\cup\star}(Y,\beta_\star)]^{\mathrm{vir}}\Big)\\
&+\sum_{m=0}^{\infty}\sum_{n=0}^{\infty}\sum_{\substack{\Gamma\in\Lambda^Y_{\beta,\vec{k},c,m,n}\\ \Gamma\text{ is stable}}}\frac{1}{m!n!}(\widetilde{ev_\star})_*\Big(\prod_{i=1}^m\frac{ev_i^*(\frac{1}{\delta_i}\mathbf{t}^{\vec{k_i}}J^Y_{\beta_i,\vec{k_i},c_i}(\mu,z)|_{z=\frac{\lambda-c_1(L)}{\delta_i}})}{-\frac{\lambda-ev_i^*c_1(L)}{r\delta_i}-\frac{\bar{\psi}_i}{r}}\\
&\cup\prod_{i=m+1}^{m+n}ev_i^*(\mathbf{t}^{\vec{k_i}}\mu_{\beta_i,\vec{k_i},c_i}(-\bar{\psi}_i))\cup\bar{\psi}_\star^b\\
&\cap\sum_{d=0}^{\infty}\epsilon_*\big(c_d(-R^\bullet\pi_*f^*L^{\frac{1}{r}})(\frac{-\lambda}{r})^{-1+m-d}\cap[\mathcal{K}_{0,\overrightarrow{m+n}\cup\star}(\sqrt[r]{L/Y},\beta_\star)]^{\mathrm{vir}}\big)\Big).
\end{aligned}
\tag{3.21}
$$

As (3.17) lies in $\mathbf{t}^{\vec{k}}H^*(\bar{I}_\mu Y,\mathbb{Q})[\lambda]$, the coefficient of $\mathbf{t}^{\vec{k}}\lambda^{-1}$ term in (3.21) must vanish. Note that the coefficients before $\lambda^{-1}$ in the first two summands in (3.21) yields

$$
\sum_{m=0}^{\infty}\sum_{\substack{(\beta_j,\vec{k_j},c_j)_{j=1}^m\in\mathrm{Adm}_Y^m,\beta_\star\in\mathrm{Eff}(Y)\\ \beta_1+\cdots+\beta_m+\beta_\star=\beta\\ \vec{k_1}+\cdots+\vec{k_m}=\vec{k}}}\frac{1}{m!}\phi^\alpha\langle\mu_{\beta_1,\vec{k_1},c_1}(-\bar{\psi}_1),\cdots,\mu_{\beta_m,\vec{k_m},c_m}(-\bar{\psi}_m),\phi_\alpha\bar{\psi}_\star^b\rangle^Y_{0,\vec{m}\cup\star,\beta_\star},
$$

which is the left-hand side of equality in (3.16). Then we extract the coefficient of the $\mathbf{t}^{\vec{k}}\lambda^{-1}$ term in the third summand in (3.21), this is equal to the term on the right-hand side of (3.16) differing only by a minus sign. This completes the proof of (3.16). $\square$

3.4. **Specializing Novikov degrees.** Now assume that $W$ is a smooth proper DM stack with projective coarse moduli with a semi-positive line bundle $L$ over $W$, and $Y$ is a smooth closed substack of $W$. We will also use $L$ to denote the restriction of $L$ to $Y$ by an abuse of notation. Given a degree $\beta\in\mathrm{Eff}(W)$, we define $\mathcal{K}_{0,\vec{m}}(Y,\beta)$ as the disjoint union[13]

$$
\bigsqcup_{\substack{d\in\mathrm{Eff}(Y)\\ i_*(d)=\beta}}\mathcal{K}_{0,\vec{m}}(Y,d)\,.
$$

An analogous convention applies to Gromov-Witten invariants: given a degree $\beta\in\mathrm{Eff}(W)$, we will denote the Gromov-witten invariants

$$
\langle\cdots\rangle^Y_{0,\vec{m},\beta}:=\sum_{d\in\mathrm{Eff}(Y):i_*(d)=\beta}\langle\cdots\rangle^Y_{0,\vec{m},d}\,.
$$

Of course, $\mathcal{K}_{0,\vec{m}}(Y,\beta)$ can be empty when there is no degree $d\in\mathrm{Eff}(W)$ such that $i_*(d)=\beta$, then the corresponding GW invariants are defined to be zero.

[13]It's a finite disjoint union as $\mathcal{K}_{0,m}(X,\beta)$ is finite type over $\mathbb{C}$, hence Noetherian.

Now we use the recursion relation (3.16) and apply the rule of specialization of degrees as above. We can show the following variation of Proposition 3.8.

**Proposition 3.10.** *Let*

$$\mu(z) = \sum_{\beta,\vec{k},c} q^\beta \mathbf{t}^{\vec{k}} \mu_{\beta,\vec{k},c}(z)$$

*be an admissible series in* $H^*(\bar{I}_\mu Y, \mathbb{C})[z][\![t_1,\cdots,t_N]\!][\![\mathrm{Eff}(W)]\!]$. *For any integer* $b \geq 0$ *and admissible triple* $(\beta,\vec{k},c) \in \mathrm{Adm}_W$ *with* $\beta(L)+(\vec{w},\vec{k}) > 0$, *when* $r$ *is a sufficiently large prime, we have the following recursive relation:*

$$\begin{aligned}
&[J^Y_{\beta,\vec{k},c}(\mu,z)]_{z^{-b-1}} \\
&= \Big[ \sum_{m=0}^{\infty} \sum_{n=0}^{\infty} \sum_{\substack{\Gamma\in\Lambda^W_{\beta,\vec{k},c,m,n} \\ \Gamma \textit{ is stable}}} \frac{1}{m!n!} (\widetilde{ev_\star})_* \Big( \prod_{i=1}^{m} \frac{ev_i^*(\frac{1}{\delta_i}\mathbf{t}^{\vec{k}_i} J^Y_{\beta_i,\vec{k}_i,c_i}(\mu,z)|_{z=\frac{\lambda-c_1(L)}{\delta_i}})}{\frac{\lambda-ev_i^*c_1(L)}{r\delta_i}+\frac{\bar{\psi}_i}{r}} \\
&\cup \prod_{i=m+1}^{m+n} ev_i^*(\mathbf{t}^{\vec{k}_i}\mu_{\beta_i,\vec{k}_i,c_i}(-\bar{\psi}_i)) \cup \bar{\psi}_\star^b \\
&\cap \sum_{d=0}^{\infty} \epsilon_*\big(c_d(-R^\bullet\pi_*f^*L^{\frac{1}{r}})(\frac{\lambda}{r})^{-1+m-d}(-1)^d \cap [\mathcal{K}_{0,\overrightarrow{m+n}\cup\star}(\sqrt[r]{L/Y},\beta_\star)]^{\mathrm{vir}}\big)\Big)\Big]_{\mathbf{t}^{\vec{k}}\lambda^{-1}}.
\end{aligned} \tag{3.22}$$

## 4. Proof of the main theorem

In this section, we will first assume that $X$ is a smooth proper Deligne-Mumford stack with projective coarse moduli and $Y \subset X$ is a smooth *hypersurface* such that the line bundle $L := \mathcal{O}_X(Y)$ is *semi-positive*, i.e., $\beta(L) \geq 0$ for any degree $\beta \in \mathrm{Eff}(X)$. Later in §4.4, $Y$ will be assumed to be a complete intersection associated to a direct sum of semi-positive line bundles.

4.1. **A root stack modification of the space of the deformation to the normal cone.** Let $P_X := \mathbb{P}_X(\mathbb{C}\oplus\mathbb{C})$ be the trivial $\mathbb{P}^1-$bundle over $X$. It has two section: one is the zero section $X_0 := \mathbb{P}(0\oplus\mathbb{C}) \subset P_X$, and the other is the $\infty-$section $X_\infty := \mathbb{P}(\mathbb{C}\oplus 0)$. We will introduce the notation $P_Y$ to the mean the divisor $\mathbb{P}_Y(\mathbb{C}\oplus\mathbb{C}) \subset P_X$.

Let $\mathfrak{Q}$ be the blow-up[14] of $P_X$ along $Y_0 := X_0 \cap P_Y$. More explicitly, $\mathfrak{Q}$ can be constructed as a hypersurface in the projective bundle $\pi : \mathfrak{F} := \mathbb{P}_{P_X}(L^\vee \oplus \mathcal{O}_{P_X}(-1)) \to P_X$ associated to the section $z_1 \cdot \pi^{-1}(s_{P_Y}) - z_2 \cdot \pi^{-1}(s_{X_0})$ of the line bundle $\mathcal{O}_{\mathfrak{F}}(1)$ , where $s_{P_Y}$ and $s_{X_0}$ are the defining equations of $P_Y$ and $X_0$ in $P_X$, and $z_1, z_2$ are the tautological sections of line bundles $\pi^*(L^\vee)\otimes\mathcal{O}_{\mathfrak{F}}(1)$ and $\pi^*\mathcal{O}_{P_X}(-1)\otimes\mathcal{O}_{\mathfrak{F}}(1)$ respectively. The space $\mathfrak{Q}$ has four special smooth divisors: (1) the strict transformation of $P_Y$, which we will still denote it to be $P_Y$; (2) the exceptional divisor, which we will denote to be $E$ and it's isomorphic to the projective bundle $\mathbb{P}_Y(L^\vee\oplus\mathbb{C})$; (3) the strict transformation of the $0-$section $X_0$, we will still denote to be $X_0$ by an abuse of notation, and it's isomorphic to $X$ with normal bundle equal to $L$; (4) the strict transformation of the $\infty-$section $X_\infty$, which we still denote it to be $X_\infty$.

Let $\mathbb{C}^*$ act on $P_X$ by scaling the $\mathbb{P}^1-$fiber so that the weight of the $\mathbb{C}^*-$action on the normal bundle of $X_\infty$ in $P_X$ is $-1$. It induces a $\mathbb{C}^*-$action on $\mathfrak{Q}$ with the $\mathbb{C}^*-$fixed loci $X_0, X_\infty$ and $D_\infty := P_Y \cap E$.

We will consider the root stack $\mathfrak{R}$ of $\mathfrak{Q}$ by taking $s-$th root of $X_0$ and $r-$th root of $P_Y$. We will still use the notation $X_0, X_\infty, E$ and $P_Y$ to mean their corresponding divisors in $\mathfrak{R}$ after taking roots.

[14] This space is known as (a compactification of) the space of the deformation to the normal cone, c.f., [Ful84, ch5].

We note that $E$ is isomorphic to $\mathbb{P}Y_{r,s}$ introduced in Section 3.1 so that $\mathcal{D}_0 := X_0 \cap E$ is isomorphic to the root stack $\sqrt[s]{L^\vee/Y}$ and $\mathcal{D}_\infty := P_Y \cap E$ is isomorphic to the root stack $\sqrt[r]{L/Y}$.

The $\mathbb{C}^*$−action on $\mathfrak{Q}$ induces a $\mathbb{C}^*$−action on $\mathfrak{R}$. We see that the normal bundle of $X_0$ in $\mathfrak{R}$ is ($\mathbb{C}^*$−equivariantly) isomorphic to $(L^\vee)^{\frac{1}{s}} \otimes \mathbb{C}_{\frac{\lambda}{s}}$, the normal bundle of $X_\infty$ in $\mathfrak{R}$ is isomorphic to $\mathbb{C}_{-\lambda}$, and the normal bundle of $\mathcal{D}_\infty$ in $\mathfrak{R}$ is isomorphic to $(L^{\frac{1}{r}} \otimes \mathbb{C}_{-\frac{\lambda}{r}}) \oplus \mathbb{C}_\lambda$.

Let $\mathrm{pr}_{r,s} : \mathfrak{R} \to X$ be the morphism induced from compositions of the following three maps: (1) the morphism $\mathfrak{R} \to \mathfrak{Q}$ forgetting root structure; (2) the blow-down from $\mathfrak{Q}$ to $P_X$; (3) and the projection from $P_X$ to the base $X$. Note that $\mathrm{pr}_{\mathrm{r,s}}$ induces a morphism from the rigidified inertia stack $\bar{I}_\mu\mathfrak{R}$ (resp. $I_\mu\mathfrak{R}$) to the rigidified inertia stack $\bar{I}_\mu X$ (resp. $I_\mu X$).

There are two morphisms

$$q_0, q_\infty : \mathfrak{R} \to \mathbb{B}\mathbb{C}^* ,$$

associated to the line bundle $\mathcal{O}(X_0)$ and $\mathcal{O}(P_Y)$ respectively such that $\mathcal{O}(X_0) \cong q_0^*(\mathbb{L})$ and $\mathcal{O}(P_Y) \cong q_\infty^*(\mathbb{L})$, here $\mathbb{L}$ is the universal line bundle over $\mathbb{B}\mathbb{C}^*$. This will induce morphisms on their *inertia stack* counterparts:

$$q_0, q_\infty : I_\mu\mathfrak{R} \to I\mathbb{B}\mathbb{C}^* .$$

Recall that $I\mathbb{B}\mathbb{C}^* \cong [\mathbb{C}^*/\mathbb{C}^*] \cong \mathbb{C}^* \times \mathbb{B}\mathbb{C}^*$, where $\mathbb{C}^*$ acts on $\mathbb{C}^*$ by conjugation. Using this isomorphism, for any number $c \in \mathbb{C}^*$, we denote $I_c\mathbb{B}\mathbb{C}^*$ to be the closed substack $\{c\} \times \mathbb{B}\mathbb{C}^*$ of $I\mathbb{B}\mathbb{C}^*$.

A $\mathbb{C}$-point of $I_c\mathbb{B}\mathbb{C}^*$ corresponds to a pair $(1_\mathbb{C}, c)$, where $1_\mathbb{C}$ is the trivial principal $\mathbb{C}$-bundle over $\mathrm{Spec}(\mathbb{C})$ and $c \in \mathbb{C}^*$ is an element in the automorphism group $\mathrm{Aut}_\mathbb{C}(1_\mathbb{C}) \cong \mathbb{C}^*$. Let $(y, g)$ be a $\mathbb{C}$−point of $I_\mu\mathfrak{R}$, where $y \in Ob(\mathfrak{R}(\mathbb{C}))$ and $g \in \mathrm{Aut}(y)$, if $q_0((y, g)) = (1_\mathbb{C}, c)$, then $g$ acts on the fiber $\mathcal{O}(X_0)|_y$ via the multiplication by $c$. We have a similar relation for $q_\infty$.

Define

$$S = \{c \in \mathbb{C}^* | \text{There exists} (y, g) \in \mathfrak{R} \text{ such that } q_0(y, g) = (1_\mathbb{C}, c) \textit{ or } q_\infty(y, g) = (1_\mathbb{C}, c)\} .$$

Then $S$ is a finite set closed under taking inversion, and the image of $q_0, q_\infty$ is contained in the substack $\bigsqcup_{s\in S} I_s\mathbb{B}\mathbb{C}^*$ of $I\mathbb{B}\mathbb{C}^*$. Now we have an induced morphism on

$$q_0, q_\infty : I\mathfrak{R} \to \bigsqcup_{s\in S} I_s\mathbb{B}\mathbb{C}^* .$$

Let $\mathscr{C}$ be a good index set for rigidified inertia stack $\bar{I}_\mu X$ satisfying the assumption 2.1. Now we will use the index set $\mathscr{C} \times \mathbb{C}^* \times \mathbb{C}^*$ to index the components of the inertia stack $I_\mu\mathfrak{R}$: for each $c \in \mathscr{C}$, $c_0, c_\infty \in \mathbb{C}^*$, we will define the inertia component $I_{(c,c_0,c_\infty)}\mathfrak{R}$ to be

$$\mathrm{pr}_{r,s}^{-1}(I_c X) \cap q_0^{-1}(I_{c_0}\mathbb{B}\mathbb{C}^*) \cap q_\infty^{-1}(I_{c_\infty}\mathbb{B}\mathbb{C}^*) .$$

Each $I_{(c,c_0,c_\infty)}\mathfrak{R}$ is an open-closed substack of $I_\mu\mathfrak{R}$. Taking rigidification of $I_{(c,c_0,c_\infty)}\mathfrak{R}$, we denote $\bar{I}_{(c,c_0,c_\infty)}\mathfrak{R}$ to be the corresponding rigidified inertia component of $\bar{I}_\mu\mathfrak{R}$. If one of $c_0, c_\infty$ is not in $S$, set $\bar{I}_{(c,c_0,c_\infty)}\mathfrak{R} = \emptyset$. Then $\bar{I}_\mu\mathfrak{R}$ can be written as a disjoint union:

$$\bar{I}_\mu\mathfrak{R} = \bigsqcup_{(c,c_0,c_\infty)\in\mathscr{C}\times\mathbb{C}^*\times\mathbb{C}^*} \bar{I}_{(c,c_0,c_\infty)}\mathfrak{R} .$$

**Definition 4.1.** *For any degree $\beta \in \mathrm{Eff}(X)$ and $\delta, d \in \mathbb{Q}$, we say a stable map $f : C \to \mathfrak{R}$ is of degree $(\beta, \frac{\delta}{r}, d)$ if $(\mathrm{pr}_{r,s} \circ f)_*[C] = \beta$, $\deg(f^*\mathcal{O}(P_Y)) = \frac{\delta}{r}$ and $\deg(f^*\mathcal{O}(X_\infty)) = d$. We will denote $\mathcal{K}_{0,\vec{m}}(\mathfrak{R}, (\beta, \frac{\delta}{r}, d))$ to be the corresponding moduli stack of twisted stable maps to $\mathfrak{R}$ of degree $(\beta, \frac{\delta}{r}, d)$. We note $\mathcal{K}_{0,\vec{m}}(\mathfrak{R}, (\beta, \frac{\delta}{r}, d))$ is proper as the degree data uniquely determines a homology* *class in $H_2(\mathfrak{R},\mathbb{Q})$. Indeed, dual to $H_2(\mathfrak{R},\mathbb{Q})$, we have $H_2(\mathfrak{R},\mathbb{Q})^\vee \cong H^2(\mathfrak{R},\mathbb{Q}) \cong H^2(\mathfrak{Q},\mathbb{Q})$, and, by the* blow-up *construction of $\mathfrak{Q}$, $H^2(\mathfrak{Q},\mathbb{Q})$ is isomorphic to the direct sum*

$$H^2(X,\mathbb{Q}) \oplus \mathbb{Q}[X_\infty] \oplus \mathbb{Q}[P_Y]\,,$$

*where $[P_Y]$, $[X_\infty]$ (and $[E]$) are the fundamental classes of $P_Y$, $X_\infty$ (and $E$) respectively and the first summand is embedded into $H^*(\mathfrak{Q},\mathbb{Q})$ via the pullback $\mathrm{pr}_{r,s}^*$. Note that we have $r[P_Y]+[E] = \mathrm{pr}_{r,s}^*([Y])$ in $H^2(\mathfrak{R},\mathbb{Q})$. We also note that $\mathcal{O}(X_\infty)$ is semi-positive, thus we can assume that $d \geq 0$.*

First we state a Lemma about $\bar{I}_\mu\mathfrak{R}$ which is parallel to Lemma 3.1.

**Lemma 4.2.** *Let $m = (c, a, b) \in \mathscr{C}\times\mathbb{C}^*\times\mathbb{C}^*$. Assume that $\bar{I}_m\mathfrak{R}$ is nonempty. Then at least one of $a, b$ is equal to one. If $a \neq 1$ (resp. $b \neq 1$), then $\bar{I}_m\mathfrak{R} = \bar{I}_m X_0$ (resp. $\bar{I}_m\mathfrak{R} = \bar{I}_m P_Y$). Furthermore, $a, b$ are roots of unity and we have $\mathrm{e}^{age_c L} = a^{-s}$ when $a \neq 1$ and $\mathrm{e}^{age_c L} = b^r$ when $b \neq 1$.*

Now we state a useful Lemma comparing $\bar{I}_\mu\mathcal{D}_\infty$ with $\bar{I}_\mu Y$.

**Lemma 4.3.** *Let $m = (c, a, b)$ be an element of $\mathscr{C}\times\mathbb{C}^*\times\mathbb{C}^*$. If $\bar{I}_m\mathcal{D}_\infty$ is nonempty, then $\mathrm{e}^{age_c L} = b^r$ and $a = 1$. Let $\theta : \mathcal{D}_\infty \to Y$ be the natural map by forgetting root ($\theta$ can also be obtained by the restriction of $\mathrm{pr}_{r,s}$ to $\mathcal{D}_\infty$), then $\theta$ induces the map on rigidified inertia stacks:*

$$\bar{\theta} : \bar{I}_\mu\mathcal{D}_\infty \to \bar{I}_\mu Y\,,$$

*and we have $\bar{\theta}(\bar{I}_m\mathcal{D}_\infty) = \bar{I}_c Y$. When $r$ is prime to all orders of isotropy groups of $\mathbb{C}-$points of $Y$(for example, $r$ is a sufficiently large prime) and $b \neq 1$, we have the restriction of $\bar{\theta}$ to $\bar{I}_\mu\mathcal{D}_\infty$ induces the isomorphism $\bar{I}_m\mathcal{D}_\infty \cong \bar{I}_c Y$.*

*Proof.* Assume that $\bar{I}_m\mathcal{D}_\infty$ is nonempty. Notice that $\mathcal{D}_\infty$ is disjoint from $X_0$, then $age_m(\mathcal{O}(X_0)) = 0$, then $a = 1$. Note that the restriction of the line bundle $\mathcal{O}(P_Y)$ to $\mathcal{D}_\infty$ is isomorphic to the root bundle $L^{\frac{1}{r}}$, then we have $age_m(L) = age_c(L)$ and $age_m(L) = age_m(\mathcal{O}(rP_Y))$, which implies that $\mathrm{e}^{age_c L} = \mathrm{e}^{r\cdot age_m \mathcal{O}(P_Y)} = b^r$.

Since we assume $Y$ is a smooth proper Deligne-Mumford stack with projective coarse moduli over $\mathbb{C}$. Then $Y$ admits a quotient stack presentation $[\widetilde{Y}/G]$ by [Kre09, Proposition 5.4], where $\widetilde{Y}$ is smooth quasi-projective variety over $\mathbb{C}$ and $G$ is a linear algebraic group over $\mathbb{C}$. Then we can assume that the line bundle $L$ comes from a $G-$equivariant line bundle $\widetilde{L}$ over $\widetilde{Y}$ by descent. We will write $G-$action on $\widetilde{L}$ as $x\cdot g$ for any $x \in \widetilde{L}$ and $g \in G$.

Without loss of generality, we take the good index set $\mathscr{C}$ to be $\mathrm{Conj}(G)$ (the conjugacy class of $G$), we can take $\bar{I}_c Y$ to be the rigidified inertia component $\bar{I}_{[g]}Y$ represented by the conjugacy class of $g \in G$. Then $\bar{I}_{[g]}Y \cong [\widetilde{Y}^g/(C_G(g)/\langle g\rangle)]$ where $\widetilde{Y}^g$ is the $g-$fixed loci of $\widetilde{Y}$, $C_G(g)$ is the centralizer subgroup of $g$ in $G$ and $\langle g\rangle$ is the (finite) group generated by $g$. Now let's describe $\bar{I}_\mu\mathcal{D}_\infty$. Recall that (see [AGV08, Appendix B]), as a root stack, $\mathcal{D}_\infty$ can be constructed as the quotient stack $[L^0/\mathbb{C}^*]$, where $L^0$ be the $\mathbb{C}^*-$principal bundle associated with $L$ (or, more explicitly, $L$ minus the zero section), and the (right) $\mathbb{C}^*-$action on $L^0$ is via the formula $x\cdot\alpha = \alpha^{-r}x$. Using the quotient stack presentation of $Y$, then $\mathcal{D}_\infty$ can be constructed as $[\widetilde{L}^0/(G\times\mathbb{C}^*)]$ via the action $x\cdot(g,\alpha) = \alpha^{-r}\cdot(x\cdot g)$, where $\widetilde{L}^0$ is the open set of $\widetilde{L}$ with zero section removed. Note that the root bundle $L^{\frac{1}{r}}$ can be constructed as the quotient stack $[(\mathbb{C}^*\times\widetilde{L}^0)/(G\times\mathbb{C}^*)]$ via the action $(t,x)\cdot(g,\alpha) = (\alpha\cdot t, \alpha^{-r}\cdot(x\cdot g))$.

For any $(g,b) \in G\times\mathbb{C}^*$, let $[(g,b)]$ be the conjugacy class of $(g,b)$ in $G\times\mathbb{C}^*$ and $m = ([g],1,b)$, then

$$\bar{I}_m\mathcal{D}_\infty = \bar{I}_{([(g,b)])}\mathcal{D}_\infty = [(\widetilde{L}^0)^{(g,b)}/\big(C_G(g)\times\mathbb{C}^*/\langle(g,b)\rangle\big)]\,.$$

Now assume that $\bar{I}_{([(g,b)])}\mathcal{D}_\infty$ is nonempty. We have that $\mathrm{e}^{age_g L} = b^r$, hence $(\widetilde{L}^0)^{(g,b)} = \widetilde{L}^0|_{\widetilde{Y}^g}$. Let $d$ be the order of $g$. As we assume that $r$ is prime to $d$, then the order of $(g,b)$ is equal to $d\cdot r$; in particular $(g^d, b^d) = (1, b^d)$ has order $r$, and the group generated by $(1, b^d)$ is isomorphic to the group $\boldsymbol{\mu}_r$ of $r-$th roots of unity. Consider the embedding of $\mathbb{C}^*$ into $C_G(g)\times\mathbb{C}^*$ by sending $\alpha$ to $(1,\alpha)$, then this induces the injection

$$f\,:\,\mathbb{C}^*/\boldsymbol{\mu_r}\to C_G(g)\times\mathbb{C}^*/\langle(g,b)\rangle\,,$$

of which the cokernel $H\,:=\,Coker(f)$ is natural isomorphic to the quotient group $C_G(g)/\langle g\rangle$. By transitivity of group actions, then we have

$$\left[(\widetilde{L}^0)^{(g,b)}/(C_G(g)\times\mathbb{C}^*/\langle(g,b)\rangle)\right]\cong\left[\left[\widetilde{L}^0|_{\widetilde{Y}^g}/(\mathbb{C}^*/\boldsymbol{\mu}_r)\right]/H\right]\cong[\widetilde{Y}^g/(C_G(g)/\langle g\rangle)]\,.$$

The above isomorphism is compatible with the morphism $\widetilde{\theta}$ (note that $\widetilde{\theta}$ is induced from the projection from $\widetilde{L}^0|_{\widetilde{Y}^g}$ to the base $\widetilde{Y}^g$). This complete the proof.

□

4.2. **Localization analysis.** The $\mathbb{C}^*-$action on $\mathfrak{R}$ induces a $\mathbb{C}^*-$action on the moduli of stable maps to $\mathfrak{R}$, we will use decorated graphs (trees) to index the components of $\mathbb{C}^*-$fixed loci of $\mathcal{K}_{0,\vec{m}}(\mathfrak{R},(\beta,\frac{\delta}{r},d))$ similar to 3.2. The decorations on a graph $\Gamma$ is given by the following data:

- Each vertex $v$ is associated with an index $j(v)\in\{X_0,\mathcal{D}_\infty,X_\infty\}$, and a degree $\beta(v)\in\mathrm{Eff}(X)$. In particular, we will also say the vertex $v$ is labeled by 0 if $j(v)=X_0$ and is labeled by $\infty$ if $j(v)=\mathcal{D}_\infty$.
- Each edge $e$ consists a pair of half-edges $\{h_j,h_{j'}\}$, and $e$ is equipped with two numbers $\delta(e),d(e)\in\mathbb{Q}$. Here we call $h_j$ and $h_{j'}$ half edges and the subscript $j$ or $j'$ is taken in the set $\{X_0,\mathcal{D}_\infty,X_\infty\}$. A half-edge $h_j$ labeled by $j$ is incident to a vertex labeled by $j$.
- Each half-edge $h$ and each leg $l$ has an element $m(h)$ or $m(l)$ in $\mathscr{C}\times\mathbb{C}^*\times\mathbb{C}^*$.
- The legs are labeled with the numbers $\{1,\dots,m\}$ and each leg is incident to a unique vertex.

For $j\in\{X_0,\mathcal{D}_\infty,X_\infty\}$, we will use the symbol $\mathfrak{R}_j$ to mean the space $j$. By the "valence" of a vertex $v$, denoted val$(v)$, we mean the total number of incident half-edges and legs. For each $\mathbb{C}^*-$fixed stable map $f\,:\,(C;q_1,\dots,q_m)\to\mathfrak{R}$ in $\mathcal{K}_{0,\vec{m}}(\mathfrak{R},(\beta_\star,\frac{\delta}{r},d))$, we can associate a decorated graph $\Gamma$ in the following way.

- Each edge $e$ corresponds to a genus-zero irreducible component $C_e$ of restricted stable-map degree $(0,\frac{\delta(e)}{r},d(e))$. Thus $f$ maps $C_e$ constantly to the base. There are two distinguished points(we will also call ramification points) $q_j$ and $q_{j'}$ on $C_e$ satisfying that $q_j$ maps to $\mathfrak{R}_j$ and $q_{j'}$ maps to $\mathcal{R}_{j'}$, respectively. There are two corresponding half-edges $h_j$ and $h_{j'}$ associated to $q_j$ and $q_{j'}$ respectively.
- Each vertex $v$ for which $j(v)$ (with unstable exceptional cases noted below) corresponds to a maximal sub-curve $C_v$ of $C$ which maps totally into $\mathfrak{R}_{j(v)}$, then the restriction of $f$ to $C_v$ defines a twisted stable map in
$$\mathcal{K}_{0,val(v)}(\mathfrak{R}_{j(v)},\beta(v))\,.$$
The label $\beta(v)$ denotes the degree coming from the composition $\mathrm{pr}_{r,s}\circ f|_{C_v}$ of the restriction and projection to the base.
- A vertex $v$ is *unstable* if stable twisted maps of the type described above do not exist. In this case, $v$ corresponds to a single point of the component $C_e$ for each incident edge $e$, which may be a node at which $C_e$ meets another edge curve $C_{e'}$, a marked point of $C_e$, or an unmarked point.
- The index $m(l)$ on a leg $l$ indicates the rigidified inertia stack component $\bar{I}_{m(l)}\mathfrak{R}$ of $\mathfrak{R}$ on which the gerby marked point corresponding to the leg $l$ is evaluated.
- A half-edge $h$ of an edge $e$ corresponds a ramification point $q\in C_e$ such that $f(q)\in\mathfrak{R}_j$ if $h$ is labeled by $j$. Then $m(h)$ indicates the rigidified inertia component $\bar{I}_{m(h)}\mathfrak{R}$ of $\mathfrak{R}$ on which the ramification point $q$ associated with $h$ is evaluated.

Moreover, the decorated graph from a fixed stable map should satisfy the following:

(1) (Monodromy constraint for edge) For each edge $e = \{h_0 := h_{X_0}, h_\infty := h_{\mathcal{D}_\infty}\}$ linking $X_0$ and $\mathcal{D}_\infty$, we have that $m(h_0) = (c^{-1}, \mathrm{e}^{\frac{\delta(e)}{s}}, 1)$ and $m(h_\infty) = (c, 1, \mathrm{e}^{\frac{\delta(e)}{r}})$ for some $c \in \mathscr{C}$. Moreover the choice of $c$ should satisfy that
$$\delta(e) = age_c(L) \bmod \mathbb{Z} ,$$
which follows from the fact that we have an isomorphism of line bundles $f|_{C_e}^* \mathcal{O}(sX_0) \cong f|_{C_e}^*(\mathcal{O}(rP_Y) \otimes \mathrm{pr}_{r,s}^* L^\vee)$ on $C_e$ and then the ages of both line bundles at $q_\infty$ should be equal.

(2) (Monodromy constraint for vertex) For each vertex $v$, let $I_v \subset [m]$ be the set of incident legs, and $H_v$ be the set of incident half-edges. For each $i \in I_v$, write $m(l_i) = (c_i, a_i, b_i)$, and for each $h \in H_v$, write $m(h) = (c_h, a_h, b_h)$. If $v$ is labeled by 0, we have that $b_i, b_h$ are equal to 1, and
$$\mathrm{e}^{\frac{\beta(v)(L)}{s}} \times \prod_{i \in I_v} a_i \times \prod_{h \in H_v} a_h^{-1} = 1 .$$
If $v$ is labeled by $\infty$, then we have $a_i$ and $a_h$ are all equal to 1 and
$$\mathrm{e}^{-\frac{\beta(v)(L)}{r}} \times \prod_{i \in I_v} b_i \times \prod_{h \in E_v} b_h^{-1} = 1 .$$

(3) (Degree constraint) We have
$$\sum_v \beta(v) = \beta , \tag{4.1}$$
and
$$\sum_e \delta(e) + \sum_{v : j(v) = \infty} \beta(v)(L) = \beta(L) . \tag{4.2}$$

In particular, we note that the decorations at each stable vertex $v$ yield a tuple
$$\overrightarrow{val}(v) \in (\mathscr{C} \times \mathbb{C}^* \times \mathbb{C}^*)^{\mathrm{val}(v)}$$
recording the multiplicities at every special point of $C_v$. Then the restriction gives a stable map in
$$\mathcal{K}_{0,\overrightarrow{val}(v)}(\mathfrak{R}_{j(v)}, \beta(v)) .$$

For each leg $l$, write $m(l) = (c_l, a_l, b_l) \in \mathscr{C} \times \mathbb{C}^* \times \mathbb{C}^*$. By Lemma 4.2, we can assume that $a_l, b_l$ are roots of unity, then we can write all $a_l$ (resp. $b_l$) in the exponential forms $\mathrm{e}^{\frac{\delta_a}{s}}$ (resp. $\mathrm{e}^{\frac{\delta_b}{r}}$) where $\delta_a$ (resp. $\delta_b$) is a rational number; we also require that $0 \le \delta_a < s$ (resp. $0 \le \delta_b < r$), then $\delta_a$ (resp. $\delta_b$) are uniquely determined.

Now, assume that $r, s$ are sufficiently large primes, i.e., $r, s$ are sufficiently large comparing to $\beta(L)$, $\delta$ and the rational numbers $\delta_a, \delta_b$ introduced above. We also require that all the edges of the localization graph $\Gamma$ link the vertex labeled by $X_0$ and the vertex labeled by $\mathcal{D}_\infty$. We note here that this happens when $\Gamma$ is localization graph for $\mathcal{K}_{0,\vec{m}}(\mathfrak{R}, (\beta, \frac{\delta}{r}, 0))$ (see Lemma 4.5 for more details). We will do a similar localization analysis as in §3.2.

4.2.1. *Vertex contribution.* Assume that $v$ is a stable vertex, the localization contributions for $X_0$ and $\mathcal{D}_\infty$ repeat the discussions for $\mathcal{D}_0$ and $\mathcal{D}_\infty$ in 3.2.1 with the replacement of the words $\mathcal{D}_0$ and $\mathcal{D}_\infty$ by $X_0$ and $\mathcal{D}_\infty$ respectively, and one additional change that the *inverse of the Euler class* of the virtual normal bundle for $\mathcal{D}_\infty$ should be equal to
$$\frac{1}{\lambda} \cdot \sum_{d \ge 0} c_d(-R^\bullet \pi_* f^*(L)^{\frac{1}{r}})(\frac{-\lambda}{r})^{|E(v)|-1-d} .$$

If $v$ is a stable vertex labeled by $X_\infty$, then the vertex moduli $\mathcal{M}_v$ is given by $\mathcal{K}_{0,\overrightarrow{val(v)}}(X_\infty, \beta(v))$ and the fixed part of perfect obstruction theory gives rise to $[\mathcal{K}_{0,\overrightarrow{val(v)}}(X_\infty, \beta(v))]^{\mathrm{vir}}$. The moving part of the perfect obstruction theory yields the *inverse of the Euler class* of the virtual normal bundle which is equal to $\frac{-1}{\lambda}$.

When $v$ is an unstable vertex over 0 (resp. $\infty$), let $h$ be the half-edge incident to $v$, the vertex moduli $\mathcal{M}_v$ is defined to be $\bar{I}_{m(h)^{-1}}X_0$ (resp. $\bar{I}_{m(h)^{-1}}\mathcal{D}_\infty$) with $[\mathcal{M}_v]^{\mathrm{vir}} = [\mathcal{M}_v]$ and the Euler class of the virtual normal bundle equal to 1 (resp. $\lambda$).

4.2.2. *Edge contribution.* Let $e = \{h_0 := h_{X_0}, h_\infty := h_{\mathcal{D}_\infty}\}$ be an edge in $\Gamma$ with decorated degree[15] $\delta(e) \in \mathbb{Q}_{>0}$ and decorated multiplicities $m(h_0)$ and $m(h_\infty)$. By the monodromy constraint for the edge (see 1), we can write $m(h_\infty) = (c, 1, \mathrm{e}^{\frac{\delta(e)}{r}})$ for some $c \in \mathscr{C}$, let $a_e := a(c)$ be the number as Assumption 2.1, we will also write $a$ as $a_e$ for simplicity if there is no confusion. Then we define the edge moduli $\mathcal{M}_e$ to be the root stack $\sqrt[a_e s\delta(e)]{L^\vee/I_cY}$. Then the virtual cycle $[\mathcal{M}_e]^{vir}$ coming from the $\mathbb{C}^*-$fixed part of the obstruction theory is the fundamental class $[\mathcal{M}_e]$ and *the inverse of the Euler class* of the virtual normal bundle is $\prod_{m=1}^{-1-\lfloor -\delta(e)\rfloor}(\lambda + \frac{m}{\delta(e)}(c_1(L)-\lambda))$.[16] In practice, we will use an equivalent form

$$\frac{1}{\lambda}\prod_{0\le m<\delta(e)}\big(c_1(L) + (\delta(e)-m)\frac{\lambda - c_1(L)}{\delta(e)}\big).$$

We note that $\mathcal{M}_e$ allows a finite étale map into the corresponding fixed-loci in $\mathcal{K}_{0,2}(\mathfrak{R}, (0, \frac{\delta(e)}{r}, 0))$ of degree $\frac{1}{a_e s}$. See appendix A.2 for more details.

4.2.3. *Node contributions.* The deformations in $\mathcal{K}_{0,\vec{m}}(\mathfrak{R}, (\beta, \frac{\delta}{r}, 0))$ smoothing a node contribute to the Euler class of the virtual normal bundle as the first Chern class of the tensor product of the two cotangent line bundles at the two branches of the node. For nodes at which a component $C_e$ meets a component $C_v$ over the stable vertex labeled by $X_0$, this contribution is

$$\frac{\lambda - c_1(L)}{a_e s\delta(e)} - \frac{\bar{\psi}_v}{a_e s}.$$

For nodes at which a component $C_e$ meets a component $C_v$ at the stable vertex labeled by $\mathcal{D}_\infty$, this contribution is

$$\frac{-\lambda + c_1(L)}{a_e r\delta(e)} - \frac{\bar{\psi}_v}{a_e r}.$$

There is also one dimensional piece from deformations of maps (corresponding to the normal direction of $E$ in $\mathfrak{R}$) at each node over $\mathcal{D}_\infty$ (including the node linking an unstable vertex and an edge), which contributes $\frac{1}{\lambda}$ as the Euler class of the virtual normal bundle to the localization.

We will not need node contributions from other types of nodes as the above types suffice the need in this paper, see Lemma 4.5 for more details.

---

[15]When $d(e) = 0$, we have $\delta(e) > 0$ by Lemma 4.5.

[16]Here, for any rational number $q$, we denote $\lfloor q\rfloor$ to be the largest integer no larger than $q$. When $-1-\lfloor -\delta(e)\rfloor = 0$, we make the convention that the product $\prod_{i=1}^{-1-\lfloor -\delta(e)\rfloor}(\lambda + \frac{i}{\delta(e)}(c_1(L)-\lambda))$ is equal to one.

4.2.4. *Total contribution.* Here we will only consider two types of graph: (1) all edges in $\Gamma$ connects $X_0$ and $\mathcal{D}_\infty$; (2) $\Gamma$ has only one vertex labeled by $X_\infty$ and no edges. They are all we need in the later localization analysis(see Lemma 4.5 for more details).

If $\Gamma$ is of type (2), let $v$ be the unique vertex of $\Gamma$. Then the localization contribution from $\Gamma$ is

$$-\frac{[\mathcal{M}_v]^{\mathrm{vir}}}{\lambda}.$$

For any decorated graph $\Gamma$ of type (1), we define $F_\Gamma$ to be

$$\prod_{v:j(v)=0}\mathcal{M}_v\times_{(\bar{I}_\mu X_0)^{|E|}}\prod_{e\in E}\mathcal{M}_e\times_{(\bar{I}_\mu\mathcal{D}_\infty)^{|E|}}\prod_{v:j(v)=\infty}\mathcal{M}_v$$

of the following diagram:

$$\begin{tikzcd}
F_\Gamma \arrow[r] \arrow[d] & \prod_{v:j(v)=0}\mathcal{M}_v\times\prod_{e\in E}\mathcal{M}_e\times\prod_{v:j(v)=\infty}\mathcal{M}_v \arrow[d, "ev_{nodes}"] \\
\prod_E(\bar{I}_\mu X_0\times\bar{I}_\mu\mathcal{D}_\infty) \arrow[r, "(\Delta^0\times\Delta^\infty)^{|E|}"] & \prod_E(\bar{I}_\mu X_0)^2\times(\bar{I}_\mu\mathcal{D}_\infty)^2,
\end{tikzcd}$$

where $\Delta^0=(id,\iota)$(resp. $\Delta^\infty=(id,\iota)$) is the diagonal map of $\bar{I}_\mu X_0$ (resp. $\bar{I}_\mu\mathcal{D}_\infty$). Here the right-hand vertical map is the product of evaluation maps at the two branches of each gluing node. Then we can define $[F_\Gamma]^{\mathrm{vir}}$ as the Gysin pull-back:

$$\left((\Delta^0\times\Delta^\infty)^{|E|}\right)^!\Big(\prod_{v:j(v)=0}[\mathcal{M}_v]^{\mathrm{vir}}\times\prod_{e\in E}[\mathcal{M}_e]^{\mathrm{vir}}\times\prod_{v:j(v)=\infty}[\mathcal{M}_v]^{\mathrm{vir}}\Big).$$

The contribution of decorated graph $\Gamma$ to the virtual localization is:

$$Cont_\Gamma=\frac{\prod_{e\in E}a_es}{|\mathrm{Aut}(\Gamma)|}(\iota_\Gamma)_*\left(\frac{[F_\Gamma]^{\mathrm{vir}}}{e^{\mathbb{C}^*}(N_\Gamma^{\mathrm{vir}})}\right). \tag{4.3}$$

Here $\iota_F:F_\Gamma\to\mathcal{K}_{0,\vec{m}}(\mathfrak{R},(\beta,\frac{\delta}{r},d))$ is a finite étale map of degree $\frac{|\mathrm{Aut}(\Gamma)|}{\prod_{e\in E}a_es}$ into the corresponding $\mathbb{C}^*$-fixed loci. The virtual normal bundle $e^{\mathbb{C}^*}(N_\Gamma^{\mathrm{vir}})$ is the product of virtual normal bundles from vertex contributions, edge contributions and node contributions.

4.3. **Auxiliary cycle.** Let $X,W$ be two smooth proper Deligne-Mumford stacks with projective coarse moduli such that $\iota:X\hookrightarrow W$ is an inclusion. We take $L_1=L$ and $r=1$ in Definition 2.4 where we replace $Y,X$ in Definition 2.4 by $X,W$, and write $\vec{w}$ to be the weight. Let $\mu^X(z)=\sum_{\beta,\vec{k},c}q^\beta\mathbf{t}^{\vec{k}}\mu^X_{\beta,\vec{k},c}(z)$ be an admissible series in $H^*_{CR}(X,\mathbb{C})[z][\![t_1,\cdots,t_N]\!][\![\mathrm{Eff}(W)]\!]$.

Recall that, for any degree $\beta\in\mathrm{Eff}(W)$, we will denote $\mathcal{K}_{0,\vec{m}}(X,\beta)$ as the disjoint union

$$\bigsqcup_{\substack{d\in\mathrm{Eff}(Y)\\ i_*(d)=\beta}}\mathcal{K}_{0,\vec{m}}(X,d).$$

and define the Gromov-witten invariants

$$\langle\cdots\rangle^X_{0,\vec{m},\beta}:=\sum_{d\in\mathrm{Eff}(X):\iota_*(d)=\beta}\langle\cdots\rangle^X_{0,\vec{m},d}.$$

as in §3.4.

For any triple $(\beta,\vec{k},c)\in \mathrm{Eff}(W)\times\mathbb{Z}_{\geq 0}^N\times\mathscr{C}$, we define $J^{X,tw}_{\beta,\vec{k},c}(\mu^X,z)$ to be

$$\begin{aligned}&\Big(\mu^X_{\beta,\vec{k},c}(z)+\mathbf{Coeff}_{\mathbf{t}^{\vec{k}}}\Big[\sum_{m\geq 0}\sum_{\substack{(\beta_j,\vec{k_j},c_j)_{j=1}^m\in\mathrm{Adm}_W^m\ \beta_\star\in\mathrm{Eff}(W)\\ \beta_1+\cdots+\beta_m+\beta_\star=\beta}}\frac{1}{m!}\phi^\alpha\langle\mathbf{t}^{\vec{k_1}}\mu^X_{\beta_1,\vec{k_1},c_1}(-\bar{\psi}_1),\\ &\cdots,\mathbf{t}^{\vec{k_m}}\mu^X_{\beta_m,\vec{k_m},c_m}(-\bar{\psi}_m),\frac{\phi_\alpha}{z-\bar{\psi}_\star}\rangle^X_{0,\vec{m}\cup\star,\beta_\star}\Big]\Big)\prod_{0\leq j<\beta(L)+(\vec{w},\vec{k})}\big(c_1(L)+(\beta(L)+(\vec{w},\vec{k})-j)z\big).\end{aligned}\tag{4.4}$$

where $\vec{m}\cup\star=\big(c_1^{-1}\cdots,c_m^{-1},c\big)\in\mathscr{C}^{m+1}$. We note that $J^{X,tw}_{0,\vec{0},c}=0$, then the formal series

$$J^{X,tw}(q,\mu^X,z)=z+\sum_{\beta,\vec{k},c}q^\beta\mathbf{t}^{\vec{k}}J^{X,tw}_{\beta,\vec{k},c}(\mu^X,z)$$

associated with the input $\mu^X(z)$ above is an admissible series near $z$.

For any admissible triple $(\beta,\vec{k},c)$ with $\beta(L)+(\vec{w},\vec{k})>0$, denote $\delta=\beta(L)+(\vec{w},\vec{k})$. Assume that $r,s$ are sufficiently large primes. For any nonnegative integer $b$, we will also compare (3.17) to the following auxiliary cycle:

$$\begin{aligned}&\sum_{m=0}^\infty\sum_{\substack{(\beta_i,\vec{k_i},c_i)\in\mathrm{Adm}_W^m,\beta_\star\in\mathrm{Eff}(W)\\ \beta_\star+\sum_{i=1}^m\beta_i=\beta\\ \vec{k_1}+\cdots+\vec{k_m}=\vec{k}}}\frac{1}{m!}(\widetilde{EV_\star})_*\Big(\prod_{i=1}^m ev_i^*\big(\mathrm{pr}_{r,s}^*(\mathbf{t}^{\vec{k_i}}\mu^X_{\beta_i,\vec{k_i},c_i}(-\bar{\psi}_i))\big)\\ &\cup\bar{\psi}_\star^b\cap[\mathcal{K}_{0,\vec{m}\cup\star}(\mathfrak{R},(\beta_\star,\frac{\delta}{r},0))]^{\mathrm{vir}}\Big)\end{aligned}\tag{4.5}$$

Here an explanation of the notations is in order:

(1) For degrees $\beta_\star,\beta_1,\cdots,\beta_m$ in $\mathrm{Eff}(W)$ and tuples $\vec{k_1},\cdots,\vec{k_m}$ in $\mathbb{Z}^N_{\geq 0}$ with $\sum_{i=1}^m\beta_i+\beta_\star=\beta$ and $\vec{k_1}+\cdots+\vec{k_m}=\vec{k}$, let $(c_1,\cdots,c_m)\in\mathscr{C}^m$ such that $(\beta_i,\vec{k_i},c_i)$ are admissible triples. Write $\delta_i=\beta_i(L)+(\vec{w},\vec{k_i})$, we define $\vec{m}\cup\star$ to be the $(m+1)$-tuple

$$\big((c_1^{-1},\mathrm{e}^{\frac{\delta_i}{s}},1),\cdots,(c_m^{-1},\mathrm{e}^{\frac{\delta_m}{s}},1),(c,1,\mathrm{e}^{\frac{\delta}{r}})\big).$$

(2) Denote $m_\star=(c,1,\mathrm{e}^{\frac{\delta}{r}})$. Consider the natural structural map $\bar{I}_{m_\star}\mathfrak{R}\to\mathfrak{R}$, of which image is contained in $P_Y$; then we can see that $\bar{I}_{m_\star}\mathfrak{R}$ is canonically isomorphic to $\bar{I}_{m_\star}\mathcal{D}_\infty\times\mathbb{P}^1$, which is further canonically isomorphic to $\bar{I}_cY\times\mathbb{P}^1$ as $\bar{I}_{m_\star}\mathcal{D}_\infty\cong\bar{I}_cY$ by Lemma 4.3. Then we have a natural projection maep $p:\bar{I}_{m_\star}\mathfrak{R}\to\bar{I}_cY$. Let $EV_\star$ be the morphism which is a composition of the following maps:

$$\mathcal{K}_{0,\vec{m}\cup\star}(\mathfrak{R},(\beta_\star,\frac{\delta}{r},0))\xrightarrow{ev_\star}\bar{I}_{m_\star}\mathfrak{R}\xrightarrow{p}\bar{I}_cY.$$

Then we define $(\widetilde{EV_\star})_*$ to be

$$\iota_*(r_\star(EV_\star)_*)$$

as in 2.1. Note here $r_\star$ is the order of the band from the gerbe structure of $\bar{I}_\mu Y$ but not $\bar{I}_\mu\mathfrak{R}$.

Applying $\mathbb{C}^*-$localization to the (4.5), we will prove the following:

**Proposition 4.4.** *With the notation as above, for any integer $b \geq 0$, we have the following recursive relation:*

$$
\begin{aligned}
&[i^*J^{X,tw}_{\beta,\vec{k},\mathscr{c}}(\mu^X, z)]_{z^{-b-1}} \\
&= \Bigg[ \sum_{m=0}^{\infty} \sum_{n=0}^{\infty} \sum_{\substack{\Gamma \in \Lambda^{W}_{\beta,\vec{k},\mathscr{c},m,n} \\ \Gamma \text{ is stable}}} \frac{1}{m!n!} (\widetilde{ev_\star})_* \Bigg( \prod_{i=1}^{m} \frac{ev_i^*(\frac{1}{\delta_i} \mathbf{t}^{\vec{k_i}} i^* J^{X,tw}_{\beta_i,\vec{k}_i,\mathscr{c}_i}(\mu^X, z)|_{z=\frac{\lambda - c_1(L)}{\delta_i}})}{\frac{\lambda - ev_i^* c_1(L)}{r\delta_i} + \frac{\bar{\psi}_i}{r}} \\
&\cup \prod_{i=m+1}^{m+n} ev_i^*(\mathbf{t}^{\vec{k_i}} i^* \mu^X_{\beta_i,\vec{k}_i,\mathscr{c}_i}(-\bar{\psi}_i)) \cup \bar{\psi}_\star^b \\
&\cap \sum_{d=0}^{\infty} \epsilon_* \big( c_d(-R^\bullet \pi_* f^* L^{\frac{1}{r}}) (\frac{\lambda}{r})^{-1+m-d} (-1)^d \cap [\mathcal{K}_{0,\overrightarrow{m+n} \cup \star}(\sqrt[r]{L/Y}, \beta_\star)]^{\mathrm{vir}} \big) \Bigg) \Bigg]_{\mathbf{t}^{\vec{k}} \lambda^{-1}} .
\end{aligned}
\tag{4.6}
$$

*Here $\delta_i = \beta_i(L) + (\vec{w}, \overrightarrow{k_i})$, and $\epsilon : \mathcal{K}_{0,\overrightarrow{m+n}\cup\star}(\sqrt[r]{L/Y}, \beta_\star) \to \mathcal{K}_{0,\overrightarrow{m+n}\cup\star}(Y, \beta_\star)$ is the natural structural morphism by forgetting root-gerbe structure of $\sqrt[r]{L/Y}$ (c.f.[TT21]), where we choose the tuple $\overrightarrow{m+n} \cup \star$ for $\mathcal{K}_{0,\overrightarrow{m+n}\cup\star}(\sqrt[r]{L/Y}, \beta_\star)$ to be*

$$
\big((\mathscr{c}_1^{-1}, 1, \mathrm{e}^{\frac{-\delta_1}{r}}), \cdots, (\mathscr{c}_{m+n}^{-1}, 1, \mathrm{e}^{\frac{-\delta_{m+n}}{r}}), (\mathscr{c}, 1, \mathrm{e}^{\frac{\delta}{r}})\big) \in (\mathscr{C} \times \mathbb{C}^* \times \mathbb{C}^*)^{m+n+1} ,
$$

*and choose the tuple $\overrightarrow{m+n} \cup \star$ for $\mathcal{K}_{0,\overrightarrow{m+n}\cup\star}(Y, \beta_\star)$ to be*

$$
(\mathscr{c}_1^{-1}, \cdots, \mathscr{c}_{m+n}^{-1}, \mathscr{c}) \in \mathscr{C}^{m+n+1} .
$$

*The notation $\Lambda^{W}_{\beta,\vec{k},\mathscr{c},m,n}$ is defined in Definition 3.7.*

Our main strategy to prove this is to apply $\mathbb{C}^*-$localization to the integral (4.5) and then extract the $\lambda^{-2}$-coefficient(or equivalently $\mathbf{t}^{\vec{k}}\lambda^{-2}$-coefficient). Using the polynomiality of (4.5), we have that the $\lambda^{-2}$ coefficient must be zero, which yields the desired result. To simplify the localization calculation, we need the following two lemmas which limit the types of localization graphs one need to consider.

**Lemma 4.5.** *Let $\Gamma$ be a decorated graph for $\mathcal{K}_{0,\vec{m}\cup\star}(\mathfrak{R}, (\beta, \frac{\delta}{r}, 0))$. If the corresponding fixed loci $F_\Gamma$ is non-empty, then there is no edge of $\Gamma$ linking $\mathcal{D}_\infty$ and $X_\infty$ and no edge of $\Gamma$ linking $X_0$ and $X_\infty$.*

*Proof.* Let $f : C \to \mathfrak{R}$ be a $\mathbb{C}^*-$fixed stable map in $\mathcal{K}_{0,\vec{m}\cup\star}(\mathfrak{R}, (\beta, \frac{\delta}{r}, 0))$, we have the pairing $(f_*([C]), [X_\infty]) = 0$. For each stable vertex $v$, obviously, we have the pairing $(f_*([C_v]), [X_\infty]) = 0$. If $e$ is an edge, there three possibilities:

(1) The edge $e$ links a vertex labeled by $\mathcal{D}_\infty$ and a vertex labeled by $X_\infty$.
(2) The edge $e$ links a vertex labeled by $X_0$ and a vertex labeled by $X_\infty$.
(3) The edge $e$ links a vertex labeled by $\mathcal{D}_\infty$ and a vertex labeled by $X_0$.

In the first two cases, we have the pairing $(f_*([C_e]), [X_\infty]) > 0$, while in the third case, the pairing $(f_*([C_e]), [X_\infty]) = 0$. As the total degree of $f^*\mathcal{O}(X_\infty)$ on $C$ is zero, we see that the degree of the restriction of $f^*\mathcal{O}(X_\infty)$ to each component $C_v$ or $C_e$ must be zero, which implies that there is no edge linking $\mathcal{D}_\infty$ and $X_\infty$ and no edge linking $X_0$ and $X_\infty$. □

**Remark 4.6.** The above lemma also works for the case when $\delta = 0$, and the proof is verbatim.

By the above the discussion, they will only be two types of graphs which will contribute to the integral (4.5): (1) all edges in $\Gamma$ connects $X_0$ and $\mathcal{D}_\infty$; (2) $\Gamma$ has only one vertex labeled by $X_\infty$ and no edges. Now let's assume the graph $\Gamma$ is of type 1, we have the following lemma by using a similar argument as in Lemma 3.9 by using the line bundle $N := f^*(\mathcal{O}(-P_Y))$.

**Lemma 4.7.** *Assume $r, s$ is sufficiently large. If localization graph $\Gamma$ has more than one vertex labeled by $\mathcal{D}_\infty$, then the corresponding $\mathbb{C}^*-$fixed loci moduli $F_\Gamma$ is empty; therefore it will contribute zero to (4.5).*

Now we are ready to prove Proposition 4.4.

*Proof.* If a localization graph $\Gamma$ has a vertex labeled by $X_\infty$, by Lemma 4.5, there will be no edges in $\Gamma$. Then $\Gamma$ is made of a single vertex. Such type of graph will contribute

$$\frac{-1}{\lambda}\sum_{m\geq 0}\sum_{\substack{(\beta_j,\vec{k}_j,c_j)_{j=1}^m\in \mathrm{Adm}_W^m\\ \beta_\star\in \mathrm{Eff}(W)\\ \beta_1+\cdots+\beta_m+\beta_\star=\beta\\ \vec{k}_1+\cdots+\vec{k}_m=\vec{k}}}\frac{1}{m!}\phi^\alpha\langle \mathbf{t}^{\vec{k}_1}\mu^X_{\beta_1,\vec{k}_1,c_1}(-\bar{\psi}_1),\cdots,\mathbf{t}^{\vec{k}_m}\mu^X_{\beta_m,\vec{k}_m,c_m}(-\bar{\psi}_m),\frac{\phi_\alpha}{z-\bar{\psi}_\star}\rangle^{X_\infty}_{0,\vec{m}\cup\star,\beta_\star} \tag{4.7}$$

to (4.5).

The other type of graph which contributes to (4.6) must satisfy that all the vertexes are labeled by either 0 or $\infty$ and all the edges only link $X_0$ and $\mathcal{D}_\infty$. Using Lemma 4.7, we can apply the same strategy of the Proposition 3.10 to do the localization computation. Denote $v_\star$ to be the only vertex of $\Gamma$ labeled by $\infty$, then the leg corresponding to $\star$ must be incident to $v_\star$ and the remaining types of localization graph is a star-shaped graph introduced in the proof of 3.10 and can be sorted as below:

(1) (Type I) The vertex $v_\star$ is unstable and there is only one edge incident to $v_\star$ in $\Gamma$, the vertex $v$ labeled by 0 is unstable;
(2) (Type II) The vertex $v_\star$ is unstable and there is only one edge incident to $v_\star$ in $\Gamma$, the vertex $v$ labeled by 0 is stable;
(3) (Type III) The vertex $v_\star$ is stable.

The graph of type I will contribute

$$\frac{1}{\delta\lambda}\mathbf{t}^{\vec{k}}\cdot\left(\frac{\lambda-c_1(L)}{\delta}\right)^b\cdot\prod_{0\leq j<\delta}\left(c_1(L)+(\delta-j)\frac{\lambda-c_1(L)}{\delta}\right)i^*\mu^X_{\beta,\vec{k},c}\left(\frac{\lambda-c_1(L)}{\delta}\right) \tag{4.8}$$

to (4.5).

The graph of type II will contribute

$$\begin{aligned}&\frac{1}{\delta\lambda}\left(\frac{\lambda-c_1(L)}{\delta}\right)^b\cdot\prod_{0\leq j<\delta}\left(c_1(L)+(\delta-j)\frac{\lambda-c_1(L)}{\delta}\right)\sum_{m\geq 0}\sum_{\substack{(\beta_i,\vec{k}_i,c_i)_{i=1}^m\in\mathrm{Adm}_W^m\\ \beta_\star\in\mathrm{Eff}(W),\ \sum_i\beta_i+\beta_\star=\beta\\ \sum_i\vec{k}_i=\vec{k}}}\frac{1}{m!}\\ &\cdot i^*\phi^\alpha\Big(\int_{[\mathcal{K}_{0,\vec{m}\cup\star}(X,\beta_\star)]^{\mathrm{vir}}}\prod_{i=1}^m ev_i^*(\mathbf{t}^{\vec{k}_i}\mu^X_{\beta_i,\vec{k}_i,c_i}(-\bar{\psi}_i))\cup\frac{ev_\star^*\phi_\alpha}{\frac{\lambda-ev_\star^*c_1(L)}{\delta}-\bar{\psi}_\star}\Big).\end{aligned} \tag{4.9}$$

to (4.5).

By an analogous analysis in the proof of Proposition 3.8 (see also Remark 3.5). The graph of type III will contribute

$$
\begin{aligned}
(4.10)\quad & \frac{1}{\lambda}\sum_{m=0}^{\infty}\sum_{n=0}^{\infty}\sum_{\substack{\Gamma\in\Lambda^{W}_{\beta,\vec{k},c,m,n}\\ \Gamma \text{ is stable}}}\frac{1}{m!}\frac{1}{n!}(\widetilde{ev_\star})_*\Bigg(\prod_{i=1}^{m}\frac{ev_i^*(\frac{1}{\delta_i}\mathbf{t}^{\vec{k}_i}i^*J^{X,tw}_{\beta_i,\vec{k}_i,c_i}(\mu^X,z)|_{z=\frac{\lambda-c_1(L)}{\delta_i}})}{\frac{-\lambda-ev_i^*c_1(L)}{r\delta_i}-\frac{\bar{\psi}_i}{r}}\\
&\cup\prod_{i=m+1}^{m+n}ev_i^*(\mathbf{t}^{\vec{k}_i}i^*\mu^X_{\beta_i,\vec{k}_i,c_i}(-\bar{\psi}_i))\cup\bar{\psi}_\star^b\\
&\cap\sum_{d=0}^{\infty}\epsilon_*\big(c_d(-R^\bullet\pi_*f^*L^{\frac{1}{r}})(\frac{\lambda}{r})^{-1+m-d}\cap[\mathcal{K}_{0,\overrightarrow{m+n}\cup\star}(\sqrt[r]{L/Y},\beta_\star)]^{\mathrm{vir}}\Bigg).
\end{aligned}
$$

to (4.5).

Note that the sum of (4.8) and (4.9) is equal to

$$\frac{1}{\delta\lambda}\mathbf{t}^{\vec{k}}(z^b\cdot i^*J^{X,tw}_{\beta,\vec{k},c}(\mu^X,z))|_{z=\frac{\lambda-c_1(L)}{\delta}},$$

of which the $\mathbf{t}^{\vec{k}}\lambda^{-2}$–coefficient is equal to

$$[i^*J^{X,tw}_{\beta,\vec{k},c}(\mu^X,z)]_{z^{-b-1}}.$$

Indeed, expand $i^*J^{X,tw}_{\beta,\vec{k},c}(\mu^X,z)$ into Laurent series in $z$:

$$i^*J^{X,tw}_{\beta,\vec{k},c}(\mu^X,z)=\cdots+zK_0+K_1+z^{-1}K_2+\cdots$$

we see that the $z^{-b-1}$–coefficient of $i^*J^{X,tw}_{\beta,\vec{k},c}(z)$ and $\mathbf{t}^{\vec{k}}\lambda^{-2}$–coefficient of

$$\frac{1}{\delta\lambda}(\mathbf{t}^{\vec{k}}\cdot z^b\cdot i^*J^{X,tw}_{\beta,\vec{k},c}(z))|_{z=\frac{\lambda-c_1(L)}{\delta}}$$

are both equal to $K_{b+2}$.

Now our auxiliary cycle (4.5) is equal to the sum of (4.7), (4.8), (4.9) and (4.10). By the polynomiality of (4.5), the $\mathbf{t}^{\vec{k}}\lambda^{-2}$–coefficient of the auxiliary cycle is zero, which implies that the sum of $\mathbf{t}^{\vec{k}}\lambda^{-2}$–coefficient of (4.7), (4.8), (4.9) and (4.10) is zero, which proves the proposition 4.4. □

4.4. **Statement of the main theorem and proof.** With the notations in this section, now we state our main theorem:

**Theorem 4.8** (Main theorem)**.** *Let $X,W$ be two smooth proper Deligne-Mumford stacks with projective coarse moduli such that $X\subset W$. Let $E:=\oplus_{j=1}^r L_j$ be a direct sum of* semi-positive *line bundles over $W$ such that the restriction $E|_X$ of $E$ to $X$ which cuts off a smooth complete intersection $Y\subset X$. Let $\mu^X(z):=\sum_{(\beta,\vec{k},c)\in\mathrm{Adm}_W}q^\beta\mathbf{t}^{\vec{k}}\mu^X_{\beta,\vec{k},c}(z)$ be an admissible series and $J^{X,tw}(q,\mu^X,z)$ be the hypergeometric modification of $J^X(q,\mu^X,z)$ as in (1.3). Let $i:\bar{I}_\mu Y\to\bar{I}_\mu X$ be the inclusion, then the series $i^*J^{X,tw}(q,\mu^X,-z)$ is a point on the Lagrange cone of $Y$. More precisely, for any admissible triple $(\beta,\vec{k},c)$, if we define*

$$\mu^{X,tw}_{\beta,\vec{k},c}(z):=[J^X_{\beta,\vec{k},c}(\mu^X,z)\prod_{j=1}^{r}\prod_{0\le m<\beta(L_j)+(\vec{w}_j,\vec{k})}(c_1(L_j)+(\beta(L_j)+(\vec{w}_j,\vec{k})-m)z)]_+$$

*to be the truncation in nonnegative $z-$powers. Here $J^X_{\beta,\vec{k},c}$ is defined in (3.15). Write*

$$\mu^{X,tw}(q,z) := \sum_{(\beta,\vec{k},c)\in \mathrm{Adm}_W} q^\beta \mathbf{t}^{\vec{k}} \mu^{X,tw}_{\beta,\vec{k},c}(z)\,.$$

*Then for any integer $b \geq 0$, we have the following relation:*

$$[J^Y_{\beta,\vec{k},c}(i^*\mu^{X,tw},z)]_{z^{-b-1}} = [i^* J^{X,tw}_{\beta,\vec{k},c}(\mu^X,z)]_{z^{-b-1}}\,. \tag{4.11}$$

*We will say that the main theorem holds for the triple $(Y,X,W)$ and the $J-$function $J^X(q,\mu^X,z)$* (where $\mu^X$ is an admissible series).

For integer $k$ with $1 \leq k \leq r$, let $Y_k \subset X$ be the complete intersection associated with the vector bundle $E_k := \oplus_{j=1}^k L_j$. Denote $Y_0 := X$, $i_k : Y_k \to Y_{k-1}$ and $\mathcal{T}_k := (Y_k, Y_{k-1}, W)$. First we assume that the main theorem holds for the triple $\mathcal{T}_1$ and $J^X(q,\mu^X,z)$. Then we get a $J-$function $J^{Y_1}(q,\mu^{Y_1},z)$ of $Y_1$, where $\mu^{Y_1} = i_1^*\mu^{X,tw}$ and $\mu^{X,tw}$ is obtained by the hypergeometric modification of $J^X(q,\mu^X,z)$ associated with the line bundle $L_1$ as explained in the main theorem. Observe that $\mu^{Y_1}$ is an admissible series in $H^*(\bar{I}_\mu Y_1)[z][\![t_1,\cdots,t_N]\!][\![\mathrm{Eff}(W)]\!]$, we can apply the main theorem to the triple $\mathcal{T}_2$ and $J^{Y_1}(q,\mu^{Y_1},z)$ to get a $J-$function $J^{Y_2}(q,\mu^{Y_2},z)$ of $Y_2$. Note that $J^{Y_2}(q,\mu^{Y_2},z)$ is the same with the hypergeometric modification of $J^X(q,\mu^X,z)$ associated with the line bundles $L_1$ and $L_2$ (after restriction to $Y_2$). Continuing this way, we can prove our main theorem for the triple $(Y,X,W)$ and $J^X(q,\mu^X,z)$. Now we see that it's sufficient to prove the main theorem when $Y$ is a *hypersurface* of $X$. Now let's assume that $Y$ is a hypersurface and use the notations in previous sections.

We will first prove (4.11) for any admissible triple $(\beta,\vec{k},c)$ with $\beta(L)+(\vec{w},\vec{k})=0$. Note that in this case we have $\mu^{X,tw}_{\beta,\vec{k},c}(z) = \mu^X_{\beta,\vec{k},c}(z)$.

We will adopt the convention that for any substack $Z$ of $\mathfrak{R}$ and $c \in \mathscr{C}$, we will write $\bar{I}_c Z := \bar{I}_\mu Z \cap \bar{I}_{(c,1,1)}\mathfrak{R}$. Now fix an admissible triple $(\beta,\vec{k},c)$ such that $\beta(L)+(\vec{w},\vec{k})=0$, we have $age_c L = 0$. Then $\bar{I}_c Y$ is proper hypersurface of $\bar{I}_c X$ with normal bundle $L$.[17] Recall that the divisor $E$ of $\mathfrak{R}$ is isomorphic to $\mathbb{P}Y_{r,s}$, Denote by $\mathrm{pr}^E_{r,s} : E \to Y$ the projection to the base, which induces a morphism on the corresponding rigidified inertia stacks, which we still denote to be $\mathrm{pr}^E_{r,s}$. We have $\bar{I}_{c^{-1}}E \cong (\mathrm{pr}^E_{r,s})^{-1}(\bar{I}_{c^{-1}}Y)$ and $\bar{I}_{c^{-1}}E$ is a hypersurface of $\bar{I}_{c^{-1}}\mathfrak{R}$.

Denote by $i_E$ the inclusion from $\bar{I}_{c^{-1}}E$ to $\bar{I}_{c^{-1}}\mathfrak{R}$. For any admissible triple $(\beta,\vec{k},c)$ with $\beta(L)+(\vec{w},\vec{k})=0$, consider the following class

$$\begin{aligned}\sum_{m=0}^{\infty} \sum_{\substack{(\beta_i,\vec{k}_i,c_i)_{i=1}^m \in \mathrm{Adm}_W^m \\ \beta_\star+\sum_{i=1}^m \beta_i=\beta,\ \beta_\star\in \mathrm{Eff}(X) \\ \vec{k_1}+\cdots+\vec{k_m}=\vec{k}}} & \frac{1}{m!}(\mathrm{pr}^E_{r,s})_* \circ i_E^* \circ (\widetilde{ev_\star})_* \\ & \left(\prod_{i=1}^m \mathrm{pr}^*_{r,s}(\mathbf{t}^{\vec{k}_i}\mu^X_{\beta_i,\vec{k}_i,c_i}(-\bar{\psi}_i)) \cup \bar{\psi}_\star^b \cap [\mathcal{K}_{0,\vec{m}\cup\star}(\mathfrak{R},(\beta_\star,0,0))]^{vir}\right).\end{aligned} \tag{4.12}$$

Here for any $\beta_\star,\beta_1,\cdots,\beta_m,\vec{k_1}\cdots,\vec{k_m}$ and $c_1,\cdots,c_m$ such that $(\beta_i,\vec{k_i},c_i)_{i=1}^m \in \mathrm{Adm}_W^m$ with $\sum_{i=1}^m \beta_i + \beta_\star = \beta$ and $\vec{k_1}+\cdots+\vec{k_m}=\vec{k}$, we associated an element in $(\mathscr{C}\times\mathbb{C}^*\times\mathbb{C}^*)^{m+1}$

$$\vec{m}\cup\star := \left((c_1^{-1},1,1),\cdots,(c_m^{-1},1,1),(c,1,1)\right).$$

[17] Here the normal bundle is a pull-back of the line bundle $L$ along the natural morphism from $\bar{I}_\mu X$ to $X$ by sending $(x,g)$ to $x$.

$i_E^*$ is the Gysin pull-back from the (equivariant) cohomology of $\bar{I}_{(c^{-1},1,1)}\mathfrak{R}$ to the (equivariant) cohomology of $\bar{I}_{(c^{-1},1,1)}E$; $(\widetilde{ev_\star})_*$ is a morphism from $H_*^{\mathbb{C}^*}(\mathcal{K}_{0,\vec{m}\cup\star}(\mathfrak{R},(\beta_\star,0,0)))$ to $H^*_{\mathbb{C}^*}(\bar{I}_{c^{-1}}\mathfrak{R})$ as defined in 2.1. Note that $\beta_i(L)+(\vec{w},\vec{k}_i)=0$ for all $1\le i\le m$ by the semi-positivity of $L$.

**Lemma 4.9.** *For any admissible triple $(\beta,\vec{k},c)\in \mathrm{Adm}_W$ with $\beta(L)+(\vec{w},\vec{k})=0$, let $b$ be an nonnegative integer, we have*

$$[J^Y_{\beta,\vec{k},c}(q,i^*\mu^X,z)]_{z^{-b-1}} = i^*[J^X_{\beta,\vec{k},c}(q,\mu^X,z)]_{z^{-b-1}}\ ,$$

*where $i^*$ is induced from the inclusion from $\bar{I}_\mu Y$ to $\bar{I}_\mu X$.*

*Proof.* We will apply $\mathbb{C}^*-$localization to (4.12). First we claim there are only two types of localization graphs contributing to the integral (4.12): 1, the localization $\Gamma_1$ has only one vertex and it's labeled by $X_0$; 2, the localization graph $\Gamma_2$ has only one vertex and it's labeled by $\mathcal{D}_\infty$. Indeed, by Lemma 4.5 (see also Remark 4.6), all vertexes must be labeled by $X_0$ or $\mathcal{D}_\infty$; otherwise the graph has only one vertex and it's labeled by $X_\infty$, but the marking $q_\star$ must go to the divisor $E$ to make nonzero localization contribution to (4.12). Furthermore, as $deg(f^*\mathcal{O}(P_Y))=0$, we claim there is no edge linking $X_0$ and $\mathcal{D}_\infty$. Indeed, recall that we only need to consider edge curve $C_e$ linking $X_0$ and $\mathcal{D}_\infty$ by Lemma 4.5 (see also Remark 4.6); then the claim comes from that the line bundle $\mathcal{O}(P_Y)$ is of nonnegative degree restricting to all vertex curves $C_v$ and of positive degree restricting to all edge curves $C_e$. The localization contribution from $\Gamma_1$ is equal to

$$\begin{aligned}(\mathrm{pr}^E_{r,s})_*\circ i_E^*\circ(\iota_{X_0})_*\Big[\sum_{m=0}^{\infty}\sum_{\substack{(\beta_i,\vec{k}_i,c_i)_{i=1}^m\in \mathrm{Adm}_W{}^m\\ \beta_\star+\sum_{i=1}^m\beta_i=\beta,\ \beta_\star\in\mathrm{Eff}(X)\\ \vec{k}_1+\cdots+\vec{k}_m=\vec{k}}}\frac{1}{m!}(\widetilde{ev_\star^{X_0}})_*\Big(\prod_{i=1}^m ev_i^*(\mathrm{pr}^*_{r,s}\mathbf{t}^{\vec{k}_i}\mu^X_{\beta_i,\vec{k}_i,c_i}(-\bar\psi_i))\\ \cup\bar\psi_\star^b\cap\sum_{d=0}^{\infty}(c_d(-R^\bullet\pi_*f^*(L^\vee)^{\frac{1}{s}})(\frac{\lambda}{s})^{-1-d}\cap[\mathcal{K}_{0,\vec{m}\cup\star}(X_0,\beta_\star)]^{\mathrm{vir}}\Big)\Big].\end{aligned}\tag{4.13}$$

Here $\iota_{X_0}:\bar{I}_{c^{-1}}X_0\to\bar{I}_{c^{-1}}\mathfrak{R}$ is the natural inclusion and $(\widetilde{ev_\star^{X_0}})_*$ is a morphism from $H_*^{\mathbb{C}^*}(\mathcal{K}_{0,\vec{m}\cup\star}(X_0,\beta_\star))$ to $H^*_{\mathbb{C}^*}(\bar{I}_{c^{-1}}X_0)$ as defined in 2.1(don't confuse $(\widetilde{ev_\star^{X_0}})_*$ used in (4.13) with $(\widetilde{ev_\star})_*$ in (4.12)).

We have the following commutative diagram where the upper square and outer square are Cartersian:

$$\begin{tikzcd}
\bar{I}_{c^{-1}}\mathcal{D}_0 \arrow[r, "i_{\mathcal{D}_0}"] \arrow[d, "\iota_{\mathcal{D}_0}"] & \bar{I}_{c^{-1}}X_0 \arrow[d, "\iota_{X_0}"] \\
\bar{I}_{c^{-1}}E \arrow[r, "i_E"] \arrow[d, "\mathrm{pr}^E_{r,s}"] & \bar{I}_{c^{-1}}\mathfrak{R} \arrow[d, "\mathrm{pr}_{r,s}"] \\
\bar{I}_{c^{-1}}Y \arrow[r, "i=i_Y"] & \bar{I}_{c^{-1}}X\,.
\end{tikzcd}$$

Here $i$ is induced from the inclusion from $Y$ to $X$ and $\iota_{\mathcal{D}_0}$ is induced from the inclusion from $\mathcal{D}_0$ to $E$. As the age $age_c(L)=0$, all rows are regular embeddings of codimension one. Then by the commutativity of the proper push-forward and Gysin pullback for the upper square, (4.13) is equal to

$$(\mathrm{pr}^E_{r,s}\circ\iota_{\mathcal{D}_0})_*i^*_{\mathcal{D}_0}[\cdots].\tag{4.14}$$

Here the dots inside the big square bracket copies the stuff in the big square bracket of (4.13). Using the outer square, (4.14) is equal to

$$i^*(\mathrm{pr}_{r,s}\circ\iota_{X_0})_*\big[\cdots\big]. \tag{4.15}$$

As the composition $\mathrm{pr}_{r,s}\circ\iota_{X_0}$ is induced from natural de-root stackfication of $X_0 \cong \sqrt[s]{L^\vee/X}$, $(\mathrm{pr}_{r,s}\circ\iota_{X_0})_*$ is the identity map on cohomology groups as it induces an identity map on their coarse moduli spaces and the cohomology group of an orbifold is identified with the cohomology group of its coarse moduli space (see [AGV08, §2.2] for more details). Therefore (4.15) is equal to

$$\begin{aligned} &i^*\Big[\sum_{m=0}^{\infty}\sum_{\substack{(\beta_i,\vec{k}_i,c_i)_{i=1}^m\in\mathrm{Adm}_W{}^m\\ \beta_\star+\sum_{i=1}^m\beta_i=\beta,\ \beta_\star\in\mathrm{Eff}(X)\\ \vec{k}_1+\cdots+\vec{k}_m=\vec{k}}}\frac{1}{m!}(\widetilde{ev_\star^X})_*\Big(\prod_{i=1}^m ev_i^*(\mathbf{t}^{\vec{k}_i}\mu^X_{\beta_i,\vec{k}_i,c_i}(-\bar{\psi}_i)\\ &\cup\bar{\psi}_\star^b\cap\sum_{d=0}^{\infty}\epsilon_*\big(c_d(-R^\bullet\pi_*f^*(L^\vee)^{\frac{1}{s}})(\frac{\lambda}{s})^{-1-d}\cap[\mathcal{K}_{0,\vec{m}\cup\star}(X_0,\beta_\star)]^{\mathrm{vir}}\big)\Big)\Big], \end{aligned} \tag{4.16}$$

where $\epsilon:\mathcal{K}_{0,\vec{m}\cup\star}(X_0,\beta_\star)\to\mathcal{K}_{0,\vec{m}\cup\star}(X,\beta_\star)$ is induced from the natural structural map forgetting the root of $X_0\cong\sqrt[s]{L^\vee/X}$.

The localization contribution from $\Gamma_2$ is equal to

$$\begin{aligned} &\frac{1}{\lambda}(\mathrm{pr}^E_{r,s})_*\circ i_E^*\circ(i_{\mathcal{D}_\infty})_*\Big[\sum_{m=0}^{\infty}\sum_{\substack{(\beta_i,\vec{k}_i,c_i)_{i=1}^m\in\mathrm{Adm}_W^m\\ \beta_\star\in\mathrm{Eff}(X),\ \beta_\star+\sum_{i=1}^m\beta_i=\beta\\ \vec{k}_1+\cdots+\vec{k}_m=\vec{k}}}\frac{1}{m!}(\widetilde{ev_\star^{\mathcal{D}_\infty}})_*\Big(\prod_{i=1}^m ev_i^*(\mathrm{pr}_{r,s}^*\mathbf{t}^{\vec{k}_i}\mu^X_{\beta_i,\vec{k}_i,c_i}(-\bar{\psi}_i))\\ &\cup\bar{\psi}_\star^b\cap\sum_{d=0}^{\infty}\big(c_d(-R^\bullet\pi_*f^*L^{\frac{1}{r}})(\frac{-\lambda}{r})^{-1-d}\cap[\mathcal{K}_{0,\vec{m}\cup\star}(\mathcal{D}_\infty,\beta_\star)]^{\mathrm{vir}}\big)\Big)\Big] \end{aligned} \tag{4.17}$$

Here $i_{\mathcal{D}_\infty}:\bar{I}_{c^{-1}}\mathcal{D}_\infty\to\bar{I}_{c^{-1}}\mathfrak{R}$ is the inclusion and $(\widetilde{ev_\star^{\mathcal{D}_\infty}})_*$ is the morphism from $H_*^{\mathbb{C}^*}(\mathcal{K}_{0,\vec{m}\cup\star}(\mathcal{D}_\infty,\beta_\star))$ to $H^*_{\mathbb{C}^*}(\bar{I}_{c^{-1}}\mathcal{D}_\infty)$ as defined in 2.1. The inclusion $i_{\mathcal{D}_\infty}$ can be rewritten as the composition of two inclusions $\iota_{\mathcal{D}_\infty}:\bar{I}_{c^{-1}}\mathcal{D}_\infty\hookrightarrow\bar{I}_{c^{-1}}E$ and $i_E:\bar{I}_{c^{-1}}E\hookrightarrow\bar{I}_{c^{-1}}\mathfrak{R}$. Then we have

$$\begin{aligned} &\frac{1}{\lambda}(\mathrm{pr}^E_{r,s})_*\circ i_E^*\circ(i_{\mathcal{D}_\infty})_*\Big[\cdots\Big]\\ &=\frac{1}{\lambda}(\mathrm{pr}^E_{r,s})_*\circ i_E^*\circ(i_E)_*\circ(\iota_{\mathcal{D}_\infty})_*\Big[\cdots\Big]\\ &=(\mathrm{pr}^E_{r,s})_*(\iota_{\mathcal{D}_\infty})_*\Big[\cdots\Big]\\ &=\Big[\cdots\Big]. \end{aligned} \tag{4.18}$$

Here the dots inside the big square bracket copies the stuff in the big square bracket of (4.17). The second equality in (4.18) uses the fact that the morphism $i_E^*(i_E)_*:H^*_{\mathbb{C}^*}(\bar{I}_{c^{-1}}E)\to H^*_{\mathbb{C}^*}(\bar{I}_{c^{-1}}E)$ is equal to the multiplication map by $e^{\mathbb{C}^*}(N_{\bar{I}_{c^{-1}}Y/\bar{I}_{c^{-1}}\mathfrak{R}})$ using excess intersection formula, which implies that $i_E^*(i_E)_*$ acts on the space $(\iota_{\mathcal{D}_\infty})_*H^*_{\mathbb{C}^*}(\mathcal{D}_\infty)$ by multiplication by $\lambda$ as $e^{\mathbb{C}^*}(N_{\bar{I}_{c^{-1}}Y/\bar{I}_{c^{-1}}\mathfrak{R}})|_{\bar{I}_{c^{-1}}\mathcal{D}_\infty}=$

$\lambda$. Finally (4.17) becomes

$$\sum_{m=0}^{\infty} \sum_{\substack{(\beta_i,\vec{k_i},c_i)_{i=1}^m \in \mathrm{Adm}_W^m, \beta_\star\in \mathrm{Eff}(X) \\ \beta_\star+\sum_{i=1}^m \beta_i=\beta \\ \vec{k_1}+\cdots+\vec{k_m}=\vec{k}}} \frac{1}{m!}(\widetilde{ev^Y_\star})_*\Big(\prod_{i=1}^m ev_i^*(i^*\mathbf{t}^{\vec{k_i}}\mu^X_{\beta_i,\vec{k_i},c_i}(-\bar{\psi}_i)) \tag{4.19}$$
$$\cup\, \bar{\psi}_\star^b \cap \sum_{d=0}^{\infty} \epsilon_*\big(c_d(-R^\bullet\pi_* f^* L^{\frac{1}{r}})(\frac{-\lambda}{r})^{-1-d} \cap [\mathcal{K}_{0,\vec{m}\cup\star}(\mathcal{D}_\infty,\beta_\star)]^{\mathrm{vir}}\big)\Big)$$

using the morphism $\epsilon : \mathcal{K}_{0,\vec{m}\cup\star}(\mathcal{D}_\infty,\beta_\star) \to \mathcal{K}_{0,\vec{m}\cup\star}(Y,\beta_\star)$ induced from forgetting the root of $\mathcal{D}_\infty \cong \sqrt[s]{L/Y}$.

Now taking $\lambda^{-1}$ coefficients of the contributions (4.16) and (4.19) from $\Gamma_1$ and $\Gamma_2$, their sum is zero by polynomiality of (4.12), this finishes the proof of (4.11) with the help of the fact

$$\epsilon_*\big([\mathcal{K}_{0,\vec{m}\cup\{\star\}}(\sqrt[s]{L^\vee/X},\beta_\star)]^{\mathrm{vir}}\big) = \frac{1}{s}[\mathcal{K}_{0,\vec{m}\cup\{\star\}}(X,\beta_\star)]^{\mathrm{vir}}$$

$$\epsilon_*\big([\mathcal{K}_{0,\vec{m}\cup\{\star\}}(\sqrt[r]{L/Y},\beta_\star)]^{\mathrm{vir}}\big) = \frac{1}{r}[\mathcal{K}_{0,\vec{m}\cup\{\star\}}(Y,\beta_\star)]^{\mathrm{vir}}$$

proved in [TT21]. □

Now we are well prepared to prove the main theorem 4.8:

*Proof of the main theorem 4.8.* Since $W$ is a smooth proper Deligne-Mumford stack with projective coarse moduli, there exists an *ample* line bundle $M$ on $W$, i.e., $\beta(M) \in \mathbb{Z}_{>0}$ for all nonzero degree $\beta \in \mathrm{Eff}(W)$. Let's prove the relation (4.11) by induction on the nonnegative integer $e := \beta(M) + |\vec{k}|$.[18] The base case is when $e = 0$, we have $(\beta,\vec{k}) = (0,\vec{0})$, then $J^Y_{0,\vec{0},c}(i^*\mu^{X,tw},z) = i^*J^{X,tw}_{0,\vec{0},c}(\mu^X,z) = 0$.

Now we suppose that the relation (4.11) holds for all admissible triple $(\beta',\vec{k'},c')$ with $\beta'(M) + |\vec{k'}| < e$ for some positive integer $e$, it remains to show the relation holds for any admissible triple $(\beta,\vec{k},c)$ with $\beta(M)+|\vec{k}| = e$. If $\beta(L)+(\vec{w},\vec{k}) = 0$, by Proposition 4.9, the relation holds (note in this case, we have $J^{X,tw}_{\beta,\vec{k},c}(\mu^X,z) = J^X_{\beta,\vec{k},c}(\mu^X,z)$). If $\beta(L) + (\vec{w},\vec{k}) \neq 0$, we choose $\mu = i^*\mu^{X,tw}$ in formula (3.22). Then formula (3.22) becomes

$$[J^Y_{\beta,\vec{k},c}(i^*\mu^{X,tw},z)]_{z^{-b-1}}$$
$$= \Big[\sum_{m=0}^{\infty}\sum_{n=0}^{\infty} \sum_{\substack{\Gamma\in\Lambda^W_{\beta,\vec{k},c,m,n} \\ \Gamma \text{ is stable}}} \frac{1}{m!n!}(\widetilde{ev_\star})_*\Big(\prod_{i=1}^m \frac{ev_i^*(\frac{1}{\delta_i}\mathbf{t}^{\vec{k_i}}J^Y_{\beta_i,\vec{k_i},c_i}(i^*\mu^{X,tw},z)|_{z=\frac{\lambda-c_1(L)}{\delta_i}})}{\frac{\lambda-ev_i^*c_1(L)}{r\delta_i}+\frac{\bar{\psi}_i}{r}} \tag{4.20}$$
$$\cup \prod_{i=m+1}^{m+n} ev_i^*(\mathbf{t}^{\vec{k_i}}i^*\mu^{X,tw}_{\beta_i,\vec{k_i},c_i}(-\bar{\psi}_i)) \cup \bar{\psi}_\star^b$$
$$\cap \sum_{d=0}^{\infty}\epsilon_*\big(c_d(-R^\bullet\pi_*f^*L^{\frac{1}{r}})(\frac{\lambda}{r})^{-1+m-d}(-1)^d \cap [\mathcal{K}_{0,\overrightarrow{m+n}\cup\star}(\sqrt[r]{L/Y},\beta_\star)]^{\mathrm{vir}}\big)\Big)\Big]_{\mathbf{t}^{\vec{k}}\lambda^{-1}},$$

[18] Here $|\vec{k}| = \sum_i k_i$.

and we have formula (4.6):

$$(4.21)\quad \begin{aligned}
&[i^*J^{X,tw}_{\beta,\vec{k},c}(\mu^X,z)]_{z^{-b-1}}\\
&=\Big[\sum_{m=0}^{\infty}\sum_{n=0}^{\infty}\sum_{\substack{\Gamma\in\Lambda^W_{\beta,\vec{k},c,m,n}\\ \Gamma\text{ is stable}}}\frac{1}{m!n!}(\widetilde{ev_\star})_*\Big(\prod_{i=1}^{m}\frac{ev_i^*(\frac{1}{\delta_i}\mathbf{t}^{\vec{k}_i}i^*J^{X,tw}_{\beta_i,\vec{k}_i,c_i}(\mu^X,z)|_{z=\frac{\lambda-c_1(L)}{\delta_i}})}{\frac{\lambda-ev_i^*c_1(L)}{r\delta_i}+\frac{\bar\psi_i}{r}}\\
&\cup\prod_{i=m+1}^{m+n}ev_i^*(\mathbf{t}^{\vec{k}_i}i^*\mu^X_{\beta_i,\vec{k}_i,c_i}(-\bar\psi_i))\cup\bar\psi_\star^b\\
&\cap\sum_{d=0}^{\infty}\epsilon_*\big(c_d(-R^\bullet\pi_*f^*L^{\frac{1}{r}})(\frac{\lambda}{r})^{-1+m-d}(-1)^d\cap[\mathcal{K}_{0,\overrightarrow{m+n}\cup\star}(\sqrt[r]{L/Y},\beta_\star)]^{\mathrm{vir}}\big)\Big)\Big]_{\mathbf{t}^{\vec{k}}\lambda^{-1}}.
\end{aligned}$$

Note that for each leg $i$ with $m+1\le i\le m+n$, the admissibility condition in the definition of $\Lambda^W_{\beta,\vec{k},c,m,n}$ forces $\beta_i(L)+(\vec{w},\vec{k}_i)=0$, so by definition of the hypergeometric modification $\mu^{X,tw}_{\beta_i,\vec{k}_i,c_i}=\mu^X_{\beta_i,\vec{k}_i,c_i}$. Then the right-hand sides of recursive relations (4.6) and (3.22) are the same once we show that

$$i^*J^{X,tw}_{\beta_i,\vec{k}_i,c_i}(\mu^X,z)=J^Y_{\beta_i,\vec{k}_i,c_i}(i^*\mu^{X,tw},z)$$

for all $(\beta_i,\vec{k}_i,c_i)$ with $1\le i\le m$. We will prove this by showing that $\beta_i(M)+|\vec{k}_i|<\beta(M)+|\vec{k}|$ and then using the induction assumption.

Let $\Gamma=\big(c,\beta_\star,((\beta_1,\vec{k_1},c_1),\cdots,(\beta_{m+n},\overrightarrow{k_{m+n}},c_{m+n}))\big)\in\Lambda^W_{\beta,\vec{k},c,m,n}$ be a stable element. Without loss of generality, we can assume that $(\beta_i,\vec{k_i})\neq(0,0)$ for all $1\le i\le m+n$. First we assume that $\beta_\star\neq0$. Then the extended degree $(\beta_i,\vec{k_i})$ must have $\beta_i(M)+|\vec{k_i}|<e$ for all $1\le i\le m$. We are left to the case when $\beta_\star=0$. Note that $m+n\ge2$ as $\Gamma$ is stable, then we still have $\beta_i(M)+|\vec{k_i}|<e$ for all $1\le i\le m$, and thus we can still use the inductive assumption. This completes the proof. □

**Remark 4.10.** The above proof can also be extended to the equivariant case, where we assume that $X$ has an algebraic torus $T$–action which preserves $Y$. Since we don't have a particular application in mind, we will leave the details to the interested readers.

Let $\{\phi_\alpha\}$ be a *graded* basis of $H^*(\bar{I}_\mu X)$ such that each $\phi_\alpha\in H^*(\bar{I}_{c_\alpha}X)$ for some $c_\alpha\in\mathscr{C}$. Let $\{t_\alpha\}$ be a coordinate system of $H^*(\bar{I}_\mu X)$ corresponding to the basis $\{\phi_\alpha\}$. We also put $\mathbb{Z}_2$-grading on $\{t_\alpha\}$ according to $\phi_\alpha$. Then we have the following:

**Corollary 4.11.** *Let $i:\bar{I}_\mu Y\to\bar{I}_\mu X$ be the inclusion of rigidified inertia stacks and $\mathbf{t}=\sum_\alpha t_\alpha\phi_\alpha$. Then the series $i^*J^{X,tw}(q,\mathbf{t},-z)$ lies on the Lagrangian cone $\mathcal{L}_Y$ of $Y$.*

## 5. A GENUS-ZERO THREE-POINT INVARIANT OF NON-AMBIENT DEGREE

Let $Y\subset\mathbb{P}(1,1,1,2)$ be the smooth cubic surface cut out by the equation

$$x_3x_0+x_1x_2(x_1+x_2)=0\,,$$

and write $j:Y\hookrightarrow\mathbb{P}(1,1,1,2)$ for the inclusion. Denote by $p=c_1(\mathcal{O}(1))$ the hyperplane class of $Y$ (or of $\mathbb{P}(1,1,1,2)$) and by $\mathbb{1}_{\frac{1}{2}}$ the fundamental class of the unique nontrivial twisted sector of $H^*(\bar{I}_\mu Y,\mathbb{Q})$ (or of $H^*(\bar{I}_\mu\mathbb{P}(1,1,1,2),\mathbb{Q})$). Let $M$ be a line in $\mathbb{P}(1,1,1,2)$ representing a generator of $\mathrm{Eff}(\mathbb{P}(1,1,1,2))$, so that the curve class $[M]$ satisfies that the pairing $(p,[M])=1/2$.

In [Wan25], we use a mirror theorem for a *non-positive* hypersurface of an $S$-extended GIT model of $Y$ to show that,

$$\langle \mathbb{1}_{\frac{1}{2}}, \mathbb{1}_{\frac{1}{2}}, \mathbb{1}_{\frac{1}{2}} \rangle^{Y}_{0,3,[M]} = \sum_{d \in \mathrm{Eff}(Y),\, j_*(d)=[M]} \langle \mathbb{1}_{\frac{1}{2}}, \mathbb{1}_{\frac{1}{2}}, \mathbb{1}_{\frac{1}{2}} \rangle^{Y}_{0,3,d} = -3/4\,.$$

However, $[M]$ is a curve class from the ambient space $\mathbb{P}(1,1,1,2)$ rather than $Y$. Our purpose here is to compute Gromov-Witten invariant of *non-ambient* degree:

$$\langle \mathbb{1}_{\frac{1}{2}}, \mathbb{1}_{\frac{1}{2}}, \mathbb{1}_{\frac{1}{2}} \rangle^{Y}_{0,3,d}\,,$$

where $d \in \mathrm{Eff}(Y)$ and $j_*(d) = [M]$. The idea of this calculation is that we will realize $Y$ as a positive complete intersection of codimension two in another ambient space $X$ rather than $\mathbb{P}(1,1,1,2)$. Then we use our main theorem with an appropriate choice of extended variables.

Let $M_1, M_2, M_3 \subset Y$ be the three lines cut out by the equations $x_0 = x_1 = 0$, $x_0 = x_2 = 0$, and $x_0 = x_1 + x_2 = 0$, respectively. We show that

$$\langle \mathbb{1}_{\frac{1}{2}}, \mathbb{1}_{\frac{1}{2}}, \mathbb{1}_{\frac{1}{2}} \rangle^{Y}_{0,3,[M_1]} = \langle \mathbb{1}_{\frac{1}{2}}, \mathbb{1}_{\frac{1}{2}}, \mathbb{1}_{\frac{1}{2}} \rangle^{Y}_{0,3,[M_2]} = \langle \mathbb{1}_{\frac{1}{2}}, \mathbb{1}_{\frac{1}{2}}, \mathbb{1}_{\frac{1}{2}} \rangle^{Y}_{0,3,[M_3]} = -1/4\,.$$

Note that $Y$ has two automorphisms:

$$\alpha_1 : (x_0, x_1, x_2, x_3) \mapsto (x_0, x_2, x_1, x_3)\,,$$
$$\alpha_2 : (x_0, x_1, x_2, x_3) \mapsto (-x_0, x_1 + x_2, -x_2, x_3)\,,$$

satisfying $\alpha_1(M_1) = M_2$ and $\alpha_2(M_1) = M_3$. Since $\alpha_1, \alpha_2$ act on $\bar{I}_\mu Y$ and fix the class $\mathbb{1}_{\frac{1}{2}}$, they preserve the relevant Gromov–Witten invariants, and we obtain

$$\langle \mathbb{1}_{\frac{1}{2}}, \mathbb{1}_{\frac{1}{2}}, \mathbb{1}_{\frac{1}{2}} \rangle^{Y}_{0,3,[M_1]} = \langle \mathbb{1}_{\frac{1}{2}}, \mathbb{1}_{\frac{1}{2}}, \mathbb{1}_{\frac{1}{2}} \rangle^{Y}_{0,3,[M_2]} = \langle \mathbb{1}_{\frac{1}{2}}, \mathbb{1}_{\frac{1}{2}}, \mathbb{1}_{\frac{1}{2}} \rangle^{Y}_{0,3,[M_3]}\,.$$

Hence it suffices to show that

$$\langle \mathbb{1}_{\frac{1}{2}}, \mathbb{1}_{\frac{1}{2}}, \mathbb{1}_{\frac{1}{2}} \rangle^{Y}_{0,3,[M_2]} = -1/4\,.$$

5.1. **The effective cone of $Y$.** We first study the cone $\mathrm{Eff}(Y)$ of effective curve classes of $Y$. By the Jacobi-ring isomorphism of the cohomology of $Y$ (see [Dol82]), we have $\dim H^2(Y,\mathbb{Q}) = 3$. More precisely, letting $M_1, M_2, M_3 \subset Y$ be the three lines above and writing $[M_1], [M_2], [M_3]$ (by abuse of notation) also for their Poincaré dual classes. Then the classes $[M_1], [M_2], [M_3]$ generate $H^2(Y,\mathbb{Q})$ by the nondegeneracy of the intersection pairing on $H^*(Y,\mathbb{Q})$ recorded below.

| | $\mathbb{1}$ | $[M_1]$ | $[M_2]$ | $[M_3]$ | $p^2$ |
|---|---|---|---|---|---|
| $\mathbb{1}$ | 0 | 0 | 0 | 0 | $\frac{3}{2}$ |
| $[M_1]$ | 0 | $-\frac{1}{2}$ | $\frac{1}{2}$ | $\frac{1}{2}$ | 0 |
| $[M_2]$ | 0 | $\frac{1}{2}$ | $-\frac{1}{2}$ | $\frac{1}{2}$ | 0 |
| $[M_3]$ | 0 | $\frac{1}{2}$ | $\frac{1}{2}$ | $-\frac{1}{2}$ | 0 |
| $p^2$ | $\frac{3}{2}$ | 0 | 0 | 0 | 0 |

We claim that the effective curve classes $d$ of $Y$ with $j_*(d) = [M]$ are exactly the classes $[M_1], [M_2], [M_3]$. Indeed, let $d$ be an effective curve class represented by a sum of irreducible curves $\sum_i C_i$ with $C_i \neq M_1, M_2, M_3$ for every $i$; then the intersection numbers $(d, [M_k]) \in \frac{1}{2}\mathbb{Z}_{\geq 0}$ for $k = 1,2,3$. Writing $d = l_1[M_1] + l_2[M_2] + l_3[M_3]$ with $l_k \in \mathbb{Q}$, we obtain

$$\frac{-l_1 + l_2 + l_3}{2} \in \frac{1}{2}\mathbb{Z}_{\geq 0},\quad \frac{l_1 - l_2 + l_3}{2} \in \frac{1}{2}\mathbb{Z}_{\geq 0},\quad \frac{l_1 + l_2 - l_3}{2} \in \frac{1}{2}\mathbb{Z}_{\geq 0}\,,$$

from which we deduce that $l_k \in \mathbb{Z}_{\geq 0}$ for all $k = 1,2,3$. Hence the monoid $\mathrm{Eff}(Y)$ is spanned by $[M_1], [M_2], [M_3]$. Since $j_*([M_k]) = [M]$ for $k = 1,2,3$, the claim follows.

5.2. **Realizing $Y$ as a complete intersection of codimension two.** To this end we realize $Y$ as a complete intersection in another ambient space. Let $X := \mathbb{P}_{\mathbb{P}(1,1,1,2)}(\mathcal{O}(-1) \oplus \mathcal{O})$ be the $\mathbb{P}^1$-bundle over $\mathbb{P}(1,1,1,2)$, and let $\pi : X \to \mathbb{P}(1,1,1,2)$ be the projection. Let $K_1 := \pi^*(\mathcal{O}(1))$ be the pullback of the line bundle $\mathcal{O}(1)$ along $\pi$, and let $K_2 := \mathcal{O}_\pi(1)$ be the relative $\mathcal{O}(1)$ line bundle over $\mathbb{P}(1,1,1,2)$. Denote by $p_i = c_1(K_i)$ for $i = 1, 2$. Then the coordinate functions $x_0, x_1, x_2, x_3, w_2, w_3$ descend to sections of $K_1, K_1, K_1, K_1^{\otimes 2}, K_1^{-1} \otimes K_2, K_2$, respectively. The space $X$ can be obtained by the toric quotient $[\mathbb{C}^6/(\mathbb{C}^*)^2]$ given by the action

$$(x_0, x_1, x_2, x_3, w_2, w_3) \cdot (t_1, t_2) = (t_1 x_0, t_1 x_1, t_1 x_2, t_1^2 x_3, t_1^{-1} t_2 w_2, t_2 w_3)\,,$$

where $(x_0, x_1, x_2, x_3, w_2, w_3) \in \mathbb{C}^6$ and $(t_1, t_2) \in (\mathbb{C}^*)^2$ and we choose the stability condition to be the character $\theta(t_1, t_2) = t_1 t_2$. The rigidified inertia stack $\bar{I}_\mu X$ of $X$ has three nonempty sectors, associated with the isotropy elements $(e^{\frac{k}{2}}, e^{\frac{l}{2}}) \in (\mathbb{C}^*)^2$, where $(k, l) = (0,0),\ (1,0),\ \text{or}\ (1,1)$. For $d_1, d_2 \in \mathbb{Z}$, we write $\mathbb{1}_{(\frac{d_1}{2}, \frac{d_2}{2})}$ for the unit of the twisted sector $H^*(\bar{I}_{(e^{\frac{d_1}{2}}, e^{\frac{d_2}{2}})} X, \mathbb{Q})$, and write $\mathbb{1} := \mathbb{1}_{(\frac{d_1}{2}, \frac{d_2}{2})}$ when $d_1, d_2$ are both even. Note that when $d_1$ is even and $d_2$ is odd, we take the convention that $\mathbb{1}_{(0, \frac{1}{2})} = 0$, since $\bar{I}_{(e^{\frac{d_1}{2}}, e^{\frac{d_2}{2}})} X$ is then empty.

The cohomology of $X$ is spanned by

$$\{\mathbb{1}, p_1, p_1^2, p_1^3, p_2, p_1 p_2, p_1^2 p_2, p_1^3 p_2, \mathbb{1}_{(\frac{1}{2}, \frac{1}{2})}, \mathbb{1}_{(\frac{1}{2}, 0)}\}\,.$$

The monoid $\mathrm{Eff}(X)$ is generated by $\beta_1, \beta_2 \in \mathrm{Hom}(H^2(X, \mathbb{Q}), \mathbb{Q})$ satisfying

$$\begin{pmatrix} \beta_1(p_1) & \beta_1(p_2) \\ \beta_2(p_1) & \beta_2(p_2) \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{pmatrix}.$$

We note that $\beta_1, \beta_2$ can be represented by torus-invariant rational curves in $X$.

For $\beta \in \mathrm{Eff}(X)$ with $\beta(p_1) = d_1, \beta(p_2) = d_2$, we also write the Novikov variable $q^\beta$ as $q_1^{2d_1} q_2^{2d_2}$ in the sequel. We note that $q^{i_*([M_1])} = q_1 q_2^2$, $q^{i_*([M_2])} = q_1$, and $q^{i_*([M_3])} = q_1 q_2^2$.

Now we realize $Y$ as a complete intersection of codimension two in $X$. Let $B$ be the blow-up of $\mathbb{P}(1,1,1,2)$ along the line given by $x_2 = x_3 = 0$; more explicitly, with $X := \mathbb{P}_{\mathbb{P}(1,1,1,2)}(\mathcal{O}(-1) \oplus \mathcal{O})$ as above, $B$ can be seen as a hypersurface in $X$ cut out by the equation $x_2 w_3 - x_3 w_2$. Then the strict transform $\widetilde{Y}$ of $Y$ in $Z$ is still isomorphic to $Y$; more explicitly, by using a standard toric affine open cover of $X$, we can check that the strict transform $i : \widetilde{Y} \to X$ is realized as the complete intersection in $X$ cut out by the two equations

$$x_2 w_3 - x_3 w_2\,, \qquad w_3 x_0 + x_1 w_2 (x_1 + x_2)\,.$$

Both equations are sections of the line bundle $K_1 \otimes K_2$ over $X$.

Let $p_1, p_2$ also denote (by abuse of notation) the restrictions $i^* p_1, i^* p_2 \in H^*(Y, \mathbb{Q})$. Then in $H^*(Y, \mathbb{Q})$ we have

$$p_1 = [M_1] + [M_2] + [M_3]\,, \qquad -p_1 + p_2 = [M_2]\,,$$

and moreover

$$p = p_1\,, \qquad p_1 p_2 = \tfrac{4}{3} p^2\,, \qquad p_2^2 = p_1 p_2\,.$$

Let $d'$ be an effective class of $Y$ such that $i_*(d') = \beta_1$. As we have shown the monoid $\mathrm{Eff}(Y)$ is spanned by $[M_1], [M_2], [M_3]$, we can write $d' = l_1[M_1] + l_2[M_2] + l_3[M_3]$ with $l_1, l_2, l_3 \in \mathbb{Z}_{\geq 0}$. From which we deduce $l_1 = l_3 = 0$ and $l_2 = 1$. Hence $d' = [M_2]$.

5.3. **The $J$-function computation.** Recall that the big $I$-function of $X$ (see [CFK16]) is given by

$$
\begin{aligned}
I^X(q_1,q_2,t,u,z) = z &\sum_{d_1,d_2\in(\mathbb{N})^3} q_1^{d_1}q_2^{d_2}\exp\left(\frac{t_0+t_1p_1+u_1p_2+t_2(p_1+d_1z/2)^2}{z}\right)\\
&\cdot\frac{\prod_{i<0}(p_1+(\frac{d_1}{2}-i)z)^3}{\prod_{i<\frac{d_1}{2}}(p_1+(\frac{d_1}{2}-i)z)^3}\cdot\frac{\prod_{i<0}(2p_1+(d_1-i)z)}{\prod_{i<d_1}(2p_1+(d_1-i)z)}\\
&\cdot\frac{\prod_{i<0}(-p_1+p_2+(\frac{d_2-d_1}{2}-i)z)}{\prod_{i<\frac{d_2-d_1}{2}}(-p_1+p_2+(\frac{d_2-d_1}{2}-i)z)}\frac{\prod_{i<0}(p_2+(\frac{d_2}{2}-i)z)}{\prod_{i<\frac{d_2}{2}}(p_2+(\frac{d_2}{2}-i)z)}\mathbb{1}_{\frac{d_1}{2},\frac{d_2}{2}}.
\end{aligned}
\tag{5.1}
$$

We now make a change of variables, replacing $t_1, u_1$ by $t_1+t$ and $u_1-7t/8$, respectively. Take $r=2, N=5$ in Theorem 4.8. Here our variables are $t_0,t_1,u_1,t_2,t$, with corresponding weights $\vec{w}_1=(0,0,0,0,1)$, $\vec{w}_2=(0,0,0,0,0)$. This gives the hypergeometric modification $i^*I^{X,tw}$ of $I^X$:

$$
\begin{aligned}
i^*I^{X,tw}(q_1,q_2,t,u,z) = z &\sum_{d_1,d_2,k\in(\mathbb{N})^3} q_1^{d_1}q_2^{d_2}\exp\left(\frac{t_0+t_1p_1+u_1p_2+t_2(p_1+d_1z/2)^2}{z}\right)\\
&\cdot\frac{t^k((p_1-7p_2/8)/z+(d_1-7d_2/8)/2)^k}{k!}\\
&\cdot\frac{\prod_{i<0}(p_1+(\frac{d_1}{2}-i)z)^3}{\prod_{i<\frac{d_1}{2}}(p_1+(\frac{d_1}{2}-i)z)^3}\cdot\frac{\prod_{i<0}(2p_1+(d_1-i)z)}{\prod_{i<d_1}(2p_1+(d_1-i)z)}\\
&\cdot\frac{\prod_{i<0}(-p_1+p_2+(\frac{d_2-d_1}{2}-i)z)}{\prod_{i<\frac{d_2-d_1}{2}}(-p_1+p_2+(\frac{d_2-d_1}{2}-i)z)}\frac{\prod_{i<0}(p_2+(\frac{d_2}{2}-i)z)}{\prod_{i<\frac{d_2}{2}}(p_2+(\frac{d_2}{2}-i)z)}\\
&\cdot\prod_{0\le i<\frac{d_1+d_2}{2}+k}(p_1+p_2+(\frac{d_1+d_2}{2}+k-i)z)\\
&\cdot\prod_{0\le i<\frac{d_1+d_2}{2}}(p_1+p_2+(\frac{d_1+d_2}{2}-i)z)i^*\mathbb{1}_{\frac{d_1}{2},\frac{d_2}{2}}.
\end{aligned}
\tag{5.2}
$$

Here $i^*\mathbb{1}_{\frac{d_1}{2},\frac{d_2}{2}}=\mathbb{1}$ if $d_1,d_2$ are both even; $i^*\mathbb{1}_{\frac{d_1}{2},\frac{d_2}{2}}=\mathbb{1}_{\frac{1}{2}}$ if $d_1$ is odd and $d_2$ is even; and otherwise $i^*\mathbb{1}_{\frac{d_1}{2},\frac{d_2}{2}}=0$. Note that we may put a $\mathbb{Z}$-grading on $q,t_i,u_1,t,z,p_1,p_2,\mathbb{1}_{\frac{1}{2}}$ such that $\deg(p_1)=\deg(p_2)=\deg(q)=\deg(z)=\deg(\mathbb{1}_{\frac{1}{2}})=\deg(t_0)=1$, $\deg(t_1)=\deg(u_1)0$, $\deg(t_2)=\deg(t)=-1$; with this grading $J^{X,tw}$ is homogeneous of degree one.

Then we have

$$
\begin{aligned}
i^*I^{X,tw} = {}& \exp(\frac{t_0 + t_1p_1 + u_1p_2}{z})\Big\{ \\
& \left(\frac{(15t^2 + 6tt_2 + t_2^2)\, q_1 \exp(\frac{t_1}{2})\mathbb{1}_{\frac{1}{2}}}{16} + 1 - \frac{5p_1^2t^2}{16} - \frac{(-8p_1 + 7p_2)\, t}{8}\right) z \\
& + \frac{(3t + t_2)\, q_1 \exp(\frac{t_1}{2})\mathbb{1}_{\frac{1}{2}}}{2} \\
& - \frac{q_1^2\, (-p_2 + p_1)\, (6t^2 + 4tt_2 + t_2^2)\exp(t_1)}{4} + p_1^2 t_2 \\
& + \mathcal{O}(q_1^4, q_2) + \mathcal{O}(t_2^3, t_2^2t, t_2t^2, t^3) + \mathcal{O}(z^{-1})\Big\}
\end{aligned}
\tag{5.3}
$$

Denote $J^{Y,1} := i^*I^{X,tw}$. Note that the following elements lie in the tangent space $T_{J^{Y,1}}\mathcal{L}^Y|_{z\mapsto -z}$:

$$
\begin{aligned}
I_{t0} &:= \frac{\partial J^{Y,1}}{\partial t_0} = \exp(\frac{t_0 + t_1p_1 + u_1p_2}{z})\big(\mathbb{1} + \mathcal{O}(z^{-1}) + \mathcal{O}(q_1^4, q_2) + \mathcal{O}(t_2, t)\big) \\
I_{t1} &:= \frac{\partial J^{Y,1}}{\partial t_1} = \exp(\frac{t_0 + t_1p_1 + u_1p_2}{z})\big(p_1 + \mathcal{O}(z^{-1}) + \mathcal{O}(q_1^4, q_2) + \mathcal{O}(t_2, t)\big) \\
I_{u1} &:= \frac{\partial J^{Y,1}}{\partial u_1} = \exp(\frac{t_0 + t_1p_1 + u_1p_2}{z})\big(p_2 + \mathcal{O}(z^{-1}) + \mathcal{O}(q_1^4, q_2) + \mathcal{O}(t_2, t)\big) \\
I_t &:= 2\cdot\frac{\partial J^{Y,1}}{\partial t_2} - \left(I_{t1} - \frac{7I_{u1}}{8}\right) z = \exp(\frac{t_0 + t_1p_1 + u_1p_2}{z})\big(q_1 exp(t_1/2)\mathbb{1}_{\frac{1}{2}} \\
&\quad + \mathcal{O}(z^{-1}) + \mathcal{O}(q_1^4, q_2) + \mathcal{O}(t_2, t)\big) \\
I_{t2} &:= \frac{\partial J^{Y,1}}{\partial t_2} - \frac{I_t}{2} = \exp(\frac{t_0 + t_1p_1 + u_1p_2}{z})\big(p_1^2 + \mathcal{O}(z^{-1}) + \mathcal{O}(q_1^4, q_2) + \mathcal{O}(t_2, t)\big)
\end{aligned}
$$

Then, we have

(5.4)

$$
\begin{aligned}
& i^*I^{X,tw} - z\cdot(\frac{128t\cdot I_{t1} - 112t\cdot I_{u1}}{128} + \frac{(-8640t^2 - 3456tt_2 - 576t_2^2 + 6912t^2 + 2304tt_2)}{9216} I_t) \\
&= \exp(\frac{t_0 + t_1p_1 + u_1p_2}{z})(z - \frac{q_1^2\exp(t_1)\,(5t^2 + 6tt_2 + 3t_2^2)\, p_1}{16} + t_2\, p_1^2 \\
&+ \frac{q_1^2\exp(t_1)\,(5t^2 + 6tt_2 + 3t_2^2)\, p_2}{16} + \frac{q_1\exp(t_1/2)\,(t + t_2)\,\mathbb{1}_{\frac{1}{2}}}{2} \\
&+ \frac{1}{z}(-\frac{q_1^2\exp(t_1)\,(8t + 8t_2)\, p_1}{16} + \frac{q_1^2\exp(t_1)\,(8t + 8t_2)\, p_2}{16} \\
&+ \frac{q_1^2\exp(t_1)\,(t^2 + 2tt_2 + t_2^2)\, p_1^2}{24} + \frac{q_1\exp(t_1/2)\,(-27q_1^2\exp(t_1)t^2 - 36q_1^2\exp(t_1)tt_2 - 18q_1^2\exp(t_1)t_2^2 + 144)\,\mathbb{1}_{\frac{1}{2}}}{72}) \\
&+ \mathcal{O}(z^{-2}) + \mathcal{O}(q_1^4, q_2) + \mathcal{O}(t^3, t^2t_2, tt_2^2, t_2^3))
\end{aligned}
$$

Now we make the substitution in which $t_0, t_1, t_2, u_1, t$ are replaced by

$$
t_0, \quad t_1 - \frac{(-64\exp(t_1)q_1^2x^2 + 256\exp(t_1)q_1^2xy - 640\exp(t_1)q_1^2y^2)}{512},
$$

$$u_1 - \frac{q_1^2\left(64\exp(t_1)x^2 - 256\exp(t_1)xy + 640\exp(t_1)y^2\right)}{512}, \quad x, \quad -x+2y\,,$$

respectively, in (5.4). We obtain

$$\begin{aligned}
&J^Y(q, t_0 + t_1 p_1 + u_1 p_2 + x p_1^2 + q_1 \exp(t_1/2) y \mathbb{1}_{\frac{1}{2}}, z) \\
&= \exp\Big(\frac{t_0 + t_1 p_1 + u_1 p_2}{z}\Big)\Big(z + q_1\exp(t_1/2)y\mathbb{1}_{\frac{1}{2}} + x\, p_1^2 \\
&+ \frac{1}{z}\Big((p_2 - p_1)\, q_1^2 \exp(t_1) y + \frac{p_1^2 q_1^2 \exp(t_1) y^2}{6} \\
&+ \frac{q_1 \exp(t_1/2)\left(-q_1^2\exp(t_1)y^2 + 8\right)\mathbb{1}_{\frac{1}{2}}}{4}\Big) \\
&+ \mathcal{O}(z^{-2}) + \mathcal{O}(q_1^4, q_2) + \mathcal{O}(x^3, x^2y, xy^2, y^3)\Big)
\end{aligned}$$

Extracting the coefficient of $q_1^3 y^2 z^{-1}$ on both sides of this identity, and matching against the genus-0 three-point function with two insertions of $q_1 y \mathbb{1}_{\frac{1}{2}}$ (each contributing one factor of $q_1$ and one of $y$) and one insertion of $\mathbb{1}_{\frac{1}{2}}$, we get

$$\frac{1}{2} q_1 \langle q_1 y \mathbb{1}_{\frac{1}{2}}, q_1 y \mathbb{1}_{\frac{1}{2}}, \mathbb{1}_{\frac{1}{2}} \rangle^Y_{0,3,\beta_1} \cdot 2\mathbb{1}_{\frac{1}{2}} = -\frac{q_1^3 y^2 \mathbb{1}_{\frac{1}{2}}}{4}\,,$$

which completes the computation.

5.4. **Further invariants.** As $i^* I^{X,tw}, I_{t0}, I_{t1}, I_{u1}, I_{t2}, I_t$ are also series in

$$H^*_{CR}(Y, \mathbb{C})[z, z^{-1}][\![t_0, t_1, u_1, t_2, q_1\exp(t_1/2)t]\!][\![\mathrm{Eff}(X)]\!]\,.$$

Using more higher-order expansions, one can obtain a closed-form expression for

$$J^Y(q_1, q_2, t_0 + t_1 p_1 + u_1 p_2 + x p^2 + y q_1 \exp(t_1/2)\mathbb{1}_{\frac{1}{2}}, z)$$

to arbitrary order in $x, y$ (note that we have $p = p_1$ in $H^*(Y)$). From this one can also compute other Gromov–Witten invariants (with the help of the automorphisms $\alpha_1, \alpha_2$); for example,

$$\langle p^2, p^2, \mathbb{1}_{\frac{1}{2}}\rangle^Y_{0,3,[M_1+M_2+M_3]} = \frac{9}{4}, \quad \langle p^2, p^2, \mathbb{1}_{\frac{1}{2}}\rangle^Y_{0,3,[M_i]+2[M_j]} = \langle p^2, p^2, \mathbb{1}_{\frac{1}{2}}\rangle^Y_{0,3,3[M_i]} = 0\,,$$

where $i, j \in \{1,2,3\}$ and $i \neq j$. This matches the computation of Corti-Martin-Hironori (see [MH14, Example 3.2]):

$$\langle p^2, p^2, \mathbb{1}_{\frac{1}{2}}\rangle^Y_{0,3,3[M]} = \sum_{\substack{d_1+d_2+d_3=3 \\ d_1,d_2,d_3 \geq 0}} \langle p^2, p^2, \mathbb{1}_{\frac{1}{2}}\rangle^Y_{0,3,[d_1M_1+d_2M_2+d_3M_3]} = \frac{9}{4}\,.$$

## Appendix A. Edge contribution

In the proof of our main theorem, we need to apply torus localization formula to the moduli of stable maps to $\mathbb{P}Y_{r,s}$ (or to $\mathfrak{R}$ which factors through $E \cong \mathbb{P}Y_{r,s}$ in §4), which is an orbi-$\mathbb{P}^1$ bundle over a stack, for which we don't find a suitable reference of the explicit localization contribution from the torus-fixed 1-dimensional orbits (named edge contribution) in our case. In this appendix, we will explain how to compute the edge contributions in §3.2.2 and §4.2.2. To achieve this, we will construct a space called $\mathcal{M}_e$ with a family of $\mathbb{C}^*$−fixed stable maps to $\mathbb{P}Y_{r,s}$ of the prefix decorated data. This space has the property that it allows a finite étale map to the corresponding substack of $\mathbb{C}^*$−fixed loci in $\mathcal{K}_{0,2}(\mathbb{P}Y_{r,s}, (0, \frac{\delta}{r}))$ (or $\mathcal{K}_{0,2}(\mathfrak{R}, (0, \frac{\delta}{r}, 0))$). Then we can use $\mathcal{M}_e$ as a substitute for

$\mathbb{C}^*-$fixed loci in the edge contribution and the explicit description of the family map helps us carry out the localization analysis concretely. To begin with, we will prove a classification result (see Lemma A.1) about $\mathbb{C}^*-$fixed stable maps to $\mathbb{P}Y_{r,s}$ which map to a fiber of $\mathbb{P}Y_{r,s}$ over $Y$ using the relation between fundamental groups and $\mathbb{C}^*-$fixed stable maps to an orbi-$\mathbb{P}^1$ curve from [LS20]. Note that we will always assume that $r, s$ are sufficiently large primes in this appendix.

A.1. **Local picture.** Write $\mathfrak{P} := \mathbb{P}Y_{r,s}$ for simplicity. For any $\mathbb{C}-$point $y \in Y$, let $G_y$ be the residual isotropy group of $y$ in $Y$. Let $\rho : G_y \to \mathbb{C}^* = Aut(L|_y)$ be the associated representation. Denote $\mathfrak{P}_y$ by the fiber product in the following square

$$\begin{tikzcd} \mathfrak{P}_y \arrow[r] \arrow[d] & \mathfrak{P} \arrow[d, "\mathrm{pr}_{r,s}"] \\ \mathbb{B}G_y \arrow[r, "i"] & Y , \end{tikzcd}$$

where $i : \mathbb{B}G_y \to Y$ is the inclusion. Let $U := \mathbb{C}^2 \backslash \{0\}$, the fiber curve $\mathfrak{P}_y$ can be represented as the quotient stack

$$[\mathbb{C}^* \times U/(G_y \times (\mathbb{C}^*)^2)]$$

via the action

$$(m, x_1, x_2)(g, t_1, t_2) = (\rho^{-1}(g) t_1^{-s} t_2^{r} m, t_1 x_1, t_2 x_2) ,$$

where $(m, x_1, x_2) \in \mathbb{C}^* \times U$, $(g, t_1, t_2) \in G_y \times (\mathbb{C}^*)^2$.

The $\mathbb{C}^*-$action on $\mathfrak{P}$ defined in 3.1 restricts to be a $\mathbb{C}^*-$action on $\mathfrak{P}_y$ by only scaling the coordinate $x_1$ with weight one. There are two $\mathbb{C}^*-$fixed points of $\mathfrak{P}_y$: (1) The point 0 corresponding to $x_1 = 0$; (2) the point $\infty$ corresponding to $x_2 = 0$. Now let's describe the isotropy groups of points 0 and $\infty$, for which we need the following definition. Let $G$ be an arbitrary finite group and $\eta$ be a character of $G$. Assume that image of $\eta$ is a finite cyclic group $\boldsymbol{\mu_t} \subset \mathbb{C}^*$ of order $t$. In other words, we have the following exact sequence from the action

$$1 \longrightarrow \ker(\eta) \longrightarrow G \xrightarrow{\eta} \boldsymbol{\mu}_t \longrightarrow 1 .$$

For any positive integer $i$, we define the central extension $G(\eta, i)$ of $G$ by the cyclic group $\boldsymbol{\mu_i}$:

$$G(\eta, i) := \{(g, b) \in G \times \mathbb{C}^* | \eta(g) = b^i\} .$$

We note $G(\eta, i)$ fits in the following Cartesian digram with all rows and all columns exact:

$$\begin{tikzcd} & & \boldsymbol{\mu}_i \arrow[r, "="] \arrow[d, hook] & \boldsymbol{\mu}_i \arrow[d, hook] & \\ 1 \arrow[r] & \ker(\eta) \arrow[r] \arrow[d, "="] & G(\eta, i) \arrow[r, "\eta_i"] \arrow[d] & \boldsymbol{\mu}_{ti} \arrow[r] \arrow[d, "a \mapsto a^i"] & 1 \\ 1 \arrow[r] & \ker(\eta) \arrow[r] & G \arrow[r, "\eta"] & \boldsymbol{\mu}_t \arrow[r] & 1 , \end{tikzcd}$$

where the morphism $\eta_i$ is induced from the projection from $G \times \mathbb{C}^*$ to the second factor. Then the isotropy group $G_0$ (resp. $G_\infty$) of 0 (resp. $\infty$) is isomorphic to $G_y(\rho^{-1}, s)$(resp. $G_y(\rho, r)$). Denote

$$\mathfrak{U}_0 := \mathfrak{P}_y \backslash \{\infty\} \cong [\mathbb{C}/G_y(\rho^{-1}, s)], \text{ and } \mathfrak{U}_\infty := \mathfrak{P}_y \backslash \{0\} \cong [\mathbb{C}/G_y(\rho, r)] ,$$

and

$$\mathfrak{o}_e := \mathfrak{P}_y \backslash \{0, \infty\} .$$

Now we quote the result from [LS20, §6](see also [NB06] ). Assume that the image of $\rho$ is $\boldsymbol{\mu}_t$. Then the curve $\mathfrak{P}_y$ is a $G_e := \mathrm{Ker}(\rho)$-gerbe over $\mathbb{P}^1_{tr,ts}$. More explicitly, note that $\mathfrak{o}_e$ is isomorphic to the quotient stack $[\mathbb{C}^*/G_y]$[19] then the morphism $\rho : G_y \to \boldsymbol{\mu}_t$ induces that $\mathfrak{o}_e$ is a $G_e-$gerbe over its coarse moduli $o_e \cong [\mathbb{C}^*/\boldsymbol{\mu}_t] = \mathbb{C}^*$. Then the morphism $\rho_s^{-1}$ induces a morphism from $\mathfrak{U}_0 \cong [\mathbb{C}/G_y(\rho^{-1}, s)]$ (resp. $\mathfrak{U}_\infty \cong [\mathbb{C}/G_y(\rho, r)]$ ) to $[\mathbb{C}/\boldsymbol{\mu}_{ts}]$ (resp. $[\mathbb{C}/\boldsymbol{\mu}_{tr}]$). The above discussion fits into the following commutative digram

$$\begin{tikzcd}
\mathfrak{U}_0 \arrow[r] & {[\mathbb{C}/\boldsymbol{\mu}_{ts}]} \\
\mathfrak{o}_e \arrow[u] \arrow[d] \arrow[r] & o_e \cong [\mathbb{C}^*/\boldsymbol{\mu}_{ts}] \cong [\mathbb{C}^*/\boldsymbol{\mu}_t] \cong [\mathbb{C}^*/\boldsymbol{\mu}_{tr}] \arrow[u] \arrow[d] \\
\mathfrak{U}_\infty \arrow[r] & {[\mathbb{C}/\boldsymbol{\mu}_{tr}]\,.}
\end{tikzcd}$$

Let $\mathfrak{p}_e \cong \mathbb{B}G_e$ be a point of $\mathfrak{o}_e$. Define

$$H_e := \pi_1(\mathfrak{o}_e, \mathfrak{p}_e)\,,$$

then the coarsification map $\mathfrak{o}_e \to o_e$ (with the corresponding base points $\mathfrak{p}_e \to p_e$) induces a surjection

$$\phi_e : H_e = \pi_1(\mathfrak{o}_e, \mathfrak{p}_e) \to \pi_1(o_e, p_e) \cong \mathbb{Z}\,,$$

whose kernel is isomorphic to $G_e$ as shown in [LS20, §6].

Choose a path from $\mathfrak{p}_e$ to 0 and a path from $\mathfrak{p}_e$ to $\infty$, the open embeddings $\mathfrak{o}_e \hookrightarrow \mathfrak{U}_0$ and $\mathfrak{o}_e \hookrightarrow \mathfrak{U}_\infty$ induce surjective group homomorphisms

$$H_e = \pi_1(\mathfrak{o}_e, \mathfrak{p}_e) \xrightarrow{\pi_{e,0}} \pi_1(\mathfrak{U}_0, 0) \cong G_y(\rho^{-1}, s), \quad H_e = \pi_1(\mathfrak{o}_e, \mathfrak{p}_e) \xrightarrow{\pi_{e,\infty}} \pi_1(\mathfrak{U}_\infty, \infty) \cong G_y(\rho, r)\,.$$

Applying $\pi_1$, the above discussion will be summarized into the following commutative diagram with all rows exact:

(A.1)
$$\begin{tikzcd}
1 \arrow[r] & G_e \arrow[r] \arrow[d, equal] & \pi_1(\mathfrak{U}_0) \cong G_y(\rho^{-1}, s) \arrow[r, "\rho_s^{-1}"] & \boldsymbol{\mu}_{ts} \arrow[r] & 1 \\
1 \arrow[r] & G_e \arrow[r] \arrow[d, equal] & \pi_1(\mathfrak{o}_e) \cong H_e \arrow[r, "\phi_e"] \arrow[u, "\pi_{e,0}"'] \arrow[d, "\pi_{e,\infty}"'] & \mathbb{Z} = \pi_1(o_e) \arrow[r] \arrow[u] \arrow[d] & 1 \\
1 \arrow[r] & G_e \arrow[r] & \pi_1(\mathfrak{U}_\infty) \cong G_y(\rho, r) \arrow[r, "\rho_r"] & \boldsymbol{\mu}_{tr} \arrow[r] & 1\,,
\end{tikzcd}$$

where in the third column $\mathbb{Z} \to \boldsymbol{\mu}_{tr}$ and $\mathbb{Z} \to \boldsymbol{\mu}_{ts}$ are given by $d \mapsto e^{\frac{d}{rt}}$ and $d \mapsto e^{\frac{d}{st}}$ respectively. By chasing diagram, we can show that $H_e$ is isomorphic to the fiber product $G_y(\rho, r) \times_{\boldsymbol{\mu}_{tr}} \mathbb{Z}$ (or $G_y(\rho^{-1}, s) \times_{\boldsymbol{\mu}_{ts}} \mathbb{Z}$ as well) using the above diagram. Furthermore using the fact $G_y(\rho, r)$ isomorphic to the fiber product $G_y \times_{\boldsymbol{\mu}_t} \boldsymbol{\mu}_{tr}$, we have $H_e$ is also isomorphic to the fiber product $G_y \times_{\boldsymbol{\mu}_t} \mathbb{Z}$.

Let $\mathbb{P}^1_{m,n}$ be the unique smooth DM stack with coarse moduli $\mathbb{P}^1$ and trivial generic stablizer and only two special points $q_\infty \cong \mathbb{B}\boldsymbol{\mu}_m$ and $q_0 \cong \mathbb{B}\boldsymbol{\mu}_n$. Here we can choose a $\mathbb{C}^*-$action on $\mathbb{P}^1_{m,n}$ such that $q_0, q_\infty$ are the only $\mathbb{C}^*-$fixed points and the $\mathbb{C}^*-$weight of the tangent space $T_{q_0}$(resp. $T_{q_\infty}$) is $\frac{1}{m}$ (resp. $\frac{-1}{n}$). Let $f : \mathbb{P}^1_{m,n} \to \mathfrak{P}$ be a $\mathbb{C}^*-$fixed stable map of degree $(0, \frac{\delta}{r})$ which factors through

[19] One can check that $\mathfrak{o}_e$ can be also represented as the quotient stack $[\mathbb{C}^*/G_y(\rho^{-1}, s)]$ or $[\mathbb{C}^*/G_y(\rho, r)]$ via $\rho_s^{-1}$ or $\rho_r$ respectively.

$\mathfrak{P}_y$. Assume that $f(q_0) = 0 \in \mathfrak{P}_y$ and $f(q_\infty) = \infty \in \mathfrak{P}_y$. Moreover, if the multiplicity[20] at $q_\infty$ is $([g], 1, e^{\frac{\delta}{r}}) \in \mathrm{Conj}(G_y) \times \mathbb{C}^* \times \mathbb{C}^*$ for some $g \in G_y$ with $(y, g) \in \bar{I}_c Y$. Let $a$ be the order of $g$, assume that $age_g(L|_y) = \frac{h}{a}$ for some integer $h$ with $0 \leq h < a$. As $\delta = r \cdot deg(f^*\mathcal{O}(\mathcal{D}_\infty))$, using orbifold Riemann-Roch for the $f^*\mathcal{O}(r\mathcal{D}_\infty)$, the condition

$$\delta - \frac{h}{a} \in \mathbb{Z}\,,$$

is a necessary condition to ensure the map $f$ to exist; thus $a\delta$ must be an integer. As $r, s$ are sufficiently large primes and $f$ is representable, we have $m = ar$.

Let $T_{ar,as}$ be the subgroup of $(\mathbb{C}^*)^2$ defined by the equation $t_1^{as} = t_2^{ar}$. Let $F : U \to \mathbb{C}^* \times U$ be the morphism sending $(x, y)$ to $(1, x^{a\delta}, y^{a\delta})$. Then $F$ is equivariant with respect to the group homomorphism from $T_{ar,as}$ to $G_y \times (\mathbb{C}^*)^2$:

$$(t_1, t_2) \mapsto (\tau(t_1^{-s} t_2^{r}), t_1^{a\delta}, t_2^{a\delta})\,,$$

where $\tau : \boldsymbol{\mu}_a \to G_y$ is the group morphism sending $\mu_a$ to $g$. We can check $F$ descents to be a morphism $\widetilde{F}$ from $\mathbb{P}^1_{ar,as}$ to $\mathfrak{P}_y$ of degree $(0, \frac{\delta}{r})$[21] with the multiplicity decoration $([g], 1, e^{\frac{\delta}{r}})$ at $q_\infty$. Moreover, if we change $g$ to any conjugate of $g$ in $G_y$, the descended morphisms are all isomorphic to each other. Conversely, we will show the following:

**Lemma A.1.** *With the notations as above, let $f$ be a $\mathbb{C}^*$−fixed morphism of degree $(0, \frac{\delta}{r})$ to $\mathfrak{P}$ factor through $\mathfrak{P}_y$ such that the multiplicity is given by $([g], 1, e^{\frac{\delta}{r}})$ at $q_\infty$. Then the morphism $f$ must be isomorphic to $\widetilde{F}$ (up to a unique 2−isomorphism) constructed as above. In particular, the multiplicity at $q_0$ is $([g^{-1}], e^{\frac{\delta}{s}}, 1) \in \mathrm{Conj}(G_y) \times \mathbb{C}^* \times \mathbb{C}^*$.*

*Proof.* Let $f_1$, $f_2$ be two stable maps with same associated decorated data as above, if we assume they are isomorphic when restricting to $O_e := \mathbb{P}^1_{m,n} \backslash \{q_0, q_\infty\}$, then $f_1$ is isomorphic to $f_2$ up to a unique 2−isomorphism(see e.g.,[CCFK15, Lemma 2.5] for a proof). Thus we only need to show the restriction $f|_{O_e}$ is uniquely determined by the degree data and multiplicity data at $q_\infty$. Note that $f|_{O_e} : O_e \to \mathfrak{o}_e$ is an étale covering map. Indeed, as $O_e$ and the coarse moduli $o_e$ of $\mathfrak{o}_e$ are both isomorphic to $\mathbb{C}^*$ and the coarsing map from $\mathfrak{o}_e$ to $o_e$ is étale, then the étaleness of $f|_{O_e}$ follows from the étaleness of the composition

$$O_e \to \mathfrak{o}_e \to o_e$$

as any $\mathbb{C}^*$−fixed morphism from $\mathbb{C}^*$ to $\mathbb{C}^*$ must be a cyclic covering. By [Noo05, Theorem 18.19], we have a bijection between conjugacy classes of subgroup $(f|_{O_e})_*(\pi_1(O_e, P_e))$ ($P_e$ is the inverse image of the point $\mathfrak{p}_e$ under $O_e \to \mathfrak{o}_e$) of $\pi_1(\mathfrak{o}_e, \mathfrak{p}_e)$ and isomorphism classes of covering map of $\mathfrak{o}_e$ (not necessarily base point preserving), it's enough to show the conjugacy class of subgroup $(f|_{O_e})_*(\pi_1(O_e, P_e))$ is uniquely determined by the degree data and the monodromy data at $q_\infty$.

We will drop the base points in the notation of fundmental group in the sequel if it's clear in the context. Let

$$U_0 := \mathbb{P}^1_{m,n} \backslash \{q_\infty\} \cong [\mathbb{C}/\boldsymbol{\mu}_n], \quad \text{and} \quad U_\infty := \mathbb{P}^1_{m,n} \backslash \{q_0\} \cong [\mathbb{C}/\boldsymbol{\mu}_m]\,.$$

Choose a path from $P_e$ to $q_0$ in $U_0$ and a path from $P_e$ to $q_\infty$ in $U_\infty$, then the open embeddings $O_e \hookrightarrow U_0$ and $O_e \hookrightarrow U_\infty$ induces surjective group homomorphisms

$$\mathbb{Z} = \pi_1(O_e) \xrightarrow{\pi_{q_0}} \pi_1(U_0) \cong \boldsymbol{\mu}_n \qquad \mathbb{Z} = \pi_1(O_e) \xrightarrow{\pi_{q_\infty}} \pi_1(U_\infty) \cong \boldsymbol{\mu}_{\boldsymbol{m}}\,.$$

[20]Here, without loss of generality, we can replace the index set $\mathscr{C}$ for $\bar{I}_\mu \mathfrak{P}_y$ by the set $\mathrm{Conj}(G_y)$ of conjugacy classes of $G_y$.

[21]We use the degree notation by viewing $\widetilde{F}$ as a morphism to $\mathfrak{P}$.

Here we choose the identification between $\pi_1(O_e)$ and $\mathbb{Z}$ such that $\pi_{q_0}(1) = e^{\frac{-1}{n}}$ and $\pi_{q_\infty}(1) = e^{\frac{1}{m}}$.[22] By the above discussion we have $m = ar$. Then the morphism

$$\pi_1(O_e) \longrightarrow \pi_1(U_\infty) \xrightarrow{(f|_{U_\infty})_*} \pi_*(\mathfrak{U}_\infty)$$

sends the generator 1 of $\pi_1(O_e)$ to the conjugacy class of $(g, e^{\frac{\delta}{r}}) \in G_y(\rho, r) \cong \pi_1(\mathfrak{U}_\infty)$ by the multiplicity data at $q_\infty$. We have a commutative digram with all vertical lines are inclusions:

$$\begin{tikzcd}
O_e \arrow[r, "f|_{O_e}"] \arrow[d] & \mathfrak{o}_e \arrow[r] \arrow[d] & o_e \arrow[d] \\
U_\infty \arrow[r] & \mathfrak{U}_\infty \arrow[r] & {[\mathbb{C}/\boldsymbol{\mu}_{tr}]} .
\end{tikzcd}$$

Using (A.1), the above diagram induces the following commutative diagram

$$\begin{tikzcd}
\mathbb{Z} = \pi_1(O_e) \arrow[r, "(f|_{O_e})_*"] \arrow[d] & \pi_1(\mathfrak{o}_e) \cong H_e \arrow[r, two heads, "\phi_e"] \arrow[d, "\pi_{e,\infty}"] & \pi_1(o_e) \cong \mathbb{Z} \arrow[d] \\
\boldsymbol{\mu}_{ar} = \pi_1(U_\infty) \arrow[r] & \pi_1(\mathfrak{U}_\infty) \cong G_y(\rho, r) \arrow[r, two heads, "\rho_r"] & \boldsymbol{\mu}_{tr} ,
\end{tikzcd} \tag{A.2}$$

where in the third column $\mathbb{Z} \to \boldsymbol{\mu}_{tr}$ is given by $d \mapsto \mathrm{e}^{\frac{d}{tr}}$. Then we see that the composition

$$\pi_1(O_e) \cong \mathbb{Z} \xrightarrow{(f|_{O_e})_*} \pi_1(\mathfrak{o}_e) = H_e \xrightarrow{\phi_e} \pi_1(o_e) \cong \mathbb{Z} .$$

from $\mathbb{Z}$ to $\mathbb{Z}$ is multiplication by $t\delta$ as the generator 1 of $\pi_1(O_e)$ sends to $e^{\frac{\delta}{r}}$ in the group $\boldsymbol{\mu}_{tr}$ in whatever path from the up-left corner to the bottom-right corner in the above commutative diagram and $r$ is sufficiently large.

Let $\gamma_e \in H_e$ be the image of generator $1 \in \pi_1(O_e)$ in $H_e$. By the previous discussion, $\gamma_e$ maps to the conjugacy class of $(g, e^{\frac{\delta}{r}})$ via $\pi_{e,\infty}$, and maps to $t\delta$ in $\pi_1(o_e)$. Conversely, using the fact $H_e$ is isomorphic to fiber product $G_y(\rho, r) \times_{\boldsymbol{\mu}_{tr}} \mathbb{Z}$ using the right square of A.2 , we see that the conjugacy class of $\gamma_e$, is uniquely determinded by the conjugacy class of the image of $\gamma_e$ in $\pi_1(\mathfrak{U}_\infty)$ and the image of $\gamma_e$ in $\pi_1(o_e)$, which are determined by the decorated data. Hence the conjugacy class of the subgroup $(f|_{O_e})_*(\pi_1(O_e))$ is also uniquely determined by the decorated data. This completes the proof. □

Next we describe the automorphism group $Aut(f)$ of $f$.

**Proposition A.2.** *Use the notation in the above proof, we have $Aut(f) \cong C_{H_e}(\gamma_e)/\langle\gamma_e\rangle$, and the order of automorphism group is equal to $|Aut(f)| = |C_{H_e}(\gamma_e)/\langle\gamma_e\rangle| = \delta \cdot |C_{G_y}(g)|$. Here $C_{H_e}(\gamma)$ (resp. $C_{G_y}(g)$) is the centralizer of $\gamma_e$(resp. $g$) in the group $H_e$(resp. $G_y$).*

*Proof.* The automorphism group of $f$ is isomorphic to the automorphism group of the restriction $f|_{O_e}$, then the first claim follows from standard fact about covering deck transformation(not necessarily basepoint preserving) as $\gamma_e$ generates the subgroup $(f|_{O_e})_*(\pi_1(O_e))$ in $\pi_1(\mathfrak{o}_e)$.

[22] Different choices of path from $P_e$ to $q_0$(resp. $q_\infty$) will only affect $\pi_{q_0}$(resp. $\pi_{q_\infty}$) up to conjugacy, then the image $\pi_{q_0}(1)$ (resp. $\pi_{q_\infty}(1)$) doesn't depend on the choice of the path.

Assume that the image $\phi_e(C_{H_e}(\gamma_e))$ in $\mathbb{Z}$ is generated by the positive integer $u$, as $H_e$ is isomorphic to the fiber product $G_y \times_{\mu_t} \mathbb{Z}$.[23] More precisely,

$$H_e \cong \{(g,d) \in G_y \times \mathbb{Z} | \rho(g) = \mathrm{e}^{\frac{d}{t}}\}\,.$$

Then $\gamma_e = (g, t\delta) \in H_e$. We can show that $t$ and $t\delta$ are both divided by $u$. Thus the image of $C_{G_y}(g)$ under $\rho$ is generated by $\mathrm{e}^{\frac{u}{t}}$.

The group $C_{H_e}(\gamma_e)$ can be represented as

$$\{(h,d) \in C_{G_y}(g) \times \mathbb{Z} | \rho(h) = \mathrm{e}^{\frac{d}{t}}\}$$

in $H_e$. Then the claim about $|Aut(f)|$ follows from the following two exact sequences:

$$1 \longrightarrow G_e \cap C_{G_y}(g) \xrightarrow{\theta} C_{H_e}(\gamma_e)/\langle \gamma_e \rangle \longrightarrow u\mathbb{Z}/t\delta\mathbb{Z} \longrightarrow 1\,,$$

and

$$1 \longrightarrow G_e \cap C_{G_y}(g) \longrightarrow C_{G_y}(g) \xrightarrow{\rho} \boldsymbol{\mu}_{\frac{t}{u}} \longrightarrow 1\,.$$

Here $u\mathbb{Z}$(resp. $t\delta\mathbb{Z}$) is the subgroup of $\mathbb{Z}$ generated by the integer $u$(resp. $t\delta$), the morphism $\theta$ is the induced from the restriction to $G_e \cap C_{G_y}(g)$ of the inclusion from $G_e$ to $H_e$. □

We can make the above proposition more concretely. Let $f = \widetilde{F}$ as in Lemma A.1. Let $(h,d) \in C_{H_e}(\gamma_e) \subset G_y \times_{\mu_t} \mathbb{Z}$ be an element, we associate $(h,d)$ an element in the automorphism group $Aut(f)$ as follows: define

$$\theta_{h,d} : U \to U : \quad (x,y) \mapsto (x, e^{\frac{d}{art\delta}} y)\,,$$

which descents to be an isomorphism of $\mathbb{P}^1_{ar,as}$ which we still denote to be $\theta_{h,d}$. Recall that $F$ is a morphism from $U \to \mathbb{C}^* \times U$ defined as in Lemma A.1, then $F \circ \theta_{h,d}$ differ from $F$ by the action of the element $(h, 1, e^{\frac{d}{rt}})$ on the target via the action of $G_y \times (\mathbb{C}^*)^2$ on $\mathbb{C}^* \times U$ which also commutes the action of $T_{ar,as}$ and $G_y \times (\mathbb{C}^*)^2$ on the source and target respectively. Then by descent, this defines an 2−isomorphism $\alpha_{h,d} : \widetilde{F} \to \widetilde{F} \circ \theta_{h,d}$. Thus the pair $(\theta_{h,d}, \alpha_{h,d})$ defines an automorphism in $Aut(f)$.

For two elements $(h_1,d_1)$ and $(h_2,d_2)$ in $C_{H_2}(\gamma_e)$, the pairs $(\theta_{h_1,d_1}, \alpha_{h_1,d_1})$ and $(\theta_{h_2,d_2}, \alpha_{h_2,d_2})$ are isomorphic to each other if there exists some integer $n$ such that $h_2 = h_1 g^n$ and $d_2 = d_1 + nt\delta$. Indeed, let $m_{(1, e^{\frac{nt\delta}{art\delta}})}$ be the automorphism of $U$ acted upon by $(1, e^{\frac{nt\delta}{art\delta}})$, we have $\theta_{h_2,d_2} = m_{(1, e^{\frac{nt\delta}{art\delta}})} \circ \theta_{h_1,d_1}$, which defines a two-isomorphism $\beta_n : \theta_{h_1,d_1} \to \theta_{h_2,d_2}$ in $\mathrm{Hom}(\mathbb{P}^1_{ar,as}, \mathbb{P}^1_{ar,as})$, then the composition $f \circ \beta_n$ defines a two-isomorphism from $\widetilde{F} \circ \theta_{h_1,d_1}$ to $\widetilde{F} \circ \theta_{h_2,d_2}$ such that $\alpha_{h_2,d_2} = f \circ \beta_n \circ \alpha_{h_1,d_1}$.

The morphism $f : \mathbb{P}^1_{ar,as} \to \mathfrak{P}_y$ defines a $\mathbb{C}$−point of $\mathcal{K}_{0,\vec{2}}(\mathbb{P}Y_{r,s}, (0, \frac{\delta}{r}))$, where we denote by $q_0$ (resp. $q_\infty$) the first (resp. second) marking, then the evaluation map

$$ev_{q_\infty} : \mathcal{K}_{0,2}(\mathbb{P}Y_{r,s}, (0, \frac{\delta}{r})) \to \bar{I}_\mu \mathbb{P}Y_{r,s}\,,$$

induces a morphism from the isotropy group $Aut(f)$ to the isotropy group $Aut((y, ([g], 1, e^{\frac{\delta}{r}})))$, which, from the concrete description of $Aut(f)$ above, coincides with the morphism from $C_{H_e}(\gamma_e)/\langle\gamma_e\rangle$

[23]We have already seen that $H_e$ is isomorphic to $G_y(\rho^{-1}, s) \times_{\mu_{st}} \mathbb{Z}$, then we use the isomorphism $G_y(\rho^{-1}, s) \cong G_y \times_{\mu_t} \boldsymbol{\mu}_{st}$

to $C_{G_y}(g)/\langle g\rangle$ induced from the projection of $G_y \times \mathbb{Z}$ to the first factor. From this concrete description, we also know the kernel of the morphism of isotropy groups:

**Proposition A.3.** *The kernel of the morphism $Aut(f) \to Aut\big((y, ([g], 1, e^{\frac{\delta}{r}}))\big)$ induced from $ev_{q_\infty}$ is isomorphic to $\mathbb{Z}/a\delta\mathbb{Z}$.*

*Proof.* This follows from by applying the snake lemma to the first two exact rows of the following diagram:

$$\begin{tikzcd}
1 \arrow[r] & \mathbb{Z} \cong \langle (1, at\delta)\rangle \arrow[r] \arrow[d, hook] & \mathbb{Z} \cong \langle \gamma_e\rangle \arrow[r, "pr_1"] \arrow[d, hook] & \mathbb{Z}/a\mathbb{Z} \cong \langle g\rangle \arrow[r] \arrow[d, hook] & 1 \\
1 \arrow[r] & \mathbb{Z} \cong \langle (1, t)\rangle \arrow[r] \arrow[d, two heads] & C_{H_e}(\gamma_e) \arrow[r, "pr_1"] \arrow[d, two heads] & C_{G_y}(g) \arrow[r] \arrow[d, two heads] & 1 \\
1 \arrow[r] & \mathbb{Z}/a\delta\mathbb{Z} \arrow[r] & C_{H_e}(\gamma_e)/\langle \gamma_e\rangle \arrow[r, "pr_1"] \arrow[d, equal] & C_{G_y}(g)/\langle g\rangle \arrow[r] \arrow[d, equal] & 1 \\
 & & Aut(f) & Aut\big((y, ([g], 1, e^{\frac{\delta}{r}}))\big) ,&
\end{tikzcd}$$

where $pr_1$ are induced from the projection from $G_y \times \mathbb{Z}$ to the first factor. □

A.2. **Construction of family stable map.** Let $e$ be an edge with decorated degree $\frac{\delta}{r}$ and decorated multiplicity $(c, 1, \mathrm{e}^{\frac{\delta}{r}})$ at the half-edge $h_\infty$ as in §3.2. Let $a := a(c)$ be the integer associated to $c$ as in the Assumption 2.1. Define the space $\mathcal{M}_e$ to be the root gerbe $\sqrt[as\delta]{L^\vee/I_cY}$. We will generalize the construction in Lemma A.1 to a family version; we will construct a family of $\mathbb{C}^*$−fixed stable map over $\mathcal{M}_e$ to $\mathbb{P}Y_{r,s}$ with the associated decorated degree. Then we get a morphism

$$g : \mathcal{M}_e \to \mathcal{K}_{0,2}(\mathbb{P}Y_{r,s}, (0, \frac{\delta}{r})) .$$

We will show the image of $g$ is a closed and open substack of the $\mathbb{C}^*$−fixed part of $\mathcal{K}_{0,2}(\mathbb{P}Y_{r,s}, (0, \frac{\delta}{r}))$, and $g$ is a finite étale map into the image of $g$. Then we can use $\mathcal{M}_e$ to do an explicit computation of the edge contribution in the virtual localization formula.

Since we assume $Y$ is a smooth proper Deligne-Mumford stack with projective coarse moduli over $\mathbb{C}$. Then $Y$ admits a quotient stack presentation $[\widetilde{Y}/G]$ by [Kre09, Proposition 5.4], where $\widetilde{Y}$ is smooth quasi-projective variety over $\mathbb{C}$ and $G$ is a linear algebraic group over $\mathbb{C}$. Then we can assume that the line bundle $L$ comes from a $G$−equivariant line bundle $\widetilde{L}$ over $\widetilde{Y}$ by descent. We will write $G$−action on $\widetilde{L}$ as $l \cdot g$ for any $l \in \widetilde{L}$ and $g \in G$.

Without loss of generality, we can take $I_cY$ to be the inertia component $I_{[g]}Y$ represented by the conjugacy class of $g \in G$. Then $I_{[g]}Y \cong [\widetilde{Y}^g/C_G(g)]$ where $\widetilde{Y}^g$ is the $g$−fixed loci of $\widetilde{Y}$ and $C_G(g)$ is the centralizer subgroup of $g$ in $G$.

By an abuse of notation, we also denote $\widetilde{L}^\vee$ to be the restriction of the line bundle $\widetilde{L}^\vee$ over $\widetilde{Y}$ to $\widetilde{Y}^g$. Then $g$ acts on each fiber of $\widetilde{L}^\vee$ over $\widetilde{Y}^g$ via the scalar multiplication by the complex number $\mathrm{e}^{\frac{-h}{a}}$ (recall that we define $age_g(L|_y) = \frac{h}{a}$ for each point $(x, g) \in \bar{I}_cY$).

Denote by $(\widetilde{L}^\vee)^0$ the open set of $\widetilde{L}^\vee$ with zero section removed. Then we note that $\mathcal{M}_e$ allows a quotient stack representation $[(\widetilde{L}^\vee)^0/C_G(g) \times \mathbb{C}^*]$ defined by the (right) action

$$l \cdot (g', t) = t^{-as\delta} l \cdot g' ,$$

where $l \in (\widetilde{L}^\vee)^0$ and $(g', t) \in C_G(g) \times \mathbb{C}^*$.

Let $U := \mathbb{C}^2 \backslash \{0\}$. Let $\mathcal{C}_e$ be the quotient stack $[(\widetilde{L}^\vee)^0 \times U / C_G(g) \times \mathbb{C}^* \times T_{ar,as}]$ defined by the (right) action

$$(l, x, y)(g', t, t_1, t_2) = (t^{-as\delta} l \cdot g', tt_1 x, t_2 y)\,,$$

where $(l, x, y) \in (\widetilde{L}^\vee)^0 \times U$ and $(g', t, t_1, t_2) \in C_G(g) \times \mathbb{C}^* \times T_{ar,as}$. Here $T_{ar,as}$ is the subgroup of $(\mathbb{C}^*)^2$ defined by$\{(t_1, t_2) \in (\mathbb{C}^*)^2 |\; t_1^{as} = t_2^{ar}\}$. Then $\mathcal{C}_e$ is a family of orbi-$\mathbb{P}^1$ bundle over $\mathcal{M}_e$ such that all fibers are isomorphic to $\mathbb{P}^1_{ar,as}$.

**Remark A.4.** Let $\bar{\psi}_\star$ be the psi-class associated with the gerbe marking corresponding to the zero loci of $y$ of $\mathcal{C}_e$. We see that $\bar{\psi}_\star$ is equal to $\frac{\lambda - c_1(L)}{\delta}$. Here $c_1(L)$ is a pullback cohomology class along the composition of natural maps $\mathcal{M}_e \to I_{[g]}Y \to Y$.

The space $\mathbb{P}Y_{r,s}$ can be represented as the quotient $[(\widetilde{L}^\vee)^0 \times U / G \times (\mathbb{C}^*)^2]$ via the action

$$(l, x, y)(g', t_1, t_2) = (t_1^{-s} t_2^r l \cdot g', t_1 x, t_2 y)\,.$$

We will define a family map $f$ as follows:

$$\begin{tikzcd} \mathcal{C}_e \arrow[r, "f"] \arrow[d, "\pi"'] & \mathbb{P}Y_{r,s} \\ \mathcal{M}_e := \sqrt[as\delta]{L^\vee / I_c Y}. & \end{tikzcd}$$

Consider the following map

$$\widetilde{f} : (\widetilde{L}^\vee)^0 \times U \to (\widetilde{L}^\vee)^0 \times U$$
$$(l, x, y) \to (l, x^{a\delta}, y^{a\delta})\,,$$

with respect to the group homomorphism $C_G(g) \times \mathbb{C}^* \times T_{ar,as} \to G \times (\mathbb{C}^*)^2$

$$(g', t, t_1, t_2) \to (g' \cdot \tau(t_1^{-s} t_2^r), (tt_1)^{a\delta}, t_2^{a\delta})\,,$$

where $\tau : \boldsymbol{\mu}_a \to G$ is the group homomorphism determined by sending $\mathrm{e}^{\frac{1}{a}}$ to $g$. Then the family map $f$ is defined by using $\widetilde{f}$ by descent. Note that we (implicitly) use the fact

$$\delta - \frac{h}{a} \in \mathbb{Z}$$

to check $\widetilde{f}$ is equivariant.

By the construction of family stable maps over $\mathcal{M}_e$. There are also two evaluation maps

$$ev_{h_0} : \mathcal{M}_e \to \bar{I}_{m(h_0)}\mathcal{D}_0, \quad ev_{h_\infty} : \mathcal{M}_e \to \bar{I}_{m(h_\infty)}\mathcal{D}_\infty$$

corresponding to the two ramification points on the edge curve $\mathcal{C}_e$. Let $\omega_e : \mathcal{M}_e \to \bar{I}_c Y$ be the natural morphism by forgetting root first and then taking rigidification. When $r, s$ are sufficiently large primes, we have two natural isomorphisms (cf. Lemma 4.3)

$$\bar{\theta}_0 : \bar{I}_{m(h_0)}\mathcal{D}_0 \to \bar{I}_{c^{-1}} Y$$

and

$$\bar{\theta}_\infty : \bar{I}_{m(h_\infty)}\mathcal{D}_\infty \to \bar{I}_c Y$$

induced by the two natural (forgetting-root) maps $\mathcal{D}_0 \to Y$ and $\mathcal{D}_\infty \to Y$ respectively. Then we have $\bar{\theta}_0 \circ ev_{h_0} = \iota \circ \omega_e$ and $\bar{\theta}_\infty \circ ev_{h_\infty} = \omega_e$.

**Remark A.5.** When $Y$ is a complete intersection in a toric stack, our construction of the family map is equivalent to the construction in [Wan25, §5.3.2].

A.3. **Computation of edge contribution.** Recall that in §3, we define a $\mathbb{C}^*-$action on $\mathbb{P}Y_{r,s}$. This can be induced from the $\mathbb{C}^*-$action on $(\widetilde{L}^\vee)^0 \times U$:

$$t \cdot (l, x, y) = (t \cdot l, x, y)\,.$$

Now we define a $\mathbb{C}^*-$action on $\mathcal{C}_e$ such that the morphism $f$ is $\mathbb{C}^*-$equivariant. Recall that $\mathcal{C}_e$ can be represented as $[(\widetilde{L}^\vee)^0 \times U/C_G(g)\times T_{ar,as}\times\mathbb{C}^*]$ given by the action

$$(l, x, y)(g', t_1, t_2, t_3) = (t_3^{-as\delta} l \cdot g', t_3 t_1 x, t_2 y)\,.$$

We define the $\mathbb{C}^*-$action on $(\widetilde{L}^\vee)^0 \times U$ in the way that

$$t \cdot (l, x, y) = (t \cdot l, x, y)\,.$$

One can see this action descends to be $\mathbb{C}^*-$action on $\mathcal{C}_e$.

Let $\chi_1$ (resp. $\chi_2$) be the character associated to $T_{ar,as}$ by sending $(t_1, t_2) \in T_{ar,as}$ to $t_1$ (resp. $t_2$). Then we can construct the line bundle $L_{\chi_1}$(resp. $L_{\chi_2}$) over $\mathcal{C}_e$ as the stack quotient $[(\widetilde{L}^\vee)^0 \times U \times \mathbb{C}/C_G(g)\times T_{ar,as}\times\mathbb{C}^*]$ using the Borel construction:

$$(l, x, y, z) \sim (t_3^{-as\delta} l \cdot g', t_3 t_1 x, t_2 y, t_1 z) \text{ (resp. } (t_3^{-as\delta} l \cdot g', t_3 t_1 x, t_2 y, t_2 z))$$

for any $(l, x, y, z) \in (\widetilde{L}^\vee)^0 \times U \times \mathbb{C}$ and $(g', t_1, t_2, t_3) \in C_G(g)\times T_{ar,as}\times\mathbb{C}^*$. Using the stack quotient representation, the line bundles $L_{\chi_1}$ and $L_{\chi_2}$ also carry a $\mathbb{C}^*-$action such that the fiber of $L_{\chi_1}$ over $D_\infty$ (given by the equation $y = 0$) is of weight $\frac{-1}{as\delta}$, the fiber of $L_{\chi_2}$ over $D_\infty$ is of weight $\frac{-1}{ar\delta}$.

**Proposition A.6.** *The coordinate $x$ of $(L^\vee)^0 \times U$ descends to be a $\mathbb{C}^*-$equivariant section of the $\mathbb{C}^*-$equivariant line bundle $L_{\chi_1} \otimes (\pi^* L^\vee)^{\frac{1}{as\delta}} \otimes \mathbb{C}_{\frac{\lambda}{as\delta}}$ over $\mathcal{C}_e$, and the coordinate $y$ of $(L^\vee)^0 \times U$ descends to be a $\mathbb{C}^*-$equivariant section of the $\mathbb{C}^*-$equivariant line bundle $L_{\chi_2}$ over $\mathcal{C}_e$.*

Fix the edge $e = \{h_0, h_\infty\}$ with the decoration as above, we will denote

$$\mathcal{K}_{0,h_0\sqcup h_\infty}(\mathbb{P}Y_{r,s}, (0, \frac{\delta}{r})) := \mathcal{K}_{0,2}(\mathbb{P}Y_{r,s}, (0, \frac{\delta}{r})) \cap ev_1^{-1}(\bar{I}_{(c^{-1}, e^{\frac{\delta}{s}}, 1)} \mathbb{P}Y_{r,s}) \cap ev_2^{-1}(\bar{I}_{(c, 1, e^{\frac{\delta}{r}})} \mathbb{P}Y_{r,s})\,.$$

Here we label the first marking by $h_0$ and the second marking by $h_\infty$. Let $\mathcal{K}_{0,h_0\sqcup h_\infty}(\mathbb{P}Y_{r,s}, (0, \frac{\delta}{r}))^{\mathbb{C}^*}$ be the $\mathbb{C}^*-$fixed loci of $\mathcal{K}_{0,h_0\sqcup h_\infty}(\mathbb{P}Y_{r,s}, (0, \frac{\delta}{r}))$, write $\mathcal{K} := \mathcal{K}_{0,h_0\sqcup h_\infty}(\mathbb{P}Y_{r,s}, (0, \frac{\delta}{r}))^{\mathbb{C}^*}$ for short, we will first show the following:

**Proposition A.7.** *Let $[\mathcal{K}]^{vir}$ be the virtual cycle associated to the $\mathbb{C}^*-$fixed part of the restriction of the perfect obstruction theory of $\mathcal{K}_{0,h_0\sqcup h_\infty}(\mathbb{P}Y_{r,s}, (0, \frac{\delta}{r}))$ to $\mathcal{K}$. Let $m = (c, 1, e^{\frac{\delta}{r}})$. We have that $[\mathcal{K}]^{\mathrm{vir}} = [\mathcal{K}]$ in $A_*(\mathcal{K})$ and the evaluation map*

$$ev_{h_\infty} : \mathcal{K} \to \bar{I}_m \mathbb{P}Y_{r,s}$$

*is étale and finite of degree $\frac{1}{a\delta}$.*

*Proof.* Let $\mathbb{E}$ be the restriction of the perfect obstruction theory[24] for $\mathcal{K}_{0,h_0\sqcup h_\infty}(\mathbb{P}Y_{r,s}, (0, \frac{\delta}{r}))$ to $\mathcal{K}$, then the $\mathbb{C}^*-$fixed part of $\mathbb{E}$, which we denote to be $\mathbb{E}^{fix}$, is a perfect obstruction theory for $\mathcal{K}$ by [GP99]. We will show that $\mathbb{E} = \mathbb{E}^{fix}$ and it's isomorphic to the locally free sheaf $ev_{h_\infty}^* \mathbb{T}_{\bar{I}_m \mathbb{P}Y_{r,s}}$. Note this implies that $\mathbb{E}^{fix} \cong \mathbb{T}_{\mathcal{K}} \cong ev_{h_\infty}^* \mathbb{T}_{\bar{I}_m \mathbb{P}Y_{r,s}}$ and therefore $\mathcal{K}$ is smooth. As $\bar{I}_m \mathbb{P}Y_{r,s}$ is also smooth and $ev_{h_\infty}$ induces isomorphisms on tangent spaces, this will implies the étaleness part; while the finiteness part will follow the properness of $ev_{h_\infty}$, Lemma A.1 and Proposition A.3.

[24]Here a perfect obstruction is a morphism $\mathbb{T} \to \mathbb{E}$ in [BF97], where $\mathbb{T} = \mathbb{L}^\vee$ is the derived dual of the cotangent complex.

It suffices to check $\mathbb{E} = \mathbb{E}^{fix} \cong ev_{h_\infty}^* \mathbb{T}_{\bar{I}_m \mathbb{P}Y_{r,s}}$ locally on every $\mathbb{C}$-point. For any $\mathbb{C}-$point $[f]$ of $\mathcal{K}$ represented by a twisted stable map $f$, note that as we assume that $r, s$ are sufficiently large primes, the source curve for $f$ must be an irreducible curve as its has no nodes. Then every such stable map $f$ can be presented as in Lemma A.1. The tangent space $T_{[f]}$ and the obstruction space $Ob_{[f]}$ of the space $\mathcal{K}_{0,h_0 \sqcup h_\infty}(\mathbb{P}Y_{r,s}, (0, \frac{\delta}{r}))$ at the point $[f]$ fit into the following exact sequence of $\mathbb{C}^*-$presentations[25]

$$
\begin{aligned}
&0 \to H^0(C, \mathbb{T}_C(-[q_0] - [q_\infty])) \to H^0(C, f^*\mathbb{T}_{\mathbb{P}Y_{r,s}}) \to T_{[f]} \\
&\to H^1(C, \mathbb{T}_C(-[q_0] - [q_\infty])) \to H^1(C, f^*\mathbb{T}_{\mathbb{P}Y_{r,s}}) \to Ob_{[f]} \to 0\,.
\end{aligned} \tag{A.3}
$$

We have the following two exact sequences of vector bundles over $\mathbb{P}Y_{r,s}$ (which are also $\mathbb{C}^*$-equivariant)

$$
0 \longrightarrow \mathbb{T}_{\mathbb{P}Y_{r,s}/Y} \longrightarrow \mathbb{T}_{\mathbb{P}Y_{r,s}} \longrightarrow \mathrm{pr}_{r,s}^* \mathbb{T}_Y \longrightarrow 0\,, \tag{A.4}
$$

and the Euler exact sequence

$$
0 \longrightarrow \mathbb{C} \longrightarrow \mathcal{O}(\mathcal{D}_0) \oplus \mathcal{O}(\mathcal{D}_\infty) \longrightarrow \mathbb{T}_{\mathbb{P}Y_{r,s}/Y} \longrightarrow 0\,. \tag{A.5}
$$

Then we have the following two right exact sequences

$$
\begin{aligned}
&0 \to H^0(C, f^*\mathbb{T}_{\mathbb{P}Y_{r,s}/Y}) \to H^0(C, f^*\mathbb{T}_{\mathbb{P}Y_{r,s}}) \to H^0(C, f^*\mathrm{pr}_{r,s}^*\mathbb{T}_Y) \\
&\to H^1(C, f^*\mathbb{T}_{\mathbb{P}Y_{r,s}/Y}) \to H^1(C, f^*\mathbb{T}_{\mathbb{P}Y_{r,s}}) \to H^1(C, f^*\mathrm{pr}_{r,s}^*\mathbb{T}_Y) \to 0\,.
\end{aligned} \tag{A.6}
$$

and

$$
\begin{aligned}
&0 \to H^0(C, \mathcal{O}) \to H^0(C, f^*\mathcal{O}(\mathcal{D}_0) \oplus f^*\mathcal{O}(\mathcal{D}_\infty)) \to H^0(C, f^*T_{\mathbb{P}Y_{r,s}/Y}) \\
&\to H^1(C, \mathcal{O}) \to H^1(C, f^*\mathcal{O}(\mathcal{D}_0) \oplus f^*\mathcal{O}(\mathcal{D}_\infty)) \to H^1(C, f^*T_{\mathbb{P}Y_{r,s}/Y}) \to 0\,.
\end{aligned} \tag{A.7}
$$

By Lemma A.1, we have

$$
f^*\mathcal{O}(\mathcal{D}_0) \oplus f^*\mathcal{O}(\mathcal{D}_\infty) \cong \mathcal{O}(a\delta[q_0] + a\delta[q_\infty])
$$

over $C \cong \mathbb{P}^1_{ar,as}$, as $r, s$ are sufficiently large primes, then we have

$$
H^1(C, f^*\mathcal{O}(\mathcal{D}_0) \oplus f^*\mathcal{O}(\mathcal{D}_\infty)) = 0\,,
$$

which implies that $H^1(C, f^*\mathbb{T}_{\mathbb{P}Y_{r,s}/Y}) = 0$ by (A.7). As $H^1(C, \mathbb{T}_C(-[q_0] - [q_\infty])) = 0$, we have

$$
Ob_{[f]} \cong H^1(C, f^*\mathrm{pr}_{r,s}^*\mathbb{T}_Y)
$$

and

$$
T_{[f]} \cong H^0(C, f^*\mathbb{T}_{\mathbb{P}Y_{r,s}})/H^0(C, \mathbb{T}_C(-[q_0] - [q_\infty]))
$$

by (A.3) and (A.6).

Moreover, the exact sequence (A.7) becomes

$$
0 \to \mathbb{C} \to \mathbb{C} \oplus \mathbb{C} \to H^0(C, f^*\mathbb{T}_{\mathbb{P}Y_{r,s}/Y}) \to 0 \to 0 \to 0 \to 0\,, \tag{A.8}
$$

which implies that $H^0(C, f^*\mathbb{T}_{\mathbb{P}Y_{r,s}/Y}) \cong \mathbb{C}$(of weight 0). Furthermore, the composition

$$
H^0(C, T_C(-[q_0] - [q_\infty])) \to H^0(C, f^*\mathbb{T}_{\mathbb{P}Y_{r,s}}) \to H^0(C, f^*\mathrm{pr}_{r,s}^*\mathbb{T}_Y)
$$

where the arrows are coming from (A.3) and (A.6), is equal to zero as the composition is induced from the composition of sheaves

$$
\mathbb{T}_C(-[q_0] - [q_\infty]) \to f^*\mathbb{T}_{\mathbb{P}Y_{r,s}} \to f^*\mathrm{pr}_{r,s}^*\mathbb{T}_Y\,,
$$

which is zero. Note $H^0(C, \mathbb{T}_C(-[q_0] - [q_\infty]))$ is one dimensional, which corresponds to the space of tangent vector field of orbi-$\mathbb{P}^1$ vanishing on the two markings, then the space $H^0(C, \mathbb{T}_C(-[q_0] -$

[25] All the arrows in this proof are $\mathbb{C}^*-$equivaraint unless otherwise stated.

$[q_\infty]$)) maps isomorphically into the subspace $H^0(C, f^*\mathbb{T}_{\mathbb{P}Y_{r,s}/Y}) \subset H^0(C, f^*\mathbb{T}_{\mathbb{P}Y_{r,s}})$ via (A.6), which implies that

$$\begin{aligned} T_{[f]} &\cong H^0(C, f^*\mathbb{T}_{\mathbb{P}Y_{r,s}})/H^0(C, \mathbb{T}_C(-[q_0]-[q_\infty])) \\ &\cong H^0(C, f^*\mathbb{T}_{\mathbb{P}Y_{r,s}})/H^0(C, f^*\mathbb{T}_{\mathbb{P}Y_{r,s}/Y}) \cong H^0(C, f^*\mathrm{pr}_{r,s}^*\mathbb{T}_{\mathbb{P}Y_{r,s}}) . \end{aligned}$$

Let $\pi : C \to \mathrm{Spec}(\mathbb{C})$ be the projection to a point, next we show that

$$R^\bullet\pi_* f^*\mathrm{pr}_{r,s}^* T_{\mathbb{P}Y_{r,s}} \cong (ev_{h_\infty}^* \mathbb{T}_{\bar{I}_m\mathbb{P}Y_{r,s}})|_{[f]} ,$$

which will imply that

$$T_{[f]} \cong (ev_{h_\infty}^* \mathbb{T}_{\bar{I}_m\mathbb{P}Y_{r,s}})|_{[f]} , \quad Ob_{[f]} = 0 ,$$

in particular $T_{[f]}$ is $\mathbb{C}^*-$fixed. Applying $R^\bullet\pi_* f^*(-)$ to the first sequence (A.4), we get a distinguished triangle

$$R^\bullet\pi_* f^*\mathbb{T}_{\mathbb{P}Y_{r,s}/Y} \longrightarrow R^\bullet\pi_* f^*\mathbb{T}_{\mathbb{P}Y_{r,s}} \longrightarrow R^\bullet\pi_* f^*\mathrm{pr}_{r,s}^*\mathbb{T}_Y \xrightarrow{+1} R^\bullet\pi_* f^*\mathbb{T}_{\mathbb{P}Y_{r,s}/Y}[1] .$$

Let $b_\infty : q_\infty \hookrightarrow C$ be the gerbe section of $C$ corresponding to the half-edge $h_\infty$. Using the divisor sequence for the section $b_\infty$

$$0 \to \mathcal{O}(-q_\infty) \to \mathcal{O}_C \to \mathcal{O}_{q_\infty} \to 0 ,$$

tensor with $f^*\mathrm{pr}_{r,s}^*\mathbb{T}_Y$ and taking $R^\bullet\pi_*$, we have

$$R^\bullet\pi_* f^*\mathrm{pr}_{r,s}^*\mathbb{T}_Y = R^\bullet(\pi\circ b_\infty)_*(f\circ b_\infty)^*(\mathrm{pr}_{r,s}^*\mathbb{T}_Y) ,$$

where we use the fact that $R^\bullet\pi_* f^*\mathrm{pr}_{r,s}^*\mathbb{T}_Y(-q_\infty) = 0$. Next we show that

$$R^\bullet(\pi\circ b_\infty)_*(f\circ b_\infty)^*(\mathrm{pr}_{r,s}^*\mathbb{T}_Y) \cong (ev_{h_\infty}^* \mathbb{T}_{\bar{I}_m\mathbb{P}Y_{r,s}})|_{[f]} ,$$

First by the tangent bundle lemma in [AGV08, §3.6.1], we have

$$(\pi\circ b_\infty)_*(f\circ b_\infty)^*(\mathbb{T}_{\mathbb{P}Y_{r,s}}) \cong (ev_{h_\infty}^* \mathbb{T}_{\bar{I}_m\mathbb{P}Y_{r,s}})|_{[f]} ,$$

thus we only need to show that

$$R^\bullet(\pi\circ b_\infty)_*(f\circ b_\infty)^*(\mathrm{pr}_{r,s}^*\mathbb{T}_Y) \cong (\pi\circ b_\infty)_*(f\circ b_\infty)^*\mathbb{T}_{\mathbb{P}Y_{r,s}} .$$

As $(\pi\circ b_\infty)_*$ is exact on coherent sheaf, we have $R^\bullet(\pi\circ b_\infty)_* = (\pi\circ b_\infty)_*$ and the following exact sequence of vector spaces:

(A.9)
$$0 \to (\pi\circ b_\infty)_*(f\circ b_\infty)^*\mathbb{T}_{\mathbb{P}Y_{r,s}/Y} \to (\pi\circ b_\infty)_*(f\circ b_\infty)^*\mathbb{T}_{\mathbb{P}Y_{r,s}} \to (\pi\circ b_\infty)_*(f\circ b_\infty)^*\mathrm{pr}_{r,s}^*\mathbb{T}_Y \to 0 ,$$

we are left to show that $(\pi\circ b_\infty)_*(f\circ b_\infty)^*\mathbb{T}_{\mathbb{P}Y_{r,s}/Y} = 0$, which can be checked by the next lemma A.8 by pulling back to every $\mathbb{C}$-point using the idea of proving the tangent bundle lemma [AGV08, §3.6.1]. □

**Lemma A.8.** *Let $x := [f]$ be a $\mathbb{C}$-point of $\mathcal{K}$ represented by a twisted stable map $f : C \to \mathbb{P}Y_{r,s}$ as in Lemma A.1. Use the notation in the proof of Proposition A.7, we have*

$$(\pi\circ b_\infty)_*(f\circ b_\infty)^*\mathbb{T}_{\mathbb{P}Y_{r,s}/Y} = 0 .$$

*Proof.* The gerbe marking $q_\infty$ of $C$ is canonically isomorphic to the classifying stack $\mathbb{B}\boldsymbol{\mu}_{ar}$, which gives a unique lift of the $\mathbb{C}-$point $x$ in the gerbe marking $q_\infty$, which we denote to be $z$. The isotropy group $Aut(z)$ of $z$ in $\mathbb{B}\boldsymbol{\mu}_{ar}$ is canonically isomorphic to the cyclic group $\boldsymbol{\mu}_{ar}$, and the generator of $\boldsymbol{\mu}_{ar}$ acts on the vector space $f^*\mathbb{T}_{\mathbb{P}Y_{r,s}/Y}|_z$ via the multiplication by $e^{\frac{\delta}{r}}$ due to the very choice of the multiplicity associated to the half-edge $h_\infty$. Note the space $(\pi\circ b_\infty)_*(f\circ b_\infty)^*\mathbb{T}_{\mathbb{P}Y_{r,s}/Y}$ is given by the $\boldsymbol{\mu}_{ar}-$invariant space $(f^*\mathbb{T}_{\mathbb{P}Y_{r,s}/Y}|_z)^{\boldsymbol{\mu}_{ar}}$, then it is equal to zero. □

The family map $f$ constructed in Section A.2 induces a morphism

$$g : \mathcal{M}_e \to \mathcal{K}$$

by the universal property of moduli stack $\mathcal{K}$. Then $g$ is surjective by Lemma A.1. Furthermore, we have:

**Proposition A.9.** *The morphism $g : \mathcal{M}_e \to \mathcal{K}$ is finite étale of degree $\frac{1}{as}$. The evaluation map $ev_{h_\infty}$ from $\mathcal{K}$ to $\bar{I}_{(c,1,e^{\frac{\delta}{r}})}\mathbb{P}Y_{r,s}$ is finite étale of degree $\frac{1}{a\delta}$.*

*Proof.* Let $\mathrm{pr}_{r,s} : \bar{I}_{(c,1,e^{\frac{\delta}{r}})}\mathbb{P}Y_{r,s} \to \bar{I}_c Y$ be the morphism induced from the projection of $\mathbb{P}Y_{r,s}$ to the base $Y$, $\mathrm{pr}_{r,s}$ is an isomorphism by Lemma 4.3. Then the composition $\mathrm{pr}_{r,s}\circ ev_{h_\infty}\circ g$ is finite étale of degree $\frac{1}{a^2 s\delta}$ as the composition can be also obtained by first forgetting root construction of $\mathcal{M}_e$ and taking rigidification afterwards. By two-of-there property for étale morphisms [Sta19, Tag 0CIR], the étaleness of $g$ comes from that $ev_{h_\infty}\circ g$ and $ev_{h_\infty}$ are étale, where the étaleness of $ev_{h_\infty}$ is proved in Proposition A.7. Finally as the composition $ev_{h_\infty}\circ g$ is of degree $\frac{1}{a^2 s\delta}$ and $ev_{h_\infty}$ is of degree $\frac{1}{a\delta}$ by Proposition A.2, we see that $g$ is of degree $\frac{1}{as}$. $\square$

**Remark A.10.** Let $ev_{h_0} : \mathcal{K} \to \bar{I}_{(c^{-1},e^{\frac{\delta}{s}},1)}\mathbb{P}Y_{r,s}$ be the evaluation map associated with the half-edge $h_0$. Using the isomorphism

$$\bar{I}_{(c^{-1},e^{\frac{\delta}{s}},1)}\mathbb{P}Y_{r,s} \cong \bar{I}_{c^{-1}}Y$$

induced from the projection from $\mathbb{P}Y_{r,s}$ to the base $Y$. We can also see that the composition $\mathrm{pr}_{r,s}\circ ev_{h_0}\circ g$ is finite étale of degree $\frac{1}{a^2 s\delta}$ as the composition can be also obtained by first forgetting root construction of $\mathcal{M}_e$ and taking rigidification afterwards.

**Lemma A.11.** *We have the following:*

(1) *When the edge moduli $\mathcal{M}_e$ arises from the localization analysis in §3.2.2, we have that $[\mathcal{M}_e]^{vir} = [\mathcal{M}_e]$ and the Euler class of the virtual normal bundle is equal to* 1.
(2) *When the edge moduli $\mathcal{M}_e$ arises from the localization analysis in §4.2.2, we have $[\mathcal{M}_e]^{vir} = [\mathcal{M}_e]$ and the inverse of Euler class of the virtual normal bundle is equal to*

$$\prod_{m=1}^{-1-\lfloor -\delta \rfloor} \left(\lambda + \frac{m}{\delta}(c_1(L) - \lambda)\right).$$

*Proof.* The first case follows from Proposition A.7. Now we calculate the edge contribution from the second case. Recall the divisor $E$ of $\mathfrak{R}$ introduced in Section 4.1 is isomorphic to $\mathbb{P}Y_{r,s}$. The normal bundle $N_{E/\mathfrak{R}}$ is isomorphic to $\mathcal{O}(-s\mathcal{D}_0)$, and the fiber of the normal bundle $N_{E/\mathfrak{R}}$ over $\mathcal{D}_\infty$ has $\mathbb{C}^*$−weight 1, thus we have $f^*N_{E/\mathfrak{R}} = L^\vee_{as\delta\chi_1} \otimes \pi^*L$ as $\mathbb{C}^*$−equivariant line bundles over $\mathcal{C}_e$. Then we see that

$$-R^*\pi_* f^* N_{E/\mathfrak{R}} = R^1\pi_*(\pi^*L \otimes L^\vee_{as\delta\chi_1})$$

in the $K$−group $K^*(\mathcal{M}_e)$.

Write $\mathfrak{P} := \mathbb{P}Y_{r,s}$ for simplicity. Let $\mathbb{E}_{\mathfrak{R}}$ (resp. $\mathbb{E}_{\mathfrak{P}}$) be the pull-back of the perfect obstruction theory of $\mathcal{K}_{0,2}(\mathfrak{R}, (0, \frac{\delta}{r}, 0))$ (resp. $\mathcal{K}_{0,2}(\mathfrak{P}, (0, \frac{\delta}{r})))$ to $\mathcal{M}_e$. As $\mathcal{M}_e$ is étale over the corresponding $\mathbb{C}^*$-fixed loci in $\mathcal{K}_{0,2}(\mathfrak{P}, (0, \frac{\delta}{r}))$ (or $\mathcal{K}_{0,2}(\mathfrak{R}, (0, \frac{\delta}{r}, 0)))$ by Proposition A.9, we see that the $\mathbb{C}^*$−fixed parts $\mathbb{E}^{fix}_{\mathfrak{P}}$ and $\mathbb{E}^{fix}_{\mathfrak{R}}$ are perfect obstruction theory for $\mathcal{M}_e$. Note we have the distinguished triangle:

$$\mathbb{E}_{\mathfrak{P}} \longrightarrow \mathbb{E}_{\mathfrak{R}} \longrightarrow R^\bullet\pi_* f^* N_{E/\mathfrak{R}} \xrightarrow{+1} \mathbb{E}_{\mathfrak{P}}[1]\,. \tag{A.10}$$

As shown in Lemma A.7, $\mathbb{E}_{\mathfrak{P}}$ is $\mathbb{C}^*-$fixed, then the moving part of $\mathbb{E}_{\mathfrak{R}}$ is equal to the moving part of $R^\bullet\pi_* f^* N_{E/\mathfrak{R}}$, which we will calculate below.

The restriction of sections $x^{-sa\delta+asi}y^{-ari}(1 \le i \le -1-\lfloor -\delta \rfloor)$ to each fiber curve $C_e$ of $\mathcal{C}_e$ over $\mathcal{M}_e$ is a basis of the vector space $H^1(C_e, f|_{C_e}^* N_{E/\mathfrak{R}})$. However they may not be sections of the vector bundle $R^1\pi_*(f^*N_{E/\mathfrak{R}})$. Instead, by Proposition A.6, one can see that $x^{-sa\delta+asi}y^{-ari}$ is a section of vector bundle $R^1\pi_*(f^*N_{E/\mathfrak{R}}) \otimes L^{-\otimes\frac{i}{\delta}} \otimes \mathbb{C}_{-\frac{\delta-i}{\delta}\lambda}$. Therefore $R^1\pi_* f^* N_{E/\mathfrak{R}}$ is isomorphic to the direct sum of line bundles $\bigoplus_{i=1}^{-1-\lfloor-\delta\rfloor} L^{\otimes\frac{i}{\delta}} \otimes \mathbb{C}_{\frac{\delta-i}{\delta}\lambda}$. Then the inverse of the Euler class of the virtual normal bundle form $\mathbb{E}_{\mathfrak{R}}$ is equal to $\prod_{m=1}^{-1-\lfloor-\delta\rfloor}(\lambda + \frac{m}{\delta}(c_1(L)-\lambda))$.

The calculation above also shows that the $\mathbb{C}^*-$fixed part of $\mathbb{E}_{\mathfrak{R}}$ is the equal to $\mathbb{E}_{\mathfrak{P}}$, then we have $[\mathcal{M}_e]^{vir} = [\mathcal{M}_e]$ in Section 4.2.2. □

**Remark A.12.** When $Y$ is of zero dimensional, our computation of the Euler class of the virtual normal bundle coincides with the computation from [LS20, §6].

## Appendix B. Virtual localization formula revisited

Let $M$ be a Deligne-Mumford stack with a $\mathbb{C}^*-$action (or an action of an algebraic torus) satisfying the assumption in [GP99] (or weaker assumption in [CKL17]). The virtual localization formula expresses the virtual cycle $[M]^{vir}$ of $M$ as

$$\sum_i (i_{M_i})_* \left( \frac{[M_i]^{vir}}{e_{\mathbb{C}^*}(N_i)} \right).$$

Here $i_{M_i} : M_i \to M$ is an inclusion map into an open and closed component of $M^{\mathbb{C}^*}$. However, in practice, the fixed loci $M_i$ (and its virtual normal bundle) may be hard to analyze directly; instead, sometimes a detailed description of a finite étale covering of $M_i$ (as well as its virtual normal bundle) is feasible. Then a (slight) generalization of the above formula is more convenient to compute things in practice.[26] Let $j_{M_i} : \widetilde{M}_i \to M_i$ be a finite étale covering of $M_i$ of degree $d_i$. Denote $[\widetilde{M}_i]^{vir}$ to be virtual cycle obtained by $\mathbb{C}^*-$fixed part of the pull-back of perfect obstruction theory of $M$ along $i_{M_i} \circ j_{M_i}$. Then we have $j_{M_i}^*([M_i]^{vir}) = [\widetilde{M}_i]^{vir}$, $(j_{M_i})_*([\widetilde{M}_i]^{vir}) = d_i[M_i]^{vir}$, and $j_{M_i}^* N_i$ is the $\mathbb{C}^*-$moving part of the pull-back of perfect obstruction theory of $M$ along $i_{M_i} \circ j_{M_i}$. Then we can obtain a generalization of the above formula.

$$\begin{aligned}
&\sum_i (i_{M_i})_* \left( \frac{\frac{1}{d_i}(j_{M_i})_*[\widetilde{M}_i]^{vir}}{e_{\mathbb{C}^*}(N_i)} \right) \\
\text{(B.1)} \qquad &= \sum_i \frac{1}{d_i}(i_{M_i})_*(j_{M_i})_* \left( \frac{[\widetilde{M}_i]^{vir}}{j_{M_i}^* e_{\mathbb{C}^*}(N_i)} \right) \\
&= \sum_i \frac{1}{d_i}(i_{\widetilde{M}_i})_* \left( \frac{[\widetilde{M}_i]^{vir}}{e_{\mathbb{C}^*}(\widetilde{N}_i)} \right)
\end{aligned}$$

where $i_{\widetilde{M}_i} := i_{M_i} \circ j_{M_i}$, and $\widetilde{N}_i := j_{M_i}^* N_i$. In practice, we don't need to compute $N_i$ first in order to get $\widetilde{N}_i$. We will apply this generalization in §3.2 and §4.2, where we take $\widetilde{M}_i = F_\Gamma$.

**Acknowledgements**: I want to express my great gratitude to the anonymous referees for the valuable comments which greatly improved this paper.

[26] This generalization is not new, and it appears in many places already, see e.g.[Liu13, §9.3] for a closed-related discussion.

## References


[AGV08] Dan Abramovich, Tom Graber, and Angelo Vistoli, *Gromov-Witten theory of Deligne-Mumford stacks*, American Journal of Mathematics **130** (2008), no. 5, 1337–1398. ↑8, 13, 28, 39, 56

[AV02] Dan Abramovich and Angelo Vistoli, *Compactifying the space of stable maps*, Journal of the American Mathematical Society **15** (2002), no. 1, 27–75. ↑6

[BF97] K. Behrend and B. Fantechi, *The Intrinsic Normal Cone*, Inventiones Mathematicae **128** (1997), no. 1, 45–88. ↑8, 54

[CDLOGP91] Philip Candelas, Xenia C. De La Ossa, Paul S. Green, and Linda Parkes, *A pair of Calabi-Yau manifolds as an exactly soluble superconformal theory*, Nuclear Physics B **359** (1991Jul), no. 1, 21–74. ↑1

[CGL21] Huai-Liang Chang, Shuai Guo, and Jun Li, *Polynomial structure of Gromov–Witten potential of quintic 3-folds*, Annals of Mathematics **194** (2021), no. 3, 585–645. ↑2

[CKL17] Huai-Liang Chang, Young-Hoon Kiem, and Jun Li, *Torus localization and wall crossing for cosection localized virtual cycles*, Advances in Mathematics **308** (2017Feb), 964–986. ↑58

[CJR21] Qile Chen, Felix Janda, and Yongbin Ruan, *The logarithmic gauged linear sigma model*, Inventiones mathematicae **225** (2021), 1077–1154. ↑2

[CJR22] ______, *Punctured logarithmic R-maps*, arXiv preprint arXiv:2208.04519 (2022). ↑2

[CCFK15] Daewoong Cheong, Ionuţ Ciocan-Fontanine, and Bumsig Kim, *Orbifold quasimap theory*, Mathematische Annalen **363** (2015Mar), no. 3-4, 777–816. ↑2, 7, 49

[CFK14] Ionut Ciocan-Fontanine and Bumsig Kim, *Wall-Crossing in Genus Zero Quasimap Theory and Mirror Maps*, Algebraic Geometry **1** (2014), no. 4, 400–448. ↑7

[CFK16] Ionuţ Ciocan-Fontanine and Bumsig Kim, *Big I-functions*, Development of Moduli Theory — Kyoto 2013, 2016, pp. 323–347. ↑44

[CFKM14] Ionuţ Ciocan-Fontanine, Bumsig Kim, and Davesh Maulik, *Stable quasimaps to GIT quotients*, Journal of Geometry and Physics **75** (2014Jan), 17–47. ↑2

[CJR17a] Emily Clader, Felix Janda, and Yongbin Ruan, *Higher-Genus Quasimap Wall-Crossing Via Localization*, ArXiv e-prints (2017), available at 1702.03427v1. ↑8

[CJR17b] ______, *Higher-Genus Wall-Crossing in the Gauged Linear Sigma Model*, CoRR (2017), available at 1706.05038. ↑8

[CG07] Thomas Coates and Alexander Givental, *Quantum Riemann-Roch, Lefschetz and Serre*, Annals of Mathematics **165** (2007), no. 1, 15–53. ↑1, 2, 4

[CCIT15] Tom Coates, Alessio Corti, Hiroshi Iritani, and Hsian-Hua Tseng, *A mirror theorem for toric stacks*, Compositio Mathematica **151** (2015Jun), no. 10, 1878–1912. ↑4

[CCIT19] ______, *Some applications of the mirror theorem for toric stacks*, Advances in Theoretical and Mathematical Physics **23** (2019), no. 3, 767–802. ↑2, 4

[CCI+20] Tom Coates, Alessio Corti, Hiroshi Iritani, Hsian-Hua Tseng, et al., *Hodge-theoretic mirror symmetry for toric stacks*, Journal of Differential Geometry **114** (2020), no. 1, 41–115. ↑4

[CGI+12] Tom Coates, Amin Gholampour, Hiroshi Iritani, Yunfeng Jiang, Paul Johnson, and Cristina Manolache, *The quantum Lefschetz hyperplane principle can fail for positive orbifold hypersurfaces*, Mathematical Research Letters **19** (2012), no. 5, 997–1005. ↑2

[Dol82] Igor V. Dolgachev, *Weighted projective varieties*, Group Actions and Vector Fields, 1982, pp. 34–71. Proceedings of a Polish-North American Seminar, Vancouver, B.C., 1981. MR704986 ↑42

[Ful84] William Fulton, *Intersection theory*, Ergebnisse der Mathematik und ihrer Grenzgebiete (3) [Results in Mathematics and Related Areas (3)], vol. 2, Springer-Verlag, Berlin, 1984. MR732620 ↑26

[Giv96] Alexander Givental, *Equivariant Gromov-Witten invariants*, International Mathematics Research Notices **1996** (1996), no. 13, 613. ↑1

[Giv98] ______, *A Mirror Theorem for Toric Complete Intersections*, Topological Field Theory, Primitive Forms and Related Topics (1998), 141–175. ↑2

[Giv04] Alexander B Givental, *Symplectic geometry of Frobenius structures* (2004), 91–112. ↑1, 7

[GP99] T. Graber and R. Pandharipande, *Localization of virtual classes*, Inventiones Mathematicae **135** (1999Jan), no. 2, 487–518. ↑10, 11, 13, 54, 58

[GJR17] Shuai Guo, Felix Janda, and Yongbin Ruan, *A Mirror Theorem for Genus Two Gromov-Witten Invariants of Quintic Threefolds*, ArXiv e-prints (2017), available at 1709.07392. ↑2

[GJR18] ______, *Structure of higher genus Gromov-Witten invariants of quintic 3-folds*, arXiv preprint arXiv:1812.11908 (2018). ↑2

[JPPZ20] Felix Janda, Rahul Pandharipande, Aaron Pixton, and Dimitri Zvonkine, *Double ramification cycles with target varieties*, Journal of Topology **13** (2020), no. 4, 1725–1766. ↑12, 13, 14

[Jia08] Yunfeng Jiang, *The orbifold cohomology ring of simplicial toric stack bundles*, Illinois Journal of Mathematics **52** (2008), no. 2, 493–514. ↑4

[KKP03] Bumsig Kim, Andrew Kresch, and Tony Pantev, *Functoriality in Intersection Theory and a Conjecture of Cox, Katz, and Lee*, Journal of Pure and Applied Algebra **179** (2003), no. 1-2, 127–136. ↑2

[Kre09] Andrew Kresch, *On the geometry of Deligne-Mumford stacks*, Algebraic geometry—Seattle 2005. Part 1, 2009, pp. 259–271. ↑28, 52

[Laz17] Robert K Lazarsfeld, *Positivity in algebraic geometry I: Classical setting: line bundles and linear series*, Vol. 48, Springer, 2017. ↑19

[Lee01] Y-P Lee, *Quantum Lefschetz hyperplane theorem*, Inventiones mathematicae **145** (2001), no. 1, 121–149. ↑2

[LT98] Jun Li and Gang Tian, *Virtual moduli cycles and Gromov-Witten invariants of algebraic varieties*, Journal of the American Mathematical Society **11** (1998), no. 1, 119–174. ↑8

[LLY99] Bong H. Lian, Kefeng Liu, and S. T. Yau, *Mirror Principle I*, Surveys in Differential Geometry **5** (1999), no. 1, 405–454. ↑1

[Liu13] CCM Liu, *Localization in Gromov-Witten theory and orbifold Gromov-Witten theory, Handbook of Moduli, Volume II, 353425, Adv. Lect. Math.,(ALM) 25*, International Press and Higher Education Press, 2013. ↑58

[LS20] Chiu-Chu Melissa Liu and Artan Sheshmani, *Stacky GKM graphs and orbifold Gromov–Witten theory*, Asian Journal of Mathematics **24** (2020), no. 5, 855–902. ↑47, 48, 58

[LR22] Zhiyu Liu and Yongbin Ruan, *Castelnuovo Bound and Higher Genus Gromov-Witten Invariants of Quintic 3-folds*, CoRR (2022), available at 2210.13411. ↑2

[MH14] Guest Martin and Sakai Hironori, *Orbifold quantum D-modules associated to weighted projective spaces*, Commentarii Mathematici Helvetici **89** (2014), no. 2, 273–297 (English). ↑46

[Noo05] Behrang Noohi, *Foundations of Topological Stacks I*, CoRR (2005), available at math/0503247. ↑49

[NB06] Behrang Noohi and Kai Behrend, *Uniformization of Deligne-Mumford curves*, Journal für die Reine und Angewandte Mathematik **599** (2006), 111–153. ↑47

[OP18] Alessandro Oneto and Andrea Petracci, *On the quantum periods of del Pezzo surfaces with $\frac{1}{3}(1,1)$ singularities*, Advances in Geometry **18** (2018), no. 3, 303–336. ↑4

[Sta19] The Stacks project authors, *The Stacks project*, 2019. ↑57

[SW22] Nawaz Sultani and Rachel Webb, *Subtleties of Quantum Lefshetz Without Convexity*, CoRR (2022), available at 2208.08987. ↑5

[TT21] Xiang Tang and Hsian-Hua Tseng, *A quantum Leray–Hirsch theorem for banded gerbes*, Journal of Differential Geometry **119** (2021), no. 3, 459–511. ↑18, 20, 22, 34, 40

[Tse10] Hsian-Hua Tseng, *Orbifold Quantum Riemann-Roch, Lefschetz and Serre*, Geometry & Topology **14** (2010), no. 1, 1–81. ↑1, 4

[Wan25] Jun Wang, *A mirror theorem for Gromov-Witten theory without convexity*, Forum of Mathematics, Sigma **13** (2025), e72. ↑2, 7, 14, 21, 42, 53

[Web18] Rachel Webb, *The Abelian-Nonabelian Correspondence for I-functions*, CoRR (2018), available at 1804.07786. ↑7

[Web21] ———, *Abelianization and Quantum Lefschetz for Orbifold Quasimap I-Functions*, CoRR (2021), available at 2109.12223. ↑7

[Zho22] Yang Zhou, *Quasimap wall-crossing for GIT quotients*, Inventiones mathematicae **227** (2022), no. 2, 581–660. ↑2

[Zin08] Aleksey Zinger, *The Reduced Genus 1 Gromov-Witten Invariants of Calabi-Yau Hypersurfaces*, Journal of the American Mathematical Society **22** (2008), no. 3, 691–737. ↑2

Beijing International Center for Mathematical Research, Peking University, Beijing,100871, China
*E-mail address*: wangjun@bicmr.pku.edu.cn